Chapter 2. 2nd year Differentiation

Average rate of change

Let

f be a real valued function the (difference quotient) $\frac{f(x_1)-f(x)}{x_1-x}$ is called average rate of cahnge.

Derivative:

let $f(x)$be a function, then its derivative is denoted by $f'(x) or \frac{df}{dx}$ and defined as;

$$f'(x) = \lim_{\delta x\to 0} \frac{f(x+\delta x)-f(x)}{\delta x}$$

- The process of finding f' is called differentiation.

Notation for Derivative:

For the function $y = f(x)$

$\Rightarrow$ $\boldsymbol{y+\delta y = f(x+\delta x) \Rightarrow \delta y = f(x+\delta x)-f(x)}$

Where δy is the increment of y(change in the valve of y corresponding to δx (increment of x)dividing $(i)$by δx on both sides, we get

$\Rightarrow$ $\frac{\delta y}{\delta x} = \frac{f(x+\delta x)-f(x)}{\delta x}$

Taking limit on both sides as $\delta x \to 0$ we get

$\Rightarrow$ $\lim_{\delta x\to o} \frac{\delta y}{\delta x} = \lim_{\delta x\to o} \frac{f(x+\delta x)-f(x)}{\delta x} \to (ii)$

$\lim_{\delta x\to o} \frac{\delta y}{\delta x}$ is denoted by $\frac{dy}{dx}$, so $(ii)$is written as $\frac{dy}{dx} = f'(x)$

Note: the symbol $\frac{dy}{dx}$ is used for the derivative of y with replace to x and here it is not a quotient of dy and dx is also denoted by y'

*Different mathematician used symbols given below.

Leibniz $\frac{dy}{dx}$ or $\frac{df}{dx}$ **Newton** $f'(x)$

Lagrange $f'(x)$ $Cauchy$ $D(fx)$

Finding $f'(x)$ from definition of derivative:

Given a function

$f, f(x) if$ it exist can be found by the following four Steps.

Step1. Find $f(x+\delta x)$

$Step2. Simplify\ f(x+\delta x - f(x)$

Step3. Divide $f(x+\delta x)-f(x) by\ \delta x$ to get $\frac{f(x+\delta x)-f(x)}{\delta x}$

And simplify it.

Step4. Find $\lim_{\delta x\to o} \frac{f(x+\delta x)-f(x)}{\delta x}$ the method of finding derivatives by this process is called differentiation by definition or by from first principle.

Derivation of x^n when $n \in Z$

a) Let $y = x^n$ when n is $+ve$ integer.

$\Rightarrow$ $y = x^n \to (i)$, $y+\delta y = (x+\delta x)^n \to (ii)$

$by\ (ii)-(i) \Rightarrow y+\delta y - \boldsymbol{y} = \boldsymbol{(x+\delta x)^n - x^n}$

$\Rightarrow$ $\boldsymbol{\delta y = (x+\delta x)^n - x^n}$

Using binomial theorem, we have

$$= \delta x\left[nx^{n-1} + \frac{(n-1)}{2!}x^{n-2}(\delta x) + \cdots + (\delta x)^n - x^n\right]$$

Dividing both side by δx

$\Rightarrow$ $\boldsymbol{\delta y = \left[x^n + nx^{n-1}.\delta x + \frac{n(n-1)(n-2)}{2!}x^{n-2}.(\delta x)^2 + \cdots + (\delta x)^{n-1}\right]}$

$\Rightarrow$ $x^n + nx^{n-1}.\delta x + \frac{n(n-1)(n-2)}{2!}x^{n-2}.(\delta x)^2 + \cdots (\delta x)^{n-1}$

$$\frac{\delta y}{\delta x} = \frac{\delta x}{\delta x}\left[nx^{(n-1)} + \frac{(n-1)}{2!}x^{n-2}(\delta x) + \cdots + (\delta x)^{n-1}\right]$$

$applying\ both\ sides\ by\ \delta x \to 0$

$$\lim_{\delta x\to o}\frac{\delta y}{\delta x} = \lim_{\delta x\to o}\frac{\delta x}{\delta x}\left[nx^{(n-1)} + \frac{(n-1)}{2!}x^{n-2}(\delta x) + \cdots + (\delta x)^{n-1}\right]$$

$$\frac{dy}{dx} = nx^{n-1} + \frac{n-1}{2!}x^{n-1}(0) + \cdots + (0)^{n-1}$$

$\Rightarrow$ $\frac{dy}{dx} = nx^{(n-1)}$

b)

Let $y = x^n$ where n is $-ve$ integer.

let $n = -m(m\ is + ve\ integer)$. then

$$y = x^{-m} \Rightarrow y = \frac{1}{x^m} \to (i)$$

$\Rightarrow$ $y+\delta y = \frac{1}{(x+\delta x)^m} \to (ii)$

$$by\ (ii)-(i) \Rightarrow y+\delta y - y = \frac{1}{(x+\delta x)^m} - \frac{1}{x^m}$$

$$\Rightarrow \delta y = \frac{x^m-(x+\delta x)^m}{x^m(x+\delta x)^m}\ using\ binomial\ theorem$$

$$= \frac{x^m - (x^m + mx^{m-1}.\delta x + \frac{m(m-1)}{2!}x^{m-2}.(\delta x)^2 + \cdots + (\delta x)^2}{x^m(x+\delta x)^m}$$

$$= \frac{x^m - x^m - mx^{m-1}.\delta x - \frac{mx^{m-1}}{2!}x^{m-2}.(\delta x)^2 - \cdots - (\delta x)^2}{x^m(x+\delta x)^m}$$

$$\delta y = \frac{-\delta x}{x^m(x+\delta x)^m}\left(mx^{m-1} + \frac{m(m-1)}{2!}x^{m-2}.(\delta x) + \cdots + (\delta x)^{m-1}\right)$$

Dividing both sides $by\ \delta x$

$$\frac{\delta y}{\delta x} = \frac{-\delta x}{x^m(x+\delta x)^m}\left(\frac{1}{\delta x}\right)\left(mx^{m-1} + \frac{m(m-1)}{2!}x^{m-2}.(\delta x) + \cdots + (\delta x)^{m-1}\right)$$

Applying limit as $\delta x \to 0$

$$\lim_{\delta x\to o}\frac{\delta y}{\delta x} = \lim_{\delta x\to o}\left[\frac{-1}{x^m(x+\delta x)^m}\left(mx^{m-1} + \frac{m(m-1)}{2!}x^{m-2}.(\delta x) + \cdots + (\delta x)^{m-1}\right)\right]$$

$$\frac{dy}{dx} = -\frac{1}{x^m.x^m}\left(mx^{m-1} + \frac{m(m-1)}{2!}x^{m-2}.(0) + \cdots + (0)^{m-1}\right)$$

$$\frac{dy}{dx} = \frac{-mx^{m-1}}{x^{2m}} = -mx^{m-1-2m} = -mx^{-m-1}$$

$$\Rightarrow \frac{dy}{dx} = -mx^{m-1}\ or \frac{dy}{dx} = nx^{n-1}$$

So for we have proved that $\frac{dy}{dx} = nx^{n-1}$, $n \in Z$

Note that $\frac{d}{dx}(x^n) = nx^{n-1}$ is called power rule and holds if $n \in Q - Z$

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Exercise 2.1

Question # 1.

Find the definition, the derivatives w.r.t ' ' x of the following functions defined as:

(i). $2x^2 + 1$

Solution.

Let $y = 2x^2 + 1 \rightarrow (i)$

$y + \delta y = 2(x + \delta x)^2 + 1 \rightarrow (ii)$

$by\ (ii) - (i)$

$\delta y = 2(x + \delta x)^2 + 1 - y$

$\delta y = 2(x + \delta x)^2 + 1 - 2x^2 - 1$

$\delta y = 2(x^2 + \delta x^2 + 2x\delta x) - 2x^2$

$\delta y = 2x^2 + 2\delta x^2 + 4x\delta x - 2x^2$

$\delta y = 2\delta x^2 + 4x\delta x$

$\delta y = \delta x(2\delta x + 4x)$

Dividing both sides by δx

$\frac{\delta y}{\delta x} = (2\delta x + 4x)$

Taking limit when $\delta x \rightarrow 0$

$\lim_{\delta x \to 0} \frac{\delta y}{\delta x} = \lim_{\delta x \to 0} (2\delta x + 4x)$

$\frac{dy}{dx} = 2(0) + 4x$

$\boldsymbol{\frac{dy}{dx} = 4x}$

Which is the required.

(ii). $2 - \sqrt{x}$

Solution.

Let $y = 2 - \sqrt{x} \rightarrow (i)$

$y + \delta y = 2 - \sqrt{x + \delta x} \rightarrow (ii)$

$eq(ii) - eq(i)$

$\delta y = 2 - \sqrt{x + \delta x} - y$

$\delta y = 2 - \sqrt{x + \delta x} - 2 + \sqrt{x}$

$\delta y = \sqrt{x} - \sqrt{x + \delta x}$

$\delta y = x^{\frac{1}{2}} - (x + \delta x)^{\frac{1}{2}}$

$\delta y = x^{\frac{1}{2}} - x^{\frac{1}{2}}\left(1 + \frac{\delta x}{x}\right)^{\frac{1}{2}}$

$\delta y = x^{\frac{1}{2}} - x^{\frac{1}{2}}\left(1 + \frac{1}{2}.\frac{\delta x}{x} + \frac{\frac{1}{2}\left(\frac{1}{2} - 1\right)}{2!}\left(\frac{\delta x}{x}\right)^2 + \cdots\right)$

$\delta y = x^{\frac{1}{2}} - x^{\frac{1}{2}} - x^{\frac{1}{2}}\left(\frac{1}{2}.\frac{\delta x}{x} + \frac{\frac{1}{2}\left(\frac{1}{2} - 1\right)}{2!}\left(\frac{\delta x}{x}\right)^2 + \cdots\right)$

$\delta y = -x^{\frac{1}{2}}\delta x\left(\frac{1}{2x} + \frac{\frac{1}{2}\left(\frac{1}{2} - 1\right)}{2!}\frac{\delta x}{x^2} + \cdots\right)$

Dividing both sides byδx , we have

$\frac{\delta y}{\delta x} = -x^{\frac{1}{2}}\left(\frac{1}{2x} + \frac{\frac{1}{2}\left(\frac{1}{2} - 1\right)}{2!}\frac{\delta x}{x^2} + \cdots\right)$

Taking limit when $\delta x \rightarrow 0$

$\lim_{\delta x \to 0} \frac{\delta y}{\delta x} = \lim_{\delta x \to 0} -x^{\frac{1}{2}}\left(\frac{1}{2x} + \frac{\frac{1}{2}\left(\frac{1}{2} - 1\right)}{2!}\frac{\delta x}{x^2} + \cdots\right)$

$\frac{dy}{dx} = \lim_{\delta x \to 0} -x^{\frac{1}{2}}\left(\frac{1}{2x} + \frac{\frac{1}{2}\left(\frac{1}{2} - 1\right)}{2!}\frac{(0)}{x^2} + \cdots\right)$

$\frac{dy}{dx} = -x^{\frac{1}{2}}\left(\frac{1}{2x}\right)$

$\frac{dy}{dx} = -\frac{1}{2}x^{\frac{1}{2}-1}$

$\frac{dy}{dx} = -\frac{1}{2}x^{-\frac{1}{2}}$

$\boldsymbol{\frac{dy}{dx} = -\frac{1}{2x^{\frac{1}{2}}}}$ **Which is the required.**

(iii). $\frac{1}{\sqrt{x}}$

Solution.

Let $y = \frac{1}{\sqrt{x}} = x^{-\frac{1}{2}} \rightarrow (i)$

$y + \delta y = (x + \delta x)^{-\frac{1}{2}} \rightarrow (ii)$

$eq(ii) - eq(i)$

$\delta y = (x + \delta x)^{-\frac{1}{2}} - y$

$\delta y = (x + \delta x)^{-\frac{1}{2}} - x^{-\frac{1}{2}}$

$\delta y = (x + \delta x)^{-\frac{1}{2}} - x^{-\frac{1}{2}}$

$\delta y = x^{-\frac{1}{2}}\left(1 + \frac{\delta x}{x}\right)^{-\frac{1}{2}} - x^{-\frac{1}{2}}$

$\delta y = x^{-\frac{1}{2}}\left(1 - \frac{1}{2}.\frac{\delta x}{x} + \frac{-\frac{1}{2}\left(-\frac{1}{2} - 1\right)}{2!}\left(\frac{\delta x}{x}\right)^2 + \cdots\right) - x^{-\frac{1}{2}}$

$\delta y = x^{-\frac{1}{2}} + x^{-\frac{1}{2}}\left(-\frac{1}{2}.\frac{\delta x}{x} + \frac{-\frac{1}{2}\left(-\frac{1}{2} - 1\right)}{2!}\left(\frac{\delta x}{x}\right)^2 + \cdots\right) - x^{-\frac{1}{2}}$

$\delta y = x^{-\frac{1}{2}}\delta x\left(-\frac{1}{2x} + \frac{-\frac{1}{2}\left(-\frac{1}{2} - 1\right)}{2!}\frac{\delta x}{x^2} + \cdots\right)$

Dividing both sides by δx , we have

$\frac{\delta y}{\delta x} = x^{-\frac{1}{2}}\left(-\frac{1}{2x} + \frac{-\frac{1}{2}\left(-\frac{1}{2} - 1\right)}{2!}\frac{\delta x}{x^2} + \cdots\right)$

Taking limit when $\delta x \rightarrow 0$

$\lim_{\delta x \to 0} \frac{\delta y}{\delta x} = \lim_{\delta x \to 0} x^{-\frac{1}{2}}\left(-\frac{1}{2x} + \frac{-\frac{1}{2}\left(-\frac{1}{2} - 1\right)}{2!}\frac{\delta x}{x^2} + \cdots\right)$

$\frac{dy}{dx} = x^{-\frac{1}{2}}\left(-\frac{1}{2x} + \frac{-\frac{1}{2}\left(-\frac{1}{2} - 1\right)}{2!}\frac{(0)}{x^2} + \cdots\right)$

$\frac{dy}{dx} = x^{-\frac{1}{2}}\left(-\frac{1}{2x}\right)$

$\frac{dy}{dx} = -\frac{1}{2}x^{-\frac{1}{2}-1}$

$\frac{dy}{dx} = -\frac{1}{2}x^{-\frac{3}{2}}$

$\boldsymbol{\frac{dy}{dx} = -\frac{1}{2x^{\frac{3}{2}}}}$

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Which is the required.

(iv) $\frac{1}{x^3}$

Solution.

Let $y = \frac{1}{x^3} = x^{-3} \to (i)$

$y + \delta y = (x + \delta x)^{-3} \to (ii)$

$eq(ii) - eq(i)$

$\delta y = (x + \delta x)^{-3} - y$

$\delta y = (x + \delta x)^{-3} - x^{-3}$

$\delta y = x^{-3}(1 + \frac{\delta x}{x})^{-3} - x^{-3}$

$\delta y = x^{-3}((1 + \frac{\delta x}{x})^{-3} - 1)$

$\delta y = x^{-3}(1 - \frac{3\delta x}{x} + \frac{-3(-3-1)}{2!}\left(\frac{\delta x}{x}\right)^2 + \cdots - 1)$

$\delta y = x^{-3}\left(-\frac{3\delta x}{x} + \frac{-3(-3-1)}{2!}\left(\frac{\delta x}{x}\right)^2 + \cdots\right)$

$\delta y = x^{-3}\delta x\left(-\frac{3}{x} + \frac{-3(-3-1)}{2!}\frac{\delta x}{x^2} + \cdots\right)$

Dividing both sides by δx , we have

$\frac{\delta y}{\delta x} = x^{-3}\left(-\frac{3}{x} + \frac{-3(-3-1)}{2!}\frac{\delta x}{x^2} + \cdots\right)$

Taking limit when $\delta x \to 0$

$\lim_{\delta x \to 0}\frac{\delta y}{\delta x} = \lim_{\delta x \to 0} x^{-3}\left(-\frac{3}{x} + \frac{-3(-3-1)}{2!}\frac{\delta x}{x^2} + \cdots\right)$

$\frac{dy}{dx} = x^{-3}\left(-\frac{3}{x} + \frac{-3(-3-1)}{2!}\frac{(0)}{x^2} + \cdots\right)$

$\frac{dy}{dx} = -\frac{3}{x^{1+3}}$

$\frac{dy}{dx} = -\frac{3}{x^4}$

Which is the required.

(v). $\frac{1}{x-a}$

Solution.

Let $y = \frac{1}{x-a} = (x-a)^{-1} \to (i)$

$y + \delta y = (x + \delta x - a)^{-1} \to (ii)$

$eq(ii) - eq(i)$

$\delta y = (x + \delta x - a)^{-3} - y$

$\delta y = (x - a + \delta x)^{-1} - (x-a)^{-1}$

$\delta y = (x-a)^{-1}(1 + \frac{\delta x}{x-a})^{-1} - (x-a)^{-1}$

$\delta y = (x-a)^{-1}((1 + \frac{\delta x}{x-a})^{-1} - 1)$

$\delta y = (x-a)^{-1}(1 - \frac{\delta x}{x-a} + \frac{-1(-1-1)}{2!}\left(\frac{\delta x}{x-a}\right)^2 + \cdots - 1)$

$\delta y = (x-a)^{-1}\left(-\frac{\delta x}{x-a} + \frac{-1(-1-1)}{2!}\left(\frac{\delta x}{x-a}\right)^2 + \cdots\right)$

$\delta y = (x-a)^{-1}\delta x\left(-\frac{1}{x-a} + \frac{-1(-1-1)}{2!}\frac{\delta x}{(x-a)^2} + \cdots\right)$

Dividing both sides by δx , we have

$\frac{\delta y}{\delta x} = (x-a)^{-1}\left(-\frac{1}{x-a} + \frac{-1(-1-1)}{2!}\frac{\delta x}{(x-a)^2} + \cdots\right)$

Taking limit when $\delta x \to 0$

$\lim_{\delta x \to 0}\frac{\delta y}{\delta x} = \lim_{\delta x \to 0}(x-a)^{-1}\left(-\frac{1}{x-a} + \frac{-1(-1-1)}{2!}\frac{\delta x}{(x-a)^2} + \cdots\right)$

$\frac{dy}{dx} = (x-a)^{-1}\left(-\frac{1}{x-a} + \frac{-1(-1-1)}{2!}\frac{(0)}{(x-a)^2} + \cdots\right)$

$\frac{dy}{dx} = -\frac{1}{(x-a)^{1+1}}$

$\frac{dy}{dx} = -\frac{1}{(x-a)^2}$

Which is the required.

(vi). $x(x-3)$

Solution.

Let $y = x(x-3) = x^2 - 3x$

$y + \delta y = (x + \delta x)^2 - 3(x + \delta x)$

$\delta y = (x + \delta x)^2 - 3x - 3\delta x - y$

$\delta y = x^2 + \delta x^2 + 2x\delta x - 3x - 3\delta x - x^2 + 3x$

$\delta y = \delta x^2 + 2x\delta x - 3\delta x$

$\delta y = \delta x(\delta x + 2x - 3)$

Dividing both sides by δx , we have

$\frac{\delta y}{\delta x} = (\delta x + 2x - 3)$

Taking limit when $\delta x \to 0$

$\lim_{\delta x \to 0}\frac{\delta y}{\delta x} = \lim_{\delta x \to 0}(\delta x + 2x - 3)$

$\frac{dy}{dx} = (0 + 2x - 3)$

$\boldsymbol{\frac{dy}{dx} = 2x - 3}$

Which is the required.

(vii) $\frac{2}{x^4}$

Solution:

Let $y = \frac{2}{x^4} = 2x^{-4} \to (i)$

$y + \delta y = 2(x + \delta x)^{-4} \to (ii)$

$e(ii) - e(i)$

$\delta y = 2(x + \delta x)^{-4} - y$

$\delta y = 2(x + \delta x)^{-4} - 2x^{-4}$

$\delta y = x^{-4}(1 + \frac{\delta x}{x})^{-4} - 2x^{-4}$

$\delta y = 2x^{-4}((1 + \frac{\delta x}{x})^{-4} - 1)$

$\delta y = 2x^{-4}(1 - \frac{4\delta x}{x} + \frac{-4(-4-1)}{2!}\left(\frac{\delta x}{x}\right)^2 + \cdots - 1)$

$\delta y = 2\,x^{-4}\left(-\frac{4\delta x}{x} + \frac{-4(-4-1)}{2!}\left(\frac{\delta x}{x}\right)^2 + \cdots\right)$

$\delta y = 2x^{-4}\delta x\left(-\frac{4}{x} + \frac{-4(-4-1)}{2!}\frac{\delta x}{x^2} + \cdots\right)$

Dividing both sides by δx , we have

$\frac{\delta y}{\delta x} = 2x^{-4}\left(-\frac{4}{x} + \frac{-4(-4-1)}{2!}\frac{\delta x}{x^2} + \cdots\right)$

Taking limit when $\delta x \to 0$

$\lim_{\delta x \to 0}\frac{\delta y}{\delta x} = \lim_{\delta x \to 0} 2x^{-4}\left(-\frac{4}{x} + \frac{-4(-4-1)}{2!}\frac{\delta x}{x^2} + \cdots\right)$

$\frac{dy}{dx} = 2x^{-4}\left(-\frac{4}{x} + \frac{-4(-4-1)}{2!}\frac{(0)}{x^2} + \cdots\right)$

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$$\frac{dy}{dx} = -\frac{8}{x^{1+4}}$$
$$\frac{dy}{dx} = -\frac{8}{x^5}$$
Which is the required.

(viii). $(x+4)^{\frac{1}{3}}$

Solution.

Let $y = (x+4)^{\frac{1}{3}} \to (i)$

$y + \delta y = (x + \delta x + 4)^{\frac{1}{3}} \to (ii)$

$eq(ii) - eq(i)$

$$\delta y = (x + \delta x + 4)^{\frac{1}{3}} - y$$
$$\delta y = (x + 4 + \delta x)^{\frac{1}{3}} - (x+4)^{\frac{1}{3}}$$
$$\delta y = (x+4)^{\frac{1}{3}}\left(1 + \frac{\delta x}{x+4}\right)^{\frac{1}{3}} - (x+4)^{\frac{1}{3}}$$
$$\delta y = (x+4)^{\frac{1}{3}}\left(\left(1 + \frac{1}{3}.\frac{\delta x}{x+4} + \frac{\frac{1}{3}\left(\frac{1}{3}-1\right)}{2!}\left(\frac{\delta x}{x+4}\right)^2 + \cdots\right) - 1\right)$$
$$\delta y = (x+4)^{\frac{1}{3}}\left(1 + \frac{1}{3}.\frac{\delta x}{x+4} + \frac{\frac{1}{3}\left(\frac{1}{3}-1\right)}{2!}\left(\frac{\delta x}{x+4}\right)^2 + \cdots - 1\right)$$
$$\delta y = (x+4)^{\frac{1}{3}}\delta x\left(\frac{1}{3}.\frac{1}{x+4} + \frac{\frac{1}{3}\left(\frac{1}{3}-1\right)}{2!}\frac{\delta x}{(x+4)^2} + \cdots\right)$$

Dividing both sides by δx , we have

$$\frac{\delta y}{\delta x} = (x+4)^{\frac{1}{3}}\left(\frac{1}{3}.\frac{1}{x+4} + \frac{\frac{1}{3}\left(\frac{1}{3}-1\right)}{2!}\frac{\delta x}{(x+4)^2} + \cdots\right)$$

Taking limit when $\delta x \to 0$

$$\lim_{\delta x \to 0}\frac{\delta y}{\delta x} = \lim_{\delta x \to 0}(x+4)^{\frac{1}{3}}\left(\frac{1}{3}.\frac{1}{x+4} + \frac{\frac{1}{3}\left(\frac{1}{3}-1\right)}{2!}\frac{\delta x}{(x+4)^2} + \cdots\right)$$
$$\frac{dy}{dx} = (x+4)^{\frac{1}{3}}\left(\frac{1}{3}.\frac{1}{x+4} + \frac{\frac{1}{3}\left(\frac{1}{3}-1\right)}{2!}\frac{(0)}{(x+4)^2} + \cdots\right)$$
$$\frac{dy}{dx} = (x+4)^{\frac{1}{3}}.\frac{1}{3(x+4)}$$
$$\frac{dy}{dx} = \frac{1}{3(x+4)^{1-\frac{1}{3}}}$$
$$\boldsymbol{\frac{dy}{dx} = \frac{1}{3(x+4)^{\frac{2}{3}}}}$$
Which is the required.

(ix). $x^{\frac{3}{2}}$

Solution.

Let $y = x^{\frac{3}{2}} \to (i)$

$y + \delta y = (x + \delta x)^{\frac{3}{2}} \to (ii)$

$eq(ii) - eq(i)$

$$\delta y = (x + \delta x)^{\frac{3}{2}} - y$$
$$\delta y = (x + \delta x)^{\frac{3}{2}} - x^{\frac{3}{2}}$$
$$\delta y = x^{\frac{3}{2}}\left(1 + \frac{\delta x}{x}\right)^{\frac{3}{2}} - x^{\frac{3}{2}}$$
$$\delta y = x^{\frac{3}{2}}\left(1 + \frac{3}{2}.\frac{\delta x}{x} + \frac{\frac{3}{2}\left(\frac{3}{2}-1\right)}{2!}\left(\frac{\delta x}{x}\right)^2 + \cdots\right) - x^{\frac{3}{2}}$$
$$\delta y = x^{\frac{3}{2}} + x^{\frac{3}{2}}\left(\frac{3}{2}.\frac{\delta x}{x} + \frac{\frac{3}{2}\left(\frac{3}{2}-1\right)}{2!}\left(\frac{\delta x}{x}\right)^2 + \cdots\right) - x^{\frac{3}{2}}$$
$$\delta y = x^{\frac{3}{2}}\delta x\left(\frac{3}{2x} + \frac{\frac{3}{2}\left(\frac{3}{2}-1\right)}{2!}\frac{\delta x}{x^2} + \cdots\right)$$

Dividing both sides by δx , we have

$$\frac{\delta y}{\delta x} = x^{\frac{3}{2}}\left(\frac{3}{2x} + \frac{\frac{3}{2}\left(\frac{3}{2}-1\right)}{2!}\frac{\delta x}{x^2} + \cdots\right)$$

Taking limit when $\delta x \to 0$

$$\lim_{\delta x \to 0}\frac{\delta y}{\delta x} = \lim_{\delta x \to 0} x^{\frac{3}{2}}\left(\frac{3}{2x} + \frac{\frac{3}{2}\left(\frac{3}{2}-1\right)}{2!}\frac{\delta x}{x^2} + \cdots\right)$$
$$\frac{dy}{dx} = x^{\frac{3}{2}}\left(\frac{3}{2x} + \frac{\frac{3}{2}\left(\frac{3}{2}-1\right)}{2!}\frac{(0)}{x^2} + \cdots\right)$$
$$\frac{dy}{dx} = x^{\frac{3}{2}}\left(\frac{3}{2x}\right)$$
$$\frac{dy}{dx} = \frac{3}{2}x^{\frac{3}{2}-1}$$
$$\boldsymbol{\frac{dy}{dx} = \frac{3}{2}x^{\frac{1}{2}}}$$
Which is the required.

(x). $x^{\frac{5}{2}}$

Solution.

Let $y = x^{\frac{5}{2}} \to (i)$

$y + \delta y = (x + \delta x)^{\frac{5}{2}} \to (ii)$

$$\delta y = (x + \delta x)^{\frac{5}{2}} - y$$
$$\delta y = (x + \delta x)^{\frac{5}{2}} - x^{\frac{5}{2}}$$
$$\delta y = x^{\frac{5}{2}}\left(1 + \frac{\delta x}{x}\right)^{\frac{5}{2}} - x^{\frac{5}{2}}$$
$$\delta y = x^{\frac{5}{2}}\left(1 + \frac{5}{2}.\frac{\delta x}{x} + \frac{\frac{5}{2}\left(\frac{5}{2}-1\right)}{2!}\left(\frac{\delta x}{x}\right)^2 + \cdots\right) - x^{\frac{5}{2}}$$
$$\delta y = x^{\frac{5}{2}} + x^{\frac{5}{2}}\left(\frac{5}{2}.\frac{\delta x}{x} + \frac{\frac{5}{2}\left(\frac{5}{2}-1\right)}{2!}\left(\frac{\delta x}{x}\right)^2 + \cdots\right) - x^{\frac{5}{2}}$$
$$\delta y = x^{\frac{5}{2}}\delta x\left(\frac{5}{2x} + \frac{\frac{5}{2}\left(\frac{5}{2}-1\right)}{2!}\frac{\delta x}{x^2} + \cdots\right)$$

Dividing both sides by δx , we have

$$\frac{\delta y}{\delta x} = x^{\frac{5}{2}}\left(\frac{5}{2x} + \frac{\frac{5}{2}\left(\frac{5}{2}-1\right)}{2!}\frac{\delta x}{x^2} + \cdots\right)$$

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Taking limit when $\delta x \to 0$

$$\lim_{\delta x\to 0}\frac{\delta y}{\delta x}=\lim_{\delta x\to 0}x^{\frac{5}{2}}\left(\frac{5}{2x}+\frac{\frac{5}{2}\left(\frac{5}{2}-1\right)}{2!}\frac{\delta x}{x^2}+\cdots\right)$$

$$\frac{dy}{dx}=x^{\frac{5}{2}}\left(\frac{5}{2x}+\frac{\frac{5}{2}\left(\frac{5}{2}-1\right)}{2!}\frac{(0)}{x^2}+\cdots\right)$$

$$\frac{dy}{dx}=x^{\frac{5}{2}}\left(\frac{5}{2x}\right)$$

$$\frac{dy}{dx}=\frac{5}{2}x^{\frac{5}{2}-1}$$

$$\boldsymbol{\frac{dy}{dx}=\frac{5}{2}x^{\frac{3}{2}}}$$

Which is the required.

(xi).x^m , $m \in N$

Solution.

Let $y=x^m \to (i)$

$y+\delta y=\ (x+\delta x)^m \to (ii)$

$eq(ii)-eq(i)$

$\delta y=\ (x+\delta x)^m-y$

$\delta y=\ (x+\delta x)^m-x^m$

$$\delta y=\ x^m(1+\frac{\delta x}{x})^m-x^m$$

$$\delta y=\ x^m((1+\frac{\delta x}{x})^m-1)$$

$$\delta y=\ x^m(1+\frac{m\delta x}{x}+\frac{m(m-1)}{2!}\left(\frac{\delta x}{x}\right)^2+\cdots-1)$$

$$\delta y=\ x^m\left(\frac{m\delta x}{x}+\frac{m(m-1)}{2!}\left(\frac{\delta x}{x}\right)^2+\cdots\right)$$

$$\delta y=\ x^m\delta x\left(\frac{m}{x}+\frac{m(m-1)}{2!}\frac{\delta x}{x^2}+\cdots\right)$$

Dividing both sides by δx , we have

$$\frac{\delta y}{\delta x}=\ x^m\left(\frac{m}{x}+\frac{m(m-1)}{2!}\frac{\delta x}{x^2}+\cdots\right)$$

Taking limit when $\delta x \to 0$

$$\lim_{\delta x\to 0}\frac{\delta y}{\delta x}=\lim_{\delta x\to 0}x^m\left(\frac{m}{x}+\frac{m(m-1)}{2!}\frac{\delta x}{x^2}+\cdots\right)$$

$$\frac{dy}{dx}=x^m\left(\frac{m}{x}+\frac{m(m-1)}{2!}\frac{(0)}{x^2}+\cdots\right)$$

$$\boldsymbol{\frac{dy}{dx}=mx^{m-1}}$$

Which is the required.

(xii). $\frac{1}{x^m}$, $m \in N$

Solution.

Let $y=x^{-m} \to (i)$

$y+\delta y=\ (x+\delta x)^{-m} \to (ii)$

$eq(ii)-eq(i)$

$\delta y=\ (x+\delta x)^{-m}-y$

$\delta y=\ (x+\delta x)^{-m}-x^{-m}$

$$\delta y=\ x^{-m}(1+\frac{\delta x}{x})^{-m}-x^{-m}$$

$$\delta y=\ x^{-m}(\left(1+\frac{\delta x}{x}\right)^{-m}-1)$$

$$\delta y=\ x^{-m}(1+\frac{-m\delta x}{x}+\frac{-m(-m-1)}{2!}\left(\frac{\delta x}{x}\right)^2+\cdots-1)$$

$$\delta y=\ x^{-m}\left(\frac{-m\delta x}{x}+\frac{-m(-m-1)}{2!}\left(\frac{\delta x}{x}\right)^2+\cdots\right)$$

$$\delta y=\ x^{-m}\delta x\left(\frac{-m}{x}+\frac{-m(-m-1)}{2!}\frac{\delta x}{x^2}+\cdots\right)$$

Dividing both sides by δx , we have

$$\frac{\delta y}{\delta x}=\ x^{-m}\left(\frac{-m}{x}+\frac{-m(-m-1)}{2!}\frac{\delta x}{x^2}+\cdots\right)$$

Taking limit when $\delta x \to 0$

$$\lim_{\delta x\to 0}\frac{\delta y}{\delta x}=\lim_{\delta x\to 0}x^{-m}\left(\frac{-m}{x}+\frac{-m(-m-1)}{2!}\frac{\delta x}{x^2}+\cdots\right)$$

$$\frac{dy}{dx}=x^{-m}\left(\frac{-m}{x}+\frac{-m(-m-1)}{2!}\frac{(0)}{x^2}+\cdots\right)$$

$$\frac{dy}{dx}=-\frac{m}{x^{1+m}}$$

$$\boldsymbol{\frac{dy}{dx}=-\frac{m}{x^{m+1}}}$$

Which is the required.

(xiii).x^{40}

Solution.

Let $y=x^{40} \to (i)$

$y+\delta y=\ (x+\delta x)^{40} \to (ii)$

$eq(ii)-eq(i)$

$\delta y=\ (x+\delta x)^{40}-y$

$\delta y=\ (x+\delta x)^{40}-x^{40}$

$$\delta y=\ x^{40}(1+\frac{\delta x}{x})^{40}-x^{40}$$

$$\delta y=\ x^{40}((1+\frac{\delta x}{x})^{40}-1)$$

$$\delta y=\ x^{40}(1+\frac{40\delta x}{x}+\frac{40(40-1)}{2!}\left(\frac{\delta x}{x}\right)^2+\cdots-1)$$

$$\delta y=\ x^{40}\left(\frac{40\delta x}{x}+\frac{40(40-1)}{2!}\left(\frac{\delta x}{x}\right)^2+\cdots\right)$$

$$\delta y=\ x^{40}\delta x\left(\frac{40}{x}+\frac{40(40-1)}{2!}\frac{\delta x}{x^2}+\cdots\right)$$

Dividing both sides by δx , we have

$$\frac{\delta y}{\delta x}=\ x^{40}\left(\frac{40}{x}+\frac{40(40-1)}{2!}\frac{\delta x}{x^2}+\cdots\right)$$

Taking limit when $\delta x \to 0$

$$\lim_{\delta x\to 0}\frac{\delta y}{\delta x}=\lim_{\delta x\to 0}x^{40}\left(\frac{40}{x}+\frac{40(40-1)}{2!}\frac{\delta x}{x^2}+\cdots\right)$$

$$\frac{dy}{dx}=x^{40}\left(\frac{40}{x}+\frac{40(40-1)}{2!}\frac{(0)}{x^2}+\cdots\right)$$

$$\frac{dy}{dx}=40x^{40-1}$$

$$\frac{dy}{dx}=40x^{39}$$

Which is the required.

(xiv).x^{-100}

Solution.

Let $y=x^{-100} \to (i)$

$y+\delta y=\ (x+\delta x)^{-100} \to (ii)$

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$$\delta y = (x+\delta x)^{-100} - y$$
$$\delta y = (x+\delta x)^{-100} - x^{-100}$$
$$\delta y = x^{-100}(1+\frac{\delta x}{x})^{-100} - x^{-100}$$
$$\delta y = x^{-100}(\left(1+\frac{\delta x}{x}\right)^{-100} - 1)$$
$$\delta y = x^{-100}(1+\frac{-100\delta x}{x}+\frac{-100(-100-1)}{2!}\left(\frac{\delta x}{x}\right)^2 + \cdots - 1)$$
$$\delta y = x^{-100}\left(\frac{-100\delta x}{x}+\frac{-100(-100-1)}{2!}\left(\frac{\delta x}{x}\right)^2+\cdots\right)$$
$$\delta y = x^{-100}\delta x\left(\frac{-100}{x}+\frac{-100(-100-1)}{2!}\frac{\delta x}{x^2}+\cdots\right)$$

Dividing both sides by δx , we have

$$\frac{\delta y}{\delta x} = x^{-100}\left(\frac{-100}{x}+\frac{-100(-100-1)}{2!}\frac{\delta x}{x^2}+\cdots\right)$$

Taking limit when $\delta x \to 0$

$$\lim_{\delta x\to 0}\frac{\delta y}{\delta x} = \lim_{\delta x\to 0} x^{-100}\left(\frac{-100}{x}+\frac{-100(-100-1)}{2!}\frac{\delta x}{x^2}+\cdots\right)$$
$$\delta y = x^{-100}\delta x\left(\frac{-100}{x}+\frac{-100(-100-1)}{2!}\frac{\delta x}{x^2}+\cdots\right)$$

Dividing both sides byδx , we have

$$\frac{\delta y}{\delta x} = x^{-100}\left(\frac{-100}{x}+\frac{-100(-100-1)}{2!}\frac{\delta x}{x^2}+\cdots\right)$$

Taking limit when $\delta x \to 0$

$$\lim_{\delta x\to 0}\frac{\delta y}{\delta x} = \lim_{\delta x\to 0} x^{-100}\left(\frac{-100}{x}+\frac{-100(-100-1)}{2!}\frac{\delta x}{x^2}+\cdots\right)$$
$$\frac{dy}{dx} = x^{-100}\left(\frac{-100}{x}+\frac{-100(-100-1)}{2!}\frac{(0)}{x^2}+\cdots\right)$$
$$\frac{dy}{dx} = -\frac{100}{x^{1+100}}$$
$$\boldsymbol{\frac{dy}{dx} = -\frac{100}{x^{101}}}$$

Which is the required.

Q2. Find $\frac{dy}{dx}$ *froom first principal if*

(i) $\sqrt{x+2}$

Solution.

Let $y = (x+2)^{\frac{1}{2}} \to (i)$

$$y+\delta y = (x+\delta x+2)^{\frac{1}{2}} \to (ii)$$
$$\delta y = (x+\delta x+2)^{\frac{1}{2}} - y$$
$$\delta y = (x+2+\delta x)^{\frac{1}{2}} - (x+2)^{\frac{1}{2}}$$
$$\delta y = (x+2)^{\frac{1}{2}}\left(1+\frac{\delta x}{x+2}\right)^{\frac{1}{2}} - (x+2)^{\frac{1}{2}}$$
$$\delta y = (x+2)^{\frac{1}{2}}(\left(1+\frac{\delta x}{x+2}\right)^{\frac{1}{2}} - 1)$$
$$\delta y = (x+2)^{\frac{1}{2}}(1+\frac{1}{2}\frac{\delta x}{(x+2)}+\frac{\frac{1}{2}\left(\frac{1}{2}-1\right)}{2!}\left(\frac{\delta x}{x+2}\right)^2+\cdots-1)$$
$$\delta y = (x+2)^{\frac{1}{2}}\delta x(\frac{1}{2}\frac{1}{(x+2)}+\frac{\frac{1}{2}\left(\frac{1}{2}-1\right)}{2!}\frac{\delta x}{(x+2)^2}+\cdots)$$

Dividing both sides by δx , we have

$$\frac{\delta y}{\delta x} = (x+2)^{\frac{1}{2}}(\frac{1}{2}\frac{1}{(x+2)}+\frac{\frac{1}{2}\left(\frac{1}{2}-1\right)}{2!}\frac{\delta x}{(x+2)^2}+\cdots)$$

Taking limit when $\delta x \to 0$

$$\lim_{\delta x\to 0}\frac{\delta y}{\delta x} = \lim_{\delta x\to 0}(x+2)^{\frac{1}{2}}(\frac{1}{2}\frac{1}{(x+2)}+\frac{\frac{1}{2}\left(\frac{1}{2}-1\right)}{2!}\frac{\delta x}{(x+2)^2}+\cdots)$$
$$\frac{dy}{dx} = (x+2)^{\frac{1}{2}}(\frac{1}{2}\frac{1}{(x+2)}+\frac{\frac{1}{2}\left(\frac{1}{2}-1\right)}{2!}\frac{(0)}{(x+2)^2}+\cdots)$$
$$\frac{dy}{dx} = (x+2)^{\frac{1}{2}}\frac{1}{2}\frac{1}{(x+2)}$$
$$\frac{dy}{dx} = \frac{1}{2(x+2)^{1-\frac{1}{2}}}$$
$$\boldsymbol{\frac{dy}{dx} = \frac{1}{2(x+2)^{\frac{1}{2}}}}$$

Which is the required.

(ii). $\frac{1}{\sqrt{x+a}}$

Solution.

Let $y = (x+a)^{-\frac{1}{2}} \to (i)$

$$y+\delta y = (x+\delta x+a)^{-\frac{1}{2}} \to (ii)$$
$$\delta y = (x+\delta x+2)^{\frac{1}{2}} - y$$
$$\delta y = (x+a+\delta x)^{-\frac{1}{2}} - (x+a)^{-\frac{1}{2}}$$
$$\delta y = (x+a)^{-\frac{1}{2}}\left(1+\frac{\delta x}{x+a}\right)^{-\frac{1}{2}} - (x+a)^{-\frac{1}{2}}$$
$$\delta y = (x+a)^{-\frac{1}{2}}(\left(1+\frac{\delta x}{x+a}\right)^{-\frac{1}{2}} - 1)$$
$$\delta y = (x+a)^{-\frac{1}{2}}(1-\frac{1}{2}\frac{\delta x}{(x+a)}+\frac{-\frac{1}{2}\left(-\frac{1}{2}-1\right)}{2!}\left(\frac{\delta x}{x+a}\right)^2+\cdots-1)$$
$$\delta y = (x+a)^{-\frac{1}{2}}\delta x(-\frac{1}{2}\frac{1}{(x+a)}+\frac{-\frac{1}{2}\left(-\frac{1}{2}-1\right)}{2!}\frac{\delta x}{(x+a)^2}+\cdots)$$

Dividing both sides by δx , we have

$$\frac{\delta y}{\delta x} = (x+a)^{-\frac{1}{2}}(-\frac{1}{2}\frac{1}{(x+a)}+\frac{-\frac{1}{2}\left(-\frac{1}{2}-1\right)}{2!}\frac{\delta x}{(x+a)^2}+\cdots)$$

Taking limit when $\delta x \to 0$

$$\lim_{\delta x\to 0}\frac{\delta y}{\delta x} = \lim_{\delta x\to 0}(x+a)^{-\frac{1}{2}}(-\frac{1}{2}\frac{1}{(x+a)}+\frac{-\frac{1}{2}\left(-\frac{1}{2}-1\right)}{2!}\frac{\delta x}{(x+a)^2}+\cdots)$$
$$\frac{dy}{dx} = (x+a)^{-\frac{1}{2}}(-\frac{1}{2}\frac{1}{(x+a)}+\frac{-\frac{1}{2}\left(-\frac{1}{2}-1\right)}{2!}\frac{(0)}{(x+a)^2}+\cdots)$$
$$\frac{dy}{dx} = (x+a)^{-\frac{1}{2}}\left(-\frac{1}{2}\frac{1}{(x+a)}\right)$$
$$\frac{dy}{dx} = -\frac{1}{2(x+a)^{1+\frac{1}{2}}}$$
$$\boldsymbol{\frac{dy}{dx} = -\frac{1}{2(x+a)^{\frac{3}{2}}}}$$

Which is the required.

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these notes are Witten by Amir Shehzad

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Differentiation of expression of the types.

$(ax+b)^n \ \ and \frac{1}{(ax+b)^n} \quad , n = 1,2,3$

Exercise 2.2

Question # 1

Find from first principles, the derivatives of the following expansions w.r.t. their respective independent variables:

(i) $\boldsymbol{(ax+b)^3}$

Solution.

Let $\boldsymbol{y = (ax+b)^3 \rightarrow (i)}$

$y + \delta y = (a(x+\delta x)+b)^3 \rightarrow (ii)$

$eq(ii) - eq(i)$

$\delta y = (ax + a\delta x + b)^3 - y$

$\delta y = ((ax+b) + a\delta x)^3 - (ax+b)^3$

$$\delta y = (ax+b)^3\left(1 + \frac{a\delta x}{(ax+b)}\right)^3 - (ax+b)^3$$

$$\delta y = (ax+b)^3\left(\left(1 + \frac{a\delta x}{(ax+b)}\right)^3 - 1\right)$$

$$\delta y = (ax+b)^3(1 + \frac{3\ a\delta x}{(ax+b)} + \frac{3(3-1)}{2!}\left(\frac{a\delta x}{ax+b}\right)^2 + \cdots - 1)$$

$$\delta y = (ax+b)^3\delta x(\frac{3\ a}{(ax+b)} + \frac{3(3-1)}{2!}\frac{a^2\delta x}{(ax+b)^2} + \cdots)$$

Dividing both sides by δx

$$\frac{\delta y}{\delta x} = (ax+b)^3(\frac{3\ a}{(ax+b)} + \frac{3(3-1)}{2!}\frac{a^2\delta x}{(ax+b)^2} + \cdots)$$

Taking limit when $\delta x \rightarrow 0$

$$\lim_{\delta x\to 0}\frac{\delta y}{\delta x} = \lim_{\delta x\to 0}(ax+b)^3(\frac{3\ a}{(ax+b)} + \frac{3(3-1)}{2!}\frac{a^2\delta x}{(ax+b)^2} + \cdots)$$

$$\frac{dy}{dx} = (ax+b)^3(\frac{3\ a}{(ax+b)} + \frac{3(3-1)}{2!}\frac{a^2(0)}{(ax+b)^2} + \cdots)$$

$$\frac{dy}{dx} = (ax+b)^3(\frac{3\ a}{(ax+b)})$$

$$\frac{dy}{dx} = 3a\ (ax+b)^{3-1}$$

$$\boldsymbol{\frac{dy}{dx} = 3a\ (ax+b)^2}$$

Which is the required.

(ii) $\boldsymbol{(2x+3)^5}$

Solution.

Let $y = (2x+3)^5 \rightarrow (i)$

$y + \delta y = (2(x+\delta x)+3)^5 \rightarrow (ii)$

$eq(ii) - eq(i)$

$\delta y = (2x + 2\delta x + 3)^5 - y$

$\delta y = ((2x+3) + 2\delta x)^5 - (2x+3)^5$

$$\delta y = (2x+3)^5\left(1 + \frac{2\delta x}{(2x+3)}\right)^5 - (2x+3)^5$$

$$\delta y = (2x+3)^5\left(\left(1 + \frac{2\delta x}{(2x+3)}\right)^5 - 1\right)$$

$$\delta y = (2x+3)^5(1 + \frac{5\ (2)\delta x}{(2x+3)} + \frac{5(5-1)}{2!}\left(\frac{2\delta x}{2x+3}\right)^2 + \cdots - 1)$$

$$\delta y = (2x+3)^5\delta x(\frac{5\ (2)}{(2x+3)} + \frac{5(5-1)}{2!}\frac{2^2\delta x}{(2x+3)^2} + \cdots)$$

Dividing both sides by δx

$$\frac{\delta y}{\delta x} = (2x+3)^5(\frac{5\ (2)}{(2x+3)} + \frac{5(5-1)}{2!}\frac{2^2\delta x}{(2x+3)^2} + \cdots)$$

Taking limit when $\delta x \rightarrow 0$

$$\lim_{\delta x\to 0}\frac{\delta y}{\delta x} = \lim_{\delta x\to 0}(2x+3)^5(\frac{5\ (2)}{(2x+3)} + \frac{5(5-1)}{2!}\frac{2^2\delta x}{(2x+3)^2} + \cdots)$$

$$\frac{dy}{dx} = (2x+3)^5(\frac{5\ (2)}{(2x+3)} + \frac{5(5-1)}{2!}\frac{2^2(0)}{(2x+3)^2} + \cdots)$$

$$\frac{dy}{dx} = (2x+3)^5(\frac{5\ (2)}{(2x+3)})$$

$$\frac{dy}{dx} = 10(2x+3)^{5-1}$$

$$\boldsymbol{\frac{dy}{dx} = 10(2x+3)^4}$$

Which is the required.

(iii) $\boldsymbol{(3t+2)^{-2}}$

Solution.

Let $y = (3t+2)^{-2} \rightarrow (i)$

$y + \delta y = (3(t+\delta t)+2)^{-2} - (ii)$

$eq(ii) - eq(i)$

$\delta y = (3t + 3\delta t + 2)^{-2} - y$

$\delta y = ((3t+2) + 3\delta t)^{-2} - (3t+2)^{-2}$

$$\delta y = (3t+2)^{-2}\left(1 + \frac{3\delta t}{(3t+2)}\right)^{-2} - (3t+2)^{-2}$$

$$\delta y = (3t+2)^{-2}(\left(1 + \frac{3\delta t}{(3t+2)}\right)^{-2} - 1)$$

$$\delta y = (3t+2)^{-2}(1 + \frac{-2\ (3)\delta t}{(3t+2)} + \frac{-2(-2-1)}{2!}\left(\frac{3\delta t}{3t+2}\right)^2 + \cdots - 1)$$

$$\delta y = (3t+2)^{-2}\delta t(-\frac{2(3)}{(3t+2)} + \frac{-2(-2-1)}{2!}\frac{3\delta t}{(3t+2)^2} + \cdots)$$

Dividing both sides by δt

$$\frac{\delta y}{\delta t} = (3t+2)^{-2}(-\frac{2(3)}{(3t+2)} + \frac{-2(-2-1)}{2!}\frac{3\delta t}{(3t+2)^2} + \cdots)$$

Taking limit when $\delta t \rightarrow 0$

$$\lim_{\delta t\to 0}\frac{\delta y}{\delta t} = \lim_{\delta t\to 0}(3t+2)^{-2}(-\frac{2(3)}{(3t+2)} + \frac{-2(-2-1)}{2!}\frac{3\delta t}{(3t+2)^2} + \cdots)$$

$$\frac{dy}{dt} = (3t+2)^{-2}(-\frac{2(3)}{(3t+2)} + \frac{-2(-2-1)}{2!}\frac{3(0)}{(3t+2)^2} + \cdots)$$

$$\frac{dy}{dt} = (3t+2)^{-2}(-\frac{2(3)}{(3t+2)})$$

$$\frac{dy}{dt} = -\frac{6}{(3t+2)^{1+2}}$$

$$\boldsymbol{\frac{dy}{dt} = -\frac{6}{(3t+2)^3}}$$

Which is the required.

(iv) $\boldsymbol{(ax+b)^{-5}}$

Solution.

Let $y = (ax+b)^{-5} \rightarrow (i)$

$y + \delta y = (a(x+\delta x)+b)^{-5} \rightarrow (ii)$

$eq(ii) - eq(i)$

$\delta y = (ax + a\delta x + b)^{-5} - y$

$\delta y = ((ax+b) + a\delta x)^{-5} - (ax+b)^{-5}$

$$\delta y = (ax+b)^{-5}\left(1 + \frac{a\delta x}{(ax+b)}\right)^{-5} - (ax+b)^{-5}$$

$$\delta y = (ax+b)^{-5}(\left(1 + \frac{a\delta x}{(ax+b)}\right)^{-5} - 1)$$

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$$\delta y = (ax+b)^{-5}(1 - \frac{5\,a\delta x}{(ax+b)} + \frac{-5(-5-1)}{2!}\left(\frac{a\delta x}{ax+b}\right)^2 + \cdots - 1)$$

$$\delta y = (ax+b)^{-5}\delta x(-\frac{5\,a}{(ax+b)} + \frac{-5(-5-1)}{2!}\frac{a^2\delta x}{(ax+b)^2} + \cdots)$$

Dividing both sides by δx

$$\frac{\delta y}{\delta x} = (ax+b)^{-5}(-\frac{5\,a}{(ax+b)} + \frac{-5(-5-1)}{2!}\frac{a^2\delta x}{(ax+b)^2} + \cdots)$$

Taking limit when $\delta x \to 0$

$$\lim_{\delta x\to 0}\frac{\delta y}{\delta x} = \lim_{\delta x\to 0}(ax+b)^{-5}(-\frac{5\,a}{(ax+b)} + \frac{-5(-5-1)}{2!}\frac{a^2\delta x}{(ax+b)^2} + \cdots)$$

$$\frac{dy}{dx} = (ax+b)^{-5}(-\frac{5\,a}{(ax+b)} + \frac{-5(-5-1)}{2!}\frac{a^2(0)}{(ax+b)^2} + \cdots)$$

$$\frac{dy}{dx} = (ax+b)^{-5}(-\frac{5\,a}{(ax+b)})$$

$$\frac{dy}{dx} = -\frac{5a}{(ax+b)^{1+5}}$$

$$\boldsymbol{\frac{dy}{dx} = -\frac{5a}{(ax+b)^6}}$$

Which is the required.

(v) $\boldsymbol{\frac{1}{(az-b)^7}}$

Solution.

Let $y = (az-b)^{-7} \to (i)$

$y + \delta y = (a(z+\delta z) - b)^{-7} \to (ii)$

$eq(ii) - eq(i)$

$\delta y = (az + a\delta z - b)^{-7} - y$

$\delta y = ((az-b) + a\delta z)^{-7} - (az-b)^{-7}$

$$\delta y = (az-b)^{-7}\left(1 + \frac{a\delta z}{(ax-b)}\right)^{-7} - (az-b)^{-7}$$

$$\delta y = (az-b)^{-7}(\left(1 + \frac{a\delta z}{(az-b)}\right)^{-7} - 1)$$

$$\delta y = (az-b)^{-7}(1 - \frac{7\,a\delta z}{(az-b)} + \frac{-7(-7-1)}{2!}\left(\frac{a\delta z}{az-b}\right)^2 + \cdots - 1)$$

$$\delta y = (az-b)^{-7}\delta z(-\frac{7\,a}{(az-b)} + \frac{-7(-7-1)}{2!}\frac{a^2\delta z}{(az-b)^2} + \cdots)$$

Dividing both sides by δz

$$\frac{\delta y}{\delta z} = (az-b)^{-7}(-\frac{7\,a}{(az-b)} + \frac{-7(-7-1)}{2!}\frac{a^2\delta z}{(az-b)^2} + \cdots)$$

Taking limit when $\delta z \to 0$

$$\lim_{\delta z\to 0}\frac{\delta y}{\delta z} = \lim_{\delta z\to 0}(az-b)^{-7}(-\frac{7\,a}{(az-b)} + \frac{-7(-7-1)}{2!}\frac{a^2\delta z}{(az-b)^2} + \cdots)$$

$$\frac{dy}{dz} = (az-b)^{-7}(-\frac{7\,a}{(az-b)} + \frac{-7(-7-1)}{2!}\frac{a^2(0)}{(az-b)^2} + \cdots)$$

$$\frac{dy}{dz} = (az-b)^{-7}(-\frac{7\,a}{(az-b)})$$

$$\frac{dy}{dz} = -\frac{7a}{(az-b)^{1+7}}$$

$$\frac{dy}{dz} = -\frac{7a}{(az-b)^8}$$

Which is the required.

Theorems on Differentiation:

We have, so far proved the following two formulas;

1. $\frac{d}{dx}(c) = 0$

$i.e$ *derivative of constant function is zero.*

2. $\frac{d}{dx}(x^n) = nx^{n-1}$

power rule when constant function is zero.

3. Derivative of $y = cf(x)\ i.e\ \frac{dy}{dx} = cf'(x)$

Proof:

$y = cf(x) \to (i) \qquad y + \delta y = cf(x+\delta x) \to (ii)$

$by\ (ii) - (i) \Rightarrow \quad y + \delta y - y = cf(x+\delta x) - cf(x)$

$\Rightarrow \delta y = c(f(x+\delta x) - f)x)$

Dividing by $\delta x\ and\ take\ limit\ \delta x \to 0$

$$\Rightarrow \lim_{\delta x\to 0}\left(\frac{\delta y}{\delta x}\right) = \lim_{\delta x\to 0}\frac{1}{\delta x}[c(f(x+\delta x) - f(x)]$$

$$\Rightarrow \frac{dy}{dx} = cf'(x)$$

Reciprocal Rule:

If $f(x)$ is differentiable $at\ x\ and\ g(x) \neq 0\ then \frac{1}{g(x)}$

$is\ differentiable\ at\ x\ and \frac{d}{dx}\left(\frac{1}{g(x)}\right) = -\frac{g'(x)}{[g(x)]^2}$

Exercise 2.3

Differentiate $w.r.t$ "x"

Question # 1. $\boldsymbol{x^4 + 2x^3 + x^2}$

Solution.

Let $y = x^4 + 2x^3 + x^2$

Differentiate w.r.t x

$$\frac{dy}{dx} = \frac{d}{dx}(x^4 + 2x^3 + x^2)$$

$$\frac{dy}{dx} = \frac{d}{dx}x^4 + 2\frac{d}{dx}x^3 + \frac{d}{dx}x^2$$

$$\frac{dy}{dx} = 4x^{4-1} + 2.3x^{3-1} + 2x^{2-1}$$

$$\frac{dy}{dx} = 4x^3 + 6x^2 + 2x$$

Which is required.

Question # 2. $\boldsymbol{x^{-3} + 2x^{-\frac{3}{2}} + 3}$

Solution.

Let $y = x^{-3} + 2x^{-\frac{3}{2}} + 3$

Differentiate w.r.t x

$$\frac{dy}{dx} = \frac{d}{dx}\left(x^{-3} + 2x^{-\frac{3}{2}} + 3\right)$$

$$\frac{dy}{dx} = \frac{d}{dx}x^{-3} + 2\frac{d}{dx}x^{-\frac{3}{2}} + \frac{d}{dx}(3)$$

$$\frac{dy}{dx} = -3x^{-3-1} - 2.\frac{3}{2}x^{-\frac{3}{2}-1} + 0$$

$$\frac{dy}{dx} = -3x^{-4} - 3x^{-\frac{5}{2}}$$

$$\boldsymbol{\frac{dy}{dx} = -3\left(\frac{1}{x^4} + \frac{1}{x^{\frac{5}{2}}}\right)}$$

Which is required.

Question#3. $\boldsymbol{x^{-3} + 2x^{-\frac{3}{2}} + 3}$

Solution.

Let $y = \frac{a+x}{a-x}$

Differentiate w.r.t x

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$$\frac{dy}{dx}=\frac{d}{dx}\left(\frac{a+x}{a-x}\right)$$
$$\frac{dy}{dx}=\frac{(a-x)\frac{d}{dx}(a+x)-(a+x)\frac{d}{dx}(a-x)}{(a-x)^2}$$
$$\frac{dy}{dx}=\frac{(a-x)(1)-(a+x)(-1)}{(a-x)^2}$$
$$\frac{dy}{dx}=\frac{a-x+a+x}{(a-x)^2}$$
$$\boldsymbol{\frac{dy}{dx}=\frac{2a}{(a-x)^2}}$$
Which is required.

Question#4. $\frac{2x-3}{2x+1}$

Solution.

Let $y=\frac{2x-3}{2x+1}$

Differentiate w.r.t x

$$\frac{dy}{dx}=\frac{d}{dx}\left(\frac{2x-3}{2x+1}\right)$$
$$\frac{dy}{dx}=\frac{(2x+1)\frac{d}{dx}(2x-3)-(2x-3)\frac{d}{dx}(2x+1)}{(2x+1)^2}$$
$$\frac{dy}{dx}=\frac{(2x+1)(2)-(2x-3)(2)}{(2x+1)^2}$$
$$\frac{dy}{dx}=\frac{2(2x+1-2x+3)}{(2x+1)^2}$$
$$\frac{dy}{dx}=\frac{2(4)}{(2x+1)^2}$$
$$\boldsymbol{\frac{dy}{dx}=\frac{8}{(2x+1)^2}}$$
Which is required.

Question#5. $(x-5)(3-x)$

Solution.

Let $y=(x-5)(3-x)$

Differentiate w.r.t x

$$\frac{dy}{dx}=\frac{d}{dx}((x-5)(3-x))$$
$$\frac{dy}{dx}=\frac{d}{dx}(3x-x^2-15+5x)$$
$$\frac{dy}{dx}=\frac{d}{dx}(-x^2+8x-15)$$
$$\frac{dy}{dx}=-\frac{d}{dx}x^2+8\frac{d}{dx}x-\frac{d}{dx}(15)$$
$$\frac{dy}{dx}=-\frac{d}{dx}x^2+8\frac{d}{dx}x-\frac{d}{dx}(15)$$
$$\frac{dy}{dx}=-2x+8+0$$
$$\boldsymbol{\frac{dy}{dx}=-2x+8}$$
Which is required.

Question#5. $\left(\sqrt{x}-\frac{1}{\sqrt{x}}\right)^2$

Solution.

Let $y=\left(\sqrt{x}-\frac{1}{\sqrt{x}}\right)^2$

Differentiate w.r.t x

$$\frac{dy}{dx}=\frac{d}{dx}\left(\sqrt{x}-\frac{1}{\sqrt{x}}\right)^2$$
$$\frac{dy}{dx}=\frac{d}{dx}\left((\sqrt{x})^2+\left(\frac{1}{\sqrt{x}}\right)^2-2(\sqrt{x})(\frac{1}{\sqrt{x}})\right)$$
$$\frac{dy}{dx}=\frac{d}{dx}\left(x+\frac{1}{x}-2\right)$$
$$\frac{dy}{dx}=\frac{d}{dx}(x+x^{-1}-2)$$
$$\frac{dy}{dx}=\frac{d}{dx}x+\frac{d}{dx}x^{-1}-\frac{d}{dx}(2)$$
$$\frac{dy}{dx}=1-x^{-2}-0$$
$$\frac{dy}{dx}=1-\frac{1}{x^2}$$
$$\boldsymbol{\frac{dy}{dx}=\frac{x^2-1}{x^2}}$$
Which is required.

Question#7. $\frac{(1+\sqrt{x})(x-x^{\frac{3}{2}})}{\sqrt{x}}$

Solution.

Let $y=\frac{(1+\sqrt{x})\left(x-x^{\frac{3}{2}}\right)}{\sqrt{x}}$

$$y=\frac{(1+\sqrt{x})x\left(1-x^{\frac{1}{2}}\right)}{\sqrt{x}}$$
$$y=\frac{x\left(1-(\sqrt{x})^2\right)}{\sqrt{x}}$$
$$y=\sqrt{x}(1-x)$$
$$y=\sqrt{x}-x\sqrt{x}$$
$$y=x^{\frac{1}{2}}-x^{\frac{3}{2}}$$

Differentiate w.r.t x

$$\frac{dy}{dx}=\frac{d}{dx}\left(x^{\frac{1}{2}}-x^{\frac{3}{2}}\right)$$
$$\frac{dy}{dx}=\frac{d}{dx}x^{\frac{1}{2}}-\frac{d}{dx}x^{\frac{3}{2}}$$
$$\frac{dy}{dx}=\frac{1}{2}x^{-\frac{1}{2}}-\frac{3}{2}x^{\frac{1}{2}}$$
$$\frac{dy}{dx}=\frac{1}{2}(\frac{1}{\sqrt{x}}-3\sqrt{x})$$
$$\boldsymbol{\frac{dy}{dx}=\frac{1}{2}(\frac{1-3x}{\sqrt{x}})}$$
Which is required.

Question#8. $\frac{(x^2+1)^2}{x^2-1}$

Solution.

Let $y=\frac{(x^2+1)^2}{x^2-1}$

Differentiate w.r.t x

$$\frac{dy}{dx}=\frac{d}{dx}\left(\frac{(x^2+1)^2}{x^2-1}\right)$$
$$\frac{dy}{dx}=\frac{(x^2-1)\frac{d}{dx}((x^2+1)^2)-(x^2+1)^2\frac{d}{dx}(x^2-1)}{(x^2-1)^2}$$
$$\frac{dy}{dx}=\frac{(x^2-1)(2(x^2+1)^1)\frac{d}{dx}(x^2+1)-(x^2+1)^2(2x)}{(x^2-1)^2}$$

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$$\frac{dy}{dx}=\frac{(x^2-1)(2(x^2+1)^1)(2x)-(x^2+1)^2(2x)}{(x^2-1)^2}$$
$$\frac{dy}{dx}=\frac{2x(x^2+1))(2(x^2-1)-x^2-1)}{(x^2-1)^2}$$
$$\frac{dy}{dx}=\frac{2x(x^2+1))(2x^2-2-x^2-1)}{(x^2-1)^2}$$
$$\boldsymbol{\frac{dy}{dx}=\frac{2x(x^2+1))(x^2-3)}{(x^2-1)^2}}$$

Which is required.

Question # 9. $\frac{x^2+1}{x^2-3}$

Solution.

Let $y=\frac{x^2+1}{x^2-3}$

Differentiate w.r.t x

$$\frac{dy}{dx}=\frac{d}{dx}\left(\frac{x^2+1}{x^2-3}\right)$$
$$\frac{dy}{dx}=\frac{(x^2-3)\frac{d}{dx}(x^2+1)-(x^2+1)\frac{d}{dx}(x^2-3)}{(x^2-3)^2}$$
$$\frac{dy}{dx}=\frac{(x^2-3)(2x)-(x^2+1)(2x)}{(x^2-3)^2}$$
$$\frac{dy}{dx}=\frac{2x((x^2-3)-(x^2+1))}{(x^2-3)^2}$$
$$\frac{dy}{dx}=\frac{2x(x^2-3-x^2-1)}{(x^2-3)^2}$$
$$\frac{dy}{dx}=\frac{2x(-4)}{(x^2-3)^2}$$
$$\boldsymbol{\frac{dy}{dx}=\frac{-8x}{(x^2-3)^2}}$$

Which is required.

Question#10. $\frac{\sqrt{1+x}}{\sqrt{1-x}}$

Solution.

Let $y=\frac{\sqrt{1+x}}{\sqrt{1-x}}=\sqrt{\frac{1+x}{1-x}}=\left(\frac{1+x}{1-x}\right)^{\frac{1}{2}}$

Differentiate w.r.t x

$$\frac{dy}{dx}=\frac{d}{dx}\left(\left(\frac{1+x}{1-x}\right)^{\frac{1}{2}}\right)$$
$$\frac{dy}{dx}=\frac{1}{2}\left(\frac{1+x}{1-x}\right)^{-\frac{1}{2}}\frac{d}{dx}\left(\frac{1+x}{1-x}\right)$$
$$\frac{dy}{dx}$$
$$=\frac{1}{2}\left(\frac{1-x}{1+x}\right)^{\frac{1}{2}}\left(\frac{(1-x)\frac{d}{dx}(1+x)-(1+x)\frac{d}{dx}(1-x)}{(1-x)^2}\right)$$
$$\frac{dy}{dx}=\frac{1}{2}\left(\frac{1-x}{1+x}\right)^{\frac{1}{2}}\left(\frac{1-x+1+x}{(1-x)^2}\right)$$
$$\frac{dy}{dx}=\frac{1}{2}\left(\frac{1-x}{1+x}\right)^{\frac{1}{2}}\left(\frac{2}{(1-x)^2}\right)$$
$$\frac{dy}{dx}=\left(\frac{1}{(1-x)^{2-\frac{1}{2}}(1+x)^{\frac{1}{2}}}\right)$$
$$\boldsymbol{\frac{dy}{dx}=\left(\frac{1}{(1-x)^{\frac{3}{2}}(1+x)^{\frac{1}{2}}}\right)}$$

Which is required.

Question # 11. $\frac{2x-1}{\sqrt{x^2+1}}$

Solution.

Let $y=\frac{2x-1}{\sqrt{x^2+1}}$

Differentiate w.r.t x

$$\frac{dy}{dx}=\frac{d}{dx}\left(\frac{2x-1}{\sqrt{x^2+1}}\right)$$
$$\frac{dy}{dx}=\left(\frac{\left(\sqrt{x^2+1}\right)\frac{d}{dx}(2x-1)-(2x-1)\frac{d}{dx}\left(\sqrt{x^2+1}\right)}{\left(\sqrt{x^2+1}\right)^2}\right)$$
$$\frac{dy}{dx}=\frac{\left(\sqrt{x^2+1}\right)(2)-(2x-1)\frac{1}{2\sqrt{x^2+1}}(2x)}{x^2+1}$$
$$\frac{dy}{dx}=\frac{(x^2+1)(2)-(2x-1)(x)}{(x^2+1)(x^2+1)^{\frac{1}{2}}}$$
$$\frac{dy}{dx}=\frac{2x^2+2-2x^2+x}{(x^2+1)^{1+\frac{1}{2}}}$$
$$\boldsymbol{\frac{dy}{dx}=\frac{x+2}{(x^2+1)^{\frac{3}{2}}}}$$

Which is required.

Question # 12. $\frac{\sqrt{a-x}}{\sqrt{a+x}}$

Solution.

Let $y=\frac{\sqrt{a-x}}{\sqrt{a+x}}=\sqrt{\frac{a-x}{a+x}}$

Differentiate w.r.t x

$$\frac{dy}{dx}=\frac{d}{dx}\left(\sqrt{\frac{a-x}{a+x}}\right)$$
$$\frac{dy}{dx}=\frac{1}{2}\left(\frac{a-x}{a+x}\right)^{-\frac{1}{2}}\frac{d}{dx}\left(\frac{a-x}{a+x}\right)$$
$$\frac{dy}{dx}=\frac{1}{2}\left(\frac{a+x}{a-x}\right)^{\frac{1}{2}}\left(\frac{(a+x)\frac{d}{dx}(a-x)-(a-x)\frac{d}{dx}(a+x)}{(a+x)^2}\right)$$
$$\frac{dy}{dx}=\frac{1}{2}\left(\frac{a+x}{a-x}\right)^{\frac{1}{2}}\left(\frac{(a+x)(-1)-(a-x)(1)}{(a+x)^2}\right)$$
$$\frac{dy}{dx}=\frac{1}{2}\left(\frac{a+x}{a-x}\right)^{\frac{1}{2}}\left(\frac{-a-x-a+x}{(a+x)^2}\right)$$
$$\frac{dy}{dx}=\frac{1}{2}\left(\frac{-2a}{(a+x)^{2-\frac{1}{2}}(a-x)^{\frac{1}{2}}}\right)$$
$$\boldsymbol{\frac{dy}{dx}=\left(\frac{-a}{(a+x)^{\frac{3}{2}}(a-x)^{\frac{1}{2}}}\right)}$$

Which is required.

Question # 13. $\frac{\sqrt{x^2+1}}{\sqrt{x^2-1}}$

Solution.

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Let $y = \frac{\sqrt{x^2+1}}{\sqrt{x^2-1}} = \sqrt{\frac{x^2+1}{x^2-1}}$

Differentiate w.r.t x

$$\frac{dy}{dx} = \frac{d}{dx}\left(\sqrt{\frac{x^2+1}{x^2-1}}\right)$$

$$\frac{dy}{dx} = \frac{1}{2}\left(\frac{x^2+1}{x^2-1}\right)^{-\frac{1}{2}} \frac{d}{dx}\left(\frac{x^2+1}{x^2-1}\right)$$

$$\frac{dy}{dx} = \frac{1}{2}\left(\frac{x^2-1}{x^2+1}\right)^{\frac{1}{2}} \left(\frac{(x^2-1)\frac{d}{dx}(x^2+1) - (x^2+1)\frac{d}{dx}(x^2-1)}{(x^2-1)^2}\right)$$

$$\frac{dy}{dx} = \frac{1}{2}\left(\frac{x^2-1}{x^2+1}\right)^{\frac{1}{2}} \left(\frac{(x^2-1)(2x) - (x^2+1)(2x)}{(x^2-1)^2}\right)$$

$$\frac{dy}{dx} = \frac{1}{2}\left(\frac{x^2-1}{x^2+1}\right)^{\frac{1}{2}} \left(\frac{2x(x^2-1-x^2-1)}{(x^2-1)^2}\right)$$

$$\frac{dy}{dx} = \frac{1}{2}\left(\frac{x^2-1}{x^2+1}\right)^{\frac{1}{2}} \left(\frac{2x(-2)}{(x^2-1)^2}\right)$$

$$\frac{dy}{dx} = \left(\frac{x(-2)}{(x^2-1)^{2-\frac{1}{2}}(x^2+1)^{\frac{1}{2}}}\right)$$

$$\boldsymbol{\frac{dy}{dx} = \frac{-2x}{(x^2-1)^{\frac{3}{2}}\sqrt{x^2+1}}}$$

Which is required.

Question # 14. $\frac{\sqrt{1+x}-\sqrt{1-x}}{\sqrt{1+x}+\sqrt{1-x}}$

Solution.

Let $y = \frac{\sqrt{1+x}-\sqrt{1-x}}{\sqrt{1+x}+\sqrt{1-x}}$

$$y = \frac{\sqrt{1+x}-\sqrt{1-x}}{\sqrt{1+x}+\sqrt{1-x}} \cdot \frac{\sqrt{1+x}-\sqrt{1-x}}{\sqrt{1+x}-\sqrt{1-x}}$$

$$y = \frac{\left(\sqrt{1+x}-\sqrt{1-x}\right)^2}{\left(\sqrt{1+x}\right)^2 - \left(\sqrt{1-x}\right)^2}$$

$$y = \frac{(\sqrt{1+x})^2 + \left(\sqrt{1-x}\right)^2 - 2(\sqrt{1+x})(\sqrt{1-x})}{\left(\sqrt{1+x}\right)^2 - \left(\sqrt{1-x}\right)^2}$$

$$y = \frac{1+x+1-x-2\sqrt{1-x^2}}{1+x-1+x}$$

$$y = \frac{2-2\sqrt{1-x^2}}{2x}$$

$$y = \frac{1-\sqrt{1-x^2}}{x}$$

Differentiate w.r.t x

$$\frac{dy}{dx} = \frac{d}{dx}\left(\frac{1-\sqrt{1-x^2}}{x}\right)$$

$$\frac{dy}{dx} = \frac{x\frac{d}{dx}\left(1-\sqrt{1-x^2}\right) - \left(1-\sqrt{1-x^2}\right)\frac{d}{dx}x}{x^2}$$

$$\frac{dy}{dx} = \frac{x\left(-\frac{-2x}{2\sqrt{1-x^2}}\right) - \left(1-\sqrt{1-x^2}\right)}{x^2}$$

$$\frac{dy}{dx} = \frac{x^2 - \sqrt{1-x^2} + 1 - x^2}{x^2\sqrt{1-x^2}}$$

$$\boldsymbol{\frac{dy}{dx} = \frac{1-\sqrt{1-x^2}}{x^2\sqrt{1-x^2}}}$$

Which is required.

Question # 15. $\frac{x\sqrt{a+x}}{\sqrt{a-x}}$

Solution.

Let $y = \frac{x\sqrt{a+x}}{\sqrt{a-x}} = x\sqrt{\frac{a+x}{a-x}}$

Differentiate w.r.t x

$$\frac{dy}{dx} = \frac{d}{dx}\left(x\sqrt{\frac{a+x}{a-x}}\right)$$

$$\frac{dy}{dx} = \sqrt{\frac{a+x}{a-x}}\frac{d}{dx}(x) + x\frac{d}{dx}\sqrt{\frac{a+x}{a-x}}$$

$$\frac{dy}{dx} = \sqrt{\frac{a+x}{a-x}} + x\frac{1}{2}\left(\frac{a+x}{a-x}\right)^{-\frac{1}{2}}\frac{d}{dx}\left(\frac{a+x}{a-x}\right)$$

$$\frac{dy}{dx}$$

$$= \sqrt{\frac{a+x}{a-x}} + x\frac{1}{2}\left(\frac{a-x}{a+x}\right)^{\frac{1}{2}}\left(\frac{(a-x)(1)-(a+x)(-1)}{(a-x)^2}\right)$$

$$\frac{dy}{dx} = \sqrt{\frac{a+x}{a-x}} + x\frac{1}{2}\left(\frac{a-x}{a+x}\right)^{\frac{1}{2}}\left(\frac{a-x+a+x}{(a-x)^2}\right)$$

$$\frac{dy}{dx} = \sqrt{\frac{a+x}{a-x}} + x\frac{1}{2}\left(\frac{a-x}{a+x}\right)^{\frac{1}{2}}\left(\frac{2a}{(a-x)^2}\right)$$

$$\frac{dy}{dx} = \sqrt{\frac{a+x}{a-x}} + \left(\frac{ax}{(a-x)^{\frac{3}{2}}((a+x)^{\frac{1}{2}}}\right)$$

$$\frac{dy}{dx} = \frac{(a+x)(a-x)+ax}{(a-x)^{\frac{3}{2}}(a+x)^{\frac{1}{2}}}$$

$$\boldsymbol{\frac{dy}{dx} = \frac{a^2-x^2+ax}{(a-x)^{\frac{3}{2}}(a+x)^{\frac{1}{2}}}}$$

Which is required.

Question # 16. $\boldsymbol{If\ y = \sqrt{x} - \frac{1}{\sqrt{x}}}$ **,** $\boldsymbol{Show\ that\ 2x\frac{dy}{dx} + y = 2\sqrt{x}}$**.**

Solution.

Since $y = \sqrt{x} - \frac{1}{\sqrt{x}}$

$$y = x^{\frac{1}{2}} - x^{-\frac{1}{2}}$$

Differentiating w.r.t "x"

$$\frac{dy}{dx} = \frac{1}{2}x^{-\frac{1}{2}} + \frac{1}{2}x^{-\frac{3}{2}}$$

Multiplying by 2x both sides

$$2x\frac{dy}{dx} = 2x\left(\frac{1}{2}x^{-\frac{1}{2}} + \frac{1}{2}x^{-\frac{3}{2}}\right)$$

$$2x\frac{dy}{dx} = x^{\frac{1}{2}} + x^{-\frac{1}{2}}$$

Adding u on both sides, we have

$$2x\frac{dy}{dx} + y = x^{\frac{1}{2}} + x^{-\frac{1}{2}} + y$$

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$$2x\frac{dy}{dx}+y=x^{\frac{1}{2}}+x^{-\frac{1}{2}}+x^{\frac{1}{2}}-x^{-\frac{1}{2}}$$
$$2x\frac{dy}{dx}+y=2x^{\frac{1}{2}}$$
$$\boldsymbol{2x\frac{dy}{dx}+y=2\sqrt{x}}$$

Hence Proved.

Question # 17. $\boldsymbol{If\ y=x^4+2x^2+2\ then\ prove\ that\ \frac{dy}{dx}=4x\sqrt{y-1}}$.

Solution.

Since $y=x^4+2x^2+2$

$Differentiate\ w.r.t.''x''$

$$\frac{dy}{dx}=4x^3+4x$$
$$\frac{dy}{dx}=4x(x^2+1)$$
$$\frac{dy}{dx}=4x\sqrt{(x^2+1)^2}$$
$$\frac{dy}{dx}=4x\sqrt{x^4+2x^2+1}$$
$$\frac{dy}{dx}=4x\sqrt{x^4+2x^2+2-1}$$
$$\boldsymbol{\frac{dy}{dx}=4x\sqrt{y-1}}\qquad \because\ \boldsymbol{y=x^4+2x^2+1}$$

Hence Proved. Which is require

The Chain Rule:

$if\ y=(fog)(x) or\ y=f\big(g(x)\big) let\ u=g(x) then$

$y=f(x)\ and\ u=g(x)\ so\ we\ find \frac{dy}{du}\ and \frac{du}{dx}\ \ then$

$\frac{dy}{dx}=\frac{dy}{du}.\frac{du}{dx}\ \ is\ called\ chain\ rule.$

Derivative of a function given in the form of parametric equation.

Differentiation of implicit Relations:

Exercise 2.4

Question # 1.

Find $\frac{dy}{dx}$ by making the suitable substitution in the following functions defined as:

Differentiate the following w.r.t "x".

(i) $\boldsymbol{y=\sqrt{\frac{1-x}{1+x}}}$

Solution.

Given $y=\sqrt{\frac{1-x}{1+x}}$

Put $u=\frac{1-x}{1+x}$

So $y=\sqrt{u}$

$$y=u^{\frac{1}{2}}$$

Differentiate ''u" w.r.t . x

$$\frac{du}{dx}=\frac{d}{dx}\left(\frac{1-x}{1+x}\right)$$
$$\frac{du}{dx}=\frac{(1+x)\frac{d}{dx}(1-x)-(1-x)\frac{d}{dx}(1+x)}{(1+x)^2}$$
$$\frac{du}{dx}=\frac{(1+x)(-1)-(1-x)(1)}{(1+x)^2}$$
$$\frac{du}{dx}=\frac{-1-x-1+x}{(1+x)^2}$$
$$\frac{du}{dx}=\frac{-2}{(1+x)^2}$$

Now differentiate y w.r.t. u

$$\frac{dy}{du}=\frac{d}{du}u^{\frac{1}{2}}$$
$$\frac{dy}{du}=\frac{1}{2}u^{-\frac{1}{2}}$$
$$\frac{dy}{du}=\frac{1}{2}\left(\frac{1-x}{1+x}\right)^{-\frac{1}{2}}$$
$$\frac{dy}{du}=\frac{1}{2}\left(\frac{1+x}{1-x}\right)^{\frac{1}{2}}$$

Now by chain rule

$$\frac{dy}{dx}=\frac{dy}{du}.\frac{du}{dx}$$
$$\frac{dy}{dx}=\frac{1}{2}\left(\frac{1+x}{1-x}\right)^{\frac{1}{2}}.\frac{-2}{(1+x)^2}$$
$$\frac{dy}{dx}=\frac{-1}{(1-x)^{\frac{1}{2}}(1+x)^{2-\frac{1}{2}}}$$
$$\frac{dy}{dx}=\frac{-1}{\sqrt{1-x}\,(1+x)^{\frac{3}{2}}}$$

Which is required.

(ii) $\boldsymbol{y=\sqrt{x+\sqrt{x}}}$

Solution.

Given $y=\sqrt{x+\sqrt{x}}$

Put $u=x+\sqrt{x}$

So $y=\sqrt{u}$

$$y=u^{\frac{1}{2}}$$

$Differentiate\ ''u"\ w.r.t\ .\ x$

$$\frac{du}{dx}=\frac{d}{dx}\left(x+\sqrt{x}\right)$$
$$\frac{du}{dx}=1+\frac{1}{2}x^{\frac{-1}{2}}$$
$$\frac{du}{dx}=1+\frac{1}{2\sqrt{x}}$$
$$\frac{du}{dx}=\frac{2\sqrt{x}+1}{2\sqrt{x}}$$

Now differentiate y w.r.t. u

$$\frac{dy}{du}=\frac{d}{du}u^{\frac{1}{2}}$$
$$\frac{dy}{du}=\frac{1}{2}u^{-\frac{1}{2}}$$
$$\frac{dy}{du}=\frac{1}{2}\left(x+\sqrt{x}\right)^{-\frac{1}{2}}$$
$$\frac{dy}{du}=\frac{1}{2\sqrt{x+\sqrt{x}}}$$

Now by chain rule

$$\frac{dy}{dx}=\frac{dy}{du}.\frac{du}{dx}$$

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$$\frac{dy}{dx} = \frac{1}{2\sqrt{x+\sqrt{x}}} \cdot \frac{2\sqrt{x}+1}{2\sqrt{x}}$$

$$\frac{dy}{dx} = \frac{2\sqrt{x}+1}{4\sqrt{x}.\sqrt{x+\sqrt{x}}}$$

Which is required.

(iii) $\boldsymbol{y = x\sqrt{\frac{a+x}{a-x}}}$

Solution.

Given $y = x\sqrt{\frac{a+x}{a-x}}$

Put $u = \frac{a+x}{a-x}$

So $y = x\sqrt{u}$

$$y = x.u^{\frac{1}{2}}$$

$Differentiate\ ''u''\ w.r.t.\ "x"$

$$\frac{du}{dx} = \frac{d}{dx}\left(\frac{a+x}{a-x}\right)\frac{du}{dx}$$

$$= \frac{(a-x)\frac{d}{dx}(a+x) - (a+x)\frac{d}{dx}(a-x)}{(a-x)^2}$$

$$\frac{du}{dx} = \frac{(a-x)(1) - (a+x)(-1)}{(a-x)^2}$$

$$\frac{du}{dx} = \frac{a-x+a+x}{(a-x)^2}$$

$$\frac{du}{dx} = \frac{2a}{(a-x)^2}$$

Now differentiate y w.r.t. x

$$\frac{dy}{du} = \frac{d}{du}\left(xu^{\frac{1}{2}}\right)$$

$$\frac{dy}{du} = x.\frac{d}{du}\left(u^{\frac{1}{2}}\right) + u^{\frac{1}{2}}\frac{d}{du}(x)$$

$$\frac{dy}{du} = x.\frac{1}{2}u^{-\frac{1}{2}}\frac{du}{dx} + u^{\frac{1}{2}}(1)$$

$$\frac{dy}{du} = x.\frac{1}{2}u^{-\frac{1}{2}}\frac{du}{dx} + u^{\frac{1}{2}}$$

Now Using Above values

$$\frac{dy}{dx} = x.\frac{1}{2}\left(\frac{a+x}{a-x}\right)^{-\frac{1}{2}} \cdot \frac{2a}{(a-x)^2} + \left(\frac{a+x}{a-x}\right)^{\frac{1}{2}}$$

$$\frac{dy}{dx} = \frac{ax\,(a-x)^{\frac{1}{2}}}{(a+x)^{\frac{1}{2}}(a-x)^2} + \frac{(a+x)^{\frac{1}{2}}}{(a-x)^{\frac{1}{2}}}$$

$$\frac{dy}{dx} = \frac{ax}{(a+x)^{\frac{1}{2}}(a-x)^{2-\frac{1}{2}}} + \frac{(a+x)^{\frac{1}{2}}}{(a-x)^{\frac{1}{2}}}$$

$$\frac{dy}{dx} = \frac{ax}{(a+x)^{\frac{1}{2}}(a-x)^{\frac{3}{2}}} + \frac{(a+x)^{\frac{1}{2}}}{(a-x)^{\frac{1}{2}}}$$

$$\frac{dy}{dx} = \frac{ax}{(a+x)^{\frac{1}{2}}(a-x)^1(a-x)^{\frac{1}{2}}} + \frac{(a+x)^{\frac{1}{2}}}{(a-x)^{\frac{1}{2}}}$$

$$\frac{dy}{dx} = \frac{ax + (a+x)(a-x)}{(a+x)^{\frac{1}{2}}(a-x)^1(a-x)^{\frac{1}{2}}}$$

$$\frac{dy}{dx} = \frac{ax + a^2 - x^2}{(a+x)^{\frac{1}{2}}(a-x)^{\frac{3}{2}}}$$

Which is required.

(iv) $\boldsymbol{y = (3x^2 - 2x + 7)^6}$

Solution.

Given $y = (3x^2 - 2x + 7)^6$

Put $u = 3x^2 - 2x + 7$

So $y = u^6$

Differentiate ''u'' w.r.t . x

$$\frac{du}{dx} = \frac{d}{dx}(3x^2 - 2x + 7)$$

$$\frac{du}{dx} = 3(2x) - 2(1) + 0$$

$$\frac{du}{dx} = 6x - 2$$

Now differentiate y w.r.t. u

$$\frac{dy}{du} = \frac{d}{du}u^6$$

$$\frac{dy}{du} = 6u^5$$

$$\frac{dy}{du} = 6(3x^2 - 2x + 7)^5$$

Now by chain rule

$$\frac{dy}{dx} = \frac{dy}{du}.\frac{du}{dx}$$

$$\frac{dy}{dx} = 6(3x^2 - 2x + 7)^5 . (6x - 2)$$

$$\frac{dy}{dx} = 12(3x - 1)(3x^2 - 2x + 7)^5$$

Which is required.

(v) $\boldsymbol{y = \sqrt{\frac{a^2+x^2}{a^2-x^2}}}$

Solution.

Given $y = \sqrt{\frac{a^2+x^2}{a^2-x^2}}$

Put $u = \frac{a^2+x^2}{a^2-x^2}$

So $y = \sqrt{u}$

$$y = u^{\frac{1}{2}}$$

$Differentiate\ ''u''\ w.r.t.\ x$

$$\frac{du}{dx} = \frac{d}{dx}\left(\frac{a^2+x^2}{a^2-x^2}\right)$$

$$\frac{du}{dx} = \frac{(a^2-x^2)\frac{d}{dx}(a^2+x^2) - (a^2+x^2)\frac{d}{dx}(a^2-x^2)}{(a^2-x^2)^2}$$

$$\frac{du}{dx} = \frac{(a^2-x^2)(2x) - (a^2+x^2)(-2x)}{(a^2-x^2)^2}$$

$$\frac{du}{dx} = \frac{2x(a^2-x^2+a^2+x^2)}{(a^2-x^2)^2}$$

$$\frac{du}{dx} = \frac{2a^2x}{(a^2-x^2)^2}$$

Now differentiate y w.r.t. u

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$$\frac{dy}{du}=\frac{d}{du}u^{\frac{1}{2}}$$
$$\frac{dy}{du}=\frac{1}{2}u^{-\frac{1}{2}}$$
$$\frac{dy}{du}=\frac{1}{2}\left(\frac{a^2+x^2}{a^2-x^2}\right)^{-\frac{1}{2}}$$
$$\frac{dy}{du}=\frac{1}{2}\left(\frac{a^2-x^2}{a^2+x^2}\right)^{\frac{1}{2}}$$

Now by chain rule

$$\frac{dy}{dx}=\frac{dy}{du}.\frac{du}{dx}$$
$$\frac{dy}{dx}=\frac{1}{2}\left(\frac{a^2-x^2}{a^2+x^2}\right)^{\frac{1}{2}}.\frac{2a^2x}{(a^2-x^2)^2}$$
$$\frac{dy}{dx}=\frac{a^2x}{(a^2+x^2)^{\frac{1}{2}}(a^2-x^2)^{2-\frac{1}{2}}}$$
$$\frac{dy}{dx}=\frac{a^2x}{(a^2+x^2)^{\frac{1}{2}}(a^2-x^2)^{\frac{3}{2}}}$$

Which is required.

Question # 2 Find $\frac{dy}{dx}$ if :

(i) $\boldsymbol{3x+4y+7=0}$

Solution:

Given $3x+4y+7=0$

Differentiate w.r.t. "x"

$$\frac{d}{dx}(3x+4y+7)=\frac{d}{dx}(0)$$
$$3+4\frac{dy}{dx}=0$$

$\frac{dy}{dx}=-\frac{3}{4}$.

Which is required.

(ii) $\boldsymbol{xy+y^2=2}$

Solution:

Given $xy+y^2=2$

Differentiate w.r.t. "x"

$$\frac{d}{dx}(xy+y^2)=\frac{d}{dx}(2)$$
$$\frac{d}{dx}(xy)+\frac{d}{dx}(y^2)=0$$
$$y.(1)+x.\frac{dy}{dx}+2y.\frac{dy}{dx}=0$$
$$(x+2y)\frac{dy}{dx}=-y$$
$$\frac{dy}{dx}=-\frac{y}{(x+2y)}$$

Which is required.

(iii) $\boldsymbol{x^2-4xy-5y=0}$

Solution:

Given $x^2-4xy-5y=0$

Differentiate w.r.t. "x"

$$\frac{d}{dx}(x^2-4xy-5y)=\frac{d}{dx}(0)$$
$$\frac{d}{dx}(x^2)-4\frac{d}{dx}(xy)-5\frac{d}{dx}(y)=0$$
$$2x-4\left(y.(1)+x.\frac{dy}{dx}\right)-5\frac{dy}{dx}=0$$
$$2x-4y-4x\frac{dy}{dx}-5\frac{dy}{dx}=0$$
$$-(4x+5)\frac{dy}{dx}=-(2x-4y)$$
$$\frac{dy}{dx}=\frac{2x-4y}{4x+5}$$

Which is required.

(iv) $\boldsymbol{4x^2+2hxy+by^2+2gx+2fy+c=0.}$

Solution:

Given $4x^2+2hxy+by^2+2gx+2fy+c=0$

Differentiate w.r.t. "x"

$$\frac{d}{dx}(4x^2+2hxy+by^2+2gx+2fy+c)=\frac{d}{dx}(0)$$
$$\frac{d}{dx}(4x^2)+2h\frac{d}{dx}(xy)+b\frac{d}{dx}(y^2)+2g\frac{d}{dx}(x)+2f\frac{d}{dx}(y)+\frac{d}{dx}(c)=0$$
$$4(2x)+2h\left(y.(1)+x.\frac{dy}{dx}\right)+2by\frac{dy}{dx}+2g(1)+2f\frac{dy}{dx}+0=0$$
$$8x+2hy+2hx.\frac{dy}{dx}+2by\frac{dy}{dx}+2g+2f\frac{dy}{dx}=0$$
$$2(hx+by+f)\frac{dy}{dx}=-2(4x+hy+g)$$
$$\frac{dy}{dx}=-\frac{4x+hy+g}{(hx+by+f)}$$

Which is required.

(v) $\boldsymbol{x\sqrt{1+y}+y\sqrt{1+x}=0}$

Solution:

Given $x\sqrt{1+y}+y\sqrt{1+x}=0$

Differentiate w.r.t. "x"

$$\frac{d}{dx}(x\sqrt{1+y}+y\sqrt{1+x})=\frac{d}{dx}(0)$$
$$\frac{d}{dx}\left(x\sqrt{1+y}\right)+\frac{d}{dx}\left(y\sqrt{1+x}\right)=0$$
$$\sqrt{1+y}.\frac{d}{dx}(x)+x.\frac{d}{dx}\left(\sqrt{1+y}\right)+y\frac{d}{dx}\left(\sqrt{1+x}\right)+\sqrt{1+x}\frac{d}{dx}(y)=0$$
$$\sqrt{1+y}.(1)+\frac{1}{2}x(1+y)^{-\frac{1}{2}}\frac{dy}{dx}+y\frac{1}{2}(1+x)^{-\frac{1}{2}}(1)+\sqrt{1+x}\frac{dy}{dx}=0$$
$$\sqrt{1+y}+\frac{1}{2}\frac{x}{\sqrt{1+y}}\frac{dy}{dx}+\frac{y}{2\sqrt{1+x}}+\sqrt{1+x}\frac{dy}{dx}=0$$
$$\left(\frac{x}{2\sqrt{1+y}}+\sqrt{1+x}\right)\frac{dy}{dx}=-\left(\frac{y}{2\sqrt{1+x}}+\sqrt{1+y}\right)$$

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$$\left(\frac{x+2\sqrt{1+y}\sqrt{1+x}}{2\sqrt{1+y}}\right)\frac{dy}{dx}=-\left(\frac{y+2\sqrt{1+y}.\sqrt{1+x}}{2\sqrt{1+x}}\right)$$

$$\frac{dy}{dx}=-\left(\frac{\sqrt{1+y}\left(y+2\sqrt{1+y}.\sqrt{1+x}\right)}{\sqrt{1+x}\left(x+2\sqrt{1+y}\sqrt{1+x}\right)}\right)$$

Which is required.

(vi) $\boldsymbol{y(x^2-1)=x\sqrt{x^2+4}=0}$

Solution:

Given $y(x^2-1)=x\sqrt{x^2+4}$

Differentiate w.r.t. "x"

$$\frac{d}{dx}y(x^2-1)=\frac{d}{dx}(x\sqrt{x^2+4})$$

$$(x^2-1)\frac{d}{dx}(y)+y\frac{d}{dx}(x^2-1)=x\frac{d}{dx}\left(\sqrt{x^2+4}\right)+\sqrt{x^2+4}\frac{d}{dx}(x)$$

$$(x^2-1)\frac{dy}{dx}+y(2x)=x\frac{1}{2}(x^2+4)^{-\frac{1}{2}}(2x)+\sqrt{x^2+4}(1)$$

$$(x^2-1)\frac{dy}{dx}+2xy=\frac{x^2}{\sqrt{x^2+4}}+\sqrt{x^2+4}$$

$$(x^2-1)\frac{dy}{dx}=\frac{x^2}{\sqrt{x^2+4}}+\sqrt{x^2+4}-2xy$$

$$(x^2-1)\frac{dy}{dx}=\frac{x^2+x^2+4-2xy\sqrt{x^2+4}}{\sqrt{x^2+4}}$$

$$\frac{dy}{dx}=\frac{2x^2+4-2xy\sqrt{x^2+4}}{(x^2-1)\sqrt{x^2+4}}$$

Which is required.

Question # 3 Find $\frac{dy}{dx}$ of the following parametric functions:

(i) $\boldsymbol{x=\theta+\frac{1}{\theta}\quad and\ y=\theta+1}$

Solution.

Since $x=\theta+\frac{1}{\theta}$

$$x=\theta+\theta^{-1}$$

$$Differentiate\ "x"\ w.r.t.\ \theta, We\ have$$

$$\frac{dx}{d\theta}=1+(-1.\theta^{-2})$$

$$\frac{dx}{d\theta}=1-\frac{1}{\theta^2}$$

$$\frac{dx}{d\theta}=\frac{\theta^2-1}{\theta^2}$$

$$\frac{d\theta}{dx}=\frac{\theta^2}{\theta^2-1}$$

Now $y=\theta+1$

$$Differentiate\ "y"\ w.r.t.\ \theta, We\ have$$

$$\frac{dy}{d\theta}=1+0$$

$$\frac{dy}{d\theta}=1$$

Now by chain rule

$$\frac{dy}{dx}=\frac{dy}{d\theta}.\frac{d\theta}{dx}$$

$$\frac{dy}{dx}=1.\frac{\theta^2}{\theta^2-1}$$

$$\frac{dy}{dx}=\frac{\theta^2}{\theta^2-1}$$

Which is required.

(ii) $x=\frac{a(1-t^2)}{1+t^2}\quad and\ y=\frac{2bt}{1+t^2}$

Solution.

Since $x=\frac{a(1-t^2)}{1+t^2}$

$$Differentiate\ "x"\ w.r.t.\ t, We\ have$$

$$\frac{dx}{dt}=\frac{d}{dt}\left(\frac{a(1-t^2)}{1+t^2}\right)$$

$$\frac{dx}{dt}=a\frac{(1+t^2)\frac{d}{dt}(1-t^2)-(1-t^2)\frac{d}{dt}(1+t^2)}{(1+t^2)^2}$$

$$\frac{dx}{dt}=a\frac{(1+t^2)(-2t)-(1-t^2)(2t)}{(1+t^2)^2}$$

$$\frac{dx}{dt}=-2at\frac{1+t^2+1-t^2}{(1+t^2)^2}$$

$$\frac{dx}{dt}=\frac{-4at}{(1+t^2)^2}$$

$$\frac{dt}{dx}=\frac{-(1+t^2)^2}{4at}$$

Now $y=\frac{2bt}{1+t^2}$

$$Differentiate\ "y"\ w.r.t.\ t, We\ have$$

$$\frac{dy}{dt}=\frac{d}{dt}\left(\frac{2bt}{1+t^2}\right)$$

$$\frac{dy}{dt}=2b\frac{(1+t^2)\frac{d}{dt}(t)-(t)\frac{d}{dt}(1+t^2)}{(1+t^2)^2}$$

$$\frac{dy}{dt}=2b\frac{(1+t^2)(1)-(t)(2t)}{(1+t^2)^2}$$

$$\frac{dy}{dt}=2b\frac{(1+t^2-2t^2)}{(1+t^2)^2}$$

$$\frac{dy}{dt}=2b\frac{(1-t^2)}{(1+t^2)^2}$$

Now by chain rule

$$\frac{dy}{dx}=\frac{dy}{dt}.\frac{dt}{dx}$$

$$\frac{dy}{dx}=2b\frac{(1-t^2)}{(1+t^2)^2}.\frac{-(1+t^2)^2}{4at}$$

$$\frac{dy}{dx}=-\frac{b(1-t^2)}{2at}$$

Which is required.

Question # 4. Prove that $\boldsymbol{y\frac{dy}{dx}+x=0\ if\ x=\frac{1-t^2}{1+t^2}\ and\ y=\frac{2t}{1+t^2}}$.

Solution.

Since $x=\frac{(1-t^2)}{1+t^2}$

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$Differentiate\ ''x''\ w.r.t.\ t\ ,We\ have$

$$\frac{dx}{dt}=\frac{d}{dt}\left(\frac{(1-t^2)}{1+t^2}\right)$$

$$\frac{dx}{dt}=\frac{(1+t^2)\frac{d}{dt}(1-t^2)-(1-t^2)\frac{d}{dt}(1+t^2)}{(1+t^2)^2}$$

$$\frac{dx}{dt}=\frac{(1+t^2)(-2t)-(1-t^2)(2t)}{(1+t^2)^2}$$

$$\frac{dx}{dt}=-2t\frac{1+t^2+1-t^2}{(1+t^2)^2}$$

$$\frac{dx}{dt}=\frac{-4t}{(1+t^2)^2}$$

$$\frac{dt}{dx}=\frac{-(1+t^2)^2}{4t}$$

Now $y=\frac{2t}{1+t^2}$

$Differentiate\ ''y''\ w.r.t.\ t, We\ have$

$$\frac{dy}{dt}=\frac{d}{dt}\left(\frac{2t}{1+t^2}\right)$$

$$\frac{dy}{dt}=2\frac{(1+t^2)\frac{d}{dt}(t)-(t)\frac{d}{dt}(1+t^2)}{(1+t^2)^2}$$

$$\frac{dy}{dt}=2\frac{(1+t^2)(1)-(t)(2t)}{(1+t^2)^2}$$

$$\frac{dy}{dt}=2\frac{(1+t^2-2t^2)}{(1+t^2)^2}$$

$$\frac{dy}{dt}=2\frac{(1-t^2)}{(1+t^2)^2}$$

Now by chain rule

$$\frac{dy}{dx}=\frac{dy}{dt}.\frac{dt}{dx}$$

$$\frac{dy}{dx}=2\frac{(1-t^2)}{(1+t^2)^2}.\frac{-(1+t^2)^2}{4t}$$

$$\frac{dy}{dx}=-\frac{(1-t^2)}{2t}$$

$$L.H.S=y\frac{dy}{dx}+x$$

$$L.H.S=\frac{2t}{1+t^2}.-\frac{(1-t^2)}{2t}+\frac{(1-t^2)}{1+t^2}$$

$$L.H.S=-\frac{(1-t^2)}{1+t^2}+\frac{(1-t^2)}{1+t^2}$$

$$L.H.S=0$$

$$L.H.S=R.H.S$$

Hence Proved.

Question # 5. Differentiate

(i) $\boldsymbol{x^2-\frac{1}{x^2}\ w.r.t.\ x^4}$

Solution.

Suppose $y=x^2-\frac{1}{x^2}\ and\ u=x^4$

$Differentiate\ ''y''\ w.r.t.\ "x"$

$$\frac{dy}{dx}=\frac{d}{dx}\left(x^2-\frac{1}{x^2}\right)$$

$$\frac{dy}{dx}=\frac{d}{dx}(x^2-x^{-2})$$

$$\frac{dy}{dx}=2x+2x^{-3}$$

$$\frac{dy}{dx}=2x+\frac{2}{x^3}$$

$$\frac{dy}{dx}=\frac{2(x^4+1)}{x^3}$$

Differentiate u w.r.t. x

$$\frac{du}{dx}=\frac{d}{dx}(x^4)$$

$$\frac{du}{dx}=4x^3$$

Now by chain rule

$$\frac{dy}{du}=\frac{dy}{dx}.\frac{dx}{du}$$

$$\frac{dy}{du}=\frac{dy}{dx}.\frac{1}{\frac{dx}{du}}$$

$$\frac{dy}{du}=\frac{2(x^4+1)}{x^3}.\frac{1}{4x^3}$$

$$\frac{dy}{du}=\frac{(x^4+1)}{2x^6}$$

Which is required.

(ii) $\boldsymbol{(1+x^2)^n\ w.r.t.\ x^2}$

Solution.

Suppose $y=(1+x^2)^n\ and\ u=x^2$

$Differentiate\ ''y''\ w.r.t.\ "x"$

$$\frac{dy}{dx}=\frac{d}{dx}((1+x^2)^n)$$

$$\frac{dy}{dx}=n(1+x^2)^{n-1}(2x)$$

$$\frac{dy}{dx}=2nx(1+x^2)^{n-1}$$

Differentiate u w.r.t. x

$$\frac{du}{dx}=\frac{d}{dx}(x^2)$$

$$\frac{du}{dx}=2x$$

Now by chain rule

$$\frac{dy}{du}=2nx(1+x^2)^{n-1}.\frac{1}{2x}$$

$$\frac{dy}{du}=n(1+x^2)^{n-1}$$

Which is required.

(iii) $\boldsymbol{\frac{x^2+1}{x^2-1}\ w.r.t.\ \frac{x-1}{x+1}}$

Solution.

Suppose $y=\frac{x^2+1}{x^2-1}\ and\ u=\frac{x-1}{x+1}$

$Differentiate\ ''y''\ w.r.t.\ "x"$

$$\frac{dy}{dx}=\frac{d}{dx}\left(\frac{x^2+1}{x^2-1}\right)$$

$$\frac{dy}{dx}=\frac{(x^2-1)\frac{d}{dx}(x^2+1)-(x^2+1)\frac{d}{dx}(x^2-1)}{(x^2-1)^2}$$

$$\frac{dy}{dx}=\frac{(x^2-1)(2x)-(x^2+1)(2x)}{(x^2-1)^2}$$

$$\frac{dy}{dx}=\frac{2x(x^2-1-x^2-1)}{(x^2-1)^2}$$

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$$\frac{dy}{dx}=\frac{2x(-2)}{(x^2-1)^2}$$

$$\frac{dy}{dx}=\frac{-4x}{(x^2-1)^2}$$

$$Differentiate\ u\ w.r.t.x$$

$$\frac{du}{dx}=\frac{d}{dx}\left(\frac{x-1}{x+1}\right)$$

$$\frac{du}{dx}=\frac{(x+1)\frac{d}{dx}(x-1)-(x-1)\frac{d}{dx}(x+1)}{(x+1)^2}$$

$$\frac{du}{dx}=\frac{(x+1)-(x-1)}{(x+1)^2}$$

$$\frac{du}{dx}=\frac{x+1-x+1}{(x+1)^2}$$

$$\frac{du}{dx}=\frac{2}{(x+1)^2}$$

Now by chain rule

$$\frac{dy}{du}=\frac{dy}{dx}.\frac{dx}{du}$$

$$\frac{dy}{du}=\frac{dy}{dx}.\frac{1}{\frac{du}{dx}}$$

$$\frac{dy}{du}=\frac{-4x}{(x^2-1)^2}.\frac{1}{\frac{2}{(x+1)^2}}$$

$$\frac{dy}{du}=\frac{-2x(x+1)^2}{(x^2-1)^2}$$

$$\frac{dy}{du}=\frac{-2x(x+1)^2}{(x-1)^2(x+1)^2}$$

$$\frac{dy}{du}=\frac{-2x}{(x-1)^2}$$

Which is required.

(iv) $\frac{ax+b}{cx+d}\ w.r.t.\frac{ax^2+b}{ax^2+d}$

Solution.

Suppose $y=\frac{ax+b}{cx+d}$ and $u=\frac{ax^2+b}{ax^2+d}$

$$Differentiate\ ''y''\ w.r.t.\ x$$

$$\frac{dy}{dx}=\frac{d}{dx}\left(\frac{ax+b}{cx+d}\right)$$

$$\frac{dy}{dx}=\frac{(cx+d)\frac{d}{dx}(ax+b)-(ax+b)\frac{d}{dx}(cx+d)}{(cx+d)^2}$$

$$\frac{dy}{dx}=\frac{(cx+d)(a)-(ax+b)(c)}{(cx+d)^2}$$

$$\frac{dy}{dx}=\frac{acx+ad-acx-bc}{(cx+d)^2}$$

$$\frac{dy}{dx}=\frac{ad-bc}{(cx+d)^2}$$

Differentiate u w.r.t. x

$$\frac{du}{dx}=\frac{d}{dx}\left(\frac{ax^2+b}{ax^2+d}\right)$$

$$\frac{du}{dx}$$

$$=\frac{(ax^2+d)\frac{d}{dx}(ax^2+b)-(ax^2+b)\frac{d}{dx}(ax^2+d)}{(ax^2+d)^2}$$

$$\frac{du}{dx}=\frac{(ax^2+d)(2ax)-(ax^2+b)(2ax)}{(ax^2+d)^2}$$

$$\frac{du}{dx}=\frac{2ax(ax^2+d-ax^2-b)}{(ax^2+d)^2}$$

$$\frac{du}{dx}=\frac{2ax(d-b)}{(ax^2+d)^2}$$

Now by chain rule

$$\frac{dy}{du}=\frac{dy}{dx}.\frac{dx}{du}$$

$$\frac{dy}{du}=\frac{dy}{dx}.\frac{1}{\frac{du}{dx}}$$

$$\frac{dy}{du}=\frac{ad-bc}{(cx+d)^2}.\frac{1}{\frac{2ax(d-b)}{(ax^2+d)^2}}$$

$$\frac{dy}{du}=\frac{(ad-bc)(ax^2+d)^2}{2ax(d-b)(cx+d)^2}$$

Which is required.

(v) $\frac{x^2+1}{x^2-1}\ w.r.t.x^3$

Solution.

Suppose $y=\frac{x^2+1}{x^2-1}$ and $u=x^3$

$$Differentiate\ ''y''\ w.r.t.\ x$$

$$\frac{dy}{dx}=\frac{d}{dx}\left(\frac{x^2+1}{x^2-1}\right)$$

$$\frac{dy}{dx}=\frac{(x^2-1)\frac{d}{dx}(x^2+1)-(x^2+1)\frac{d}{dx}(x^2-1)}{(x^2-1)^2}$$

$$\frac{dy}{dx}=\frac{(x^2-1)(2x)-(x^2+1)(2x)}{(x^2-1)^2}$$

$$\frac{dy}{dx}=2x\frac{x^2-1-x^2-1}{(x^2-1)^2}$$

$$\frac{dy}{dx}=2x\frac{-2}{(x^2-1)^2}$$

$$\frac{dy}{dx}=\frac{-4x}{(x^2-1)^2}$$

Differentiate u w.r.t. x

$$\frac{du}{dx}=\frac{d}{dx}(x^3)$$

$$\frac{du}{dx}=3x^2$$

Now by chain rule

$$\frac{dy}{du}=\frac{dy}{dx}.\frac{dx}{du}$$

$$\frac{dy}{du}=\frac{dy}{dx}.\frac{1}{\frac{du}{dx}}$$

$$\frac{dy}{du}=\frac{-4x}{(x^2-1)^2}.\frac{1}{3x^2}$$

$$\frac{dy}{du}=\frac{-4}{3x(x^2-1)^2}$$

Which is re

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Derivative of Trigonometric function:

1. $\frac{d}{dx}(\boldsymbol{sinx}) = \boldsymbol{cosx}$

Proof: *let* $y = sinx \to (i)$

$\Rightarrow$ $\boldsymbol{y + \delta y = sin(x + \delta x)} \to \boldsymbol{(ii)}$

$by\ (ii) - (i) \Rightarrow y + \delta y - y = \sin(x + \delta x) - sinx$

$(\because sinA - sinB = 2cos\frac{A+B}{2} sin\frac{A-B}{2}$

$\Rightarrow \delta y = 2\cos\left(\frac{x+\delta x+x}{2}\right)\sin\frac{(x+\delta x-x)}{2}$

$\Rightarrow \delta y = 2\cos\left(\frac{2x+\delta x}{2}\right)\sin\left(\frac{\delta x}{2}\right)$

$dividing\ by\ \delta x\ and\ take\ limit\ \delta x \to 0$

$$\lim_{\delta x\to 0}\frac{\delta y}{\delta x} = \lim_{\delta x\to 0}\left[2\cos\left(x+\frac{\delta x}{2}\right).\frac{\left(\sin\left(\frac{\delta x}{2}\right)\right)}{\delta x}\right]$$

$\Rightarrow \frac{dy}{dx} = 2\lim_{\delta x\to 0}\left[\cos\left(x+\frac{\delta x}{2}\right).\frac{\left(\sin\left(\frac{\delta x}{2}\right)\right)}{2\times\frac{\delta x}{2}}\right]$

$\Rightarrow \frac{dy}{dx} = cosx \qquad (\because)\lim_{\theta\to 0}\frac{sin\theta}{2} = 1$

Thus $\frac{d}{dx}sinx = cosx$

2. $\frac{d}{dx}(\boldsymbol{cosx}) = \boldsymbol{-sinx}$

Proof: *let* $y = cosx \to (i)$

$\Rightarrow$ $\boldsymbol{y + \delta y = cos(x + \delta x)} \to \boldsymbol{(ii)}$

$by\ (ii) - (i) \Rightarrow y + \delta y - y = \cos(x + \delta x) - cosx$

$\Rightarrow \delta y = \cos(x + \delta x) - cosx$

$(\because cosA - cosB = -2sin\frac{A+B}{2} sin\frac{A-B}{2}$

$\Rightarrow \delta y = -2\sin\left(\frac{x+\delta x+x}{2}\right)\sin\frac{(x+\delta x-x)}{2}$

$\Rightarrow \delta y = -2\sin\left(\frac{2x+\delta x}{2}\right)\sin\left(\frac{\delta x}{2}\right)$

$dividing\ by\ \delta x\ and\ take\ limit\ \delta x \to 0$

$$\lim_{\delta x\to 0}\frac{\delta y}{\delta x} = \lim_{\delta x\to 0}\left[-2\sin\left(x+\frac{\delta x}{2}\right).\frac{\left(\sin\left(\frac{\delta x}{2}\right)\right)}{\delta x}\right]$$

$\Rightarrow \frac{dy}{dx} = -2\lim_{\delta x\to 0}\left[-2\left(x+\frac{\delta x}{2}\right).\frac{\left(\sin\left(\frac{\delta x}{2}\right)\right)}{2\times\frac{\delta x}{2}}\right]$

$\Rightarrow \frac{dy}{dx} = -sinx \qquad (\because)\lim_{\theta\to 0}\frac{sin\theta}{2} = 1$

Thus $\frac{d}{dx}cosx = -sinx$

3. $\frac{d}{dx}(\boldsymbol{Tanx}) = \boldsymbol{sec^2x}$

Proof: *let* $y = Tanx \to (i)$

$\Rightarrow$ $y + \delta y = Tan(x + \delta x) \to (ii)$

$by\ (ii) - (i) \Rightarrow y + \delta y - y = \mathrm{Tan}(x + \delta x) - Tanx$

$\Rightarrow \delta y = \frac{\sin(x+\delta x)}{\cos(x+\delta x)} - \frac{sinx}{cosx}$

$\Rightarrow \frac{\sin(x+\delta x-x)cosx - \cos(x+\delta x).sinx}{\cos(x+\delta x).cosx}$

$(\because sin\alpha cos\beta - cos\alpha sin\beta = \sin(\alpha - \beta)$

$\Rightarrow \delta y = \frac{\sin(x+\delta s-x)}{\cos(x+\delta x).cosx} = \frac{sin\delta x}{\cos(x+\delta x).cosx}$

$dividing\ by\ \delta x\ and\ take\ limit\ \delta x \to 0$

$$\lim_{\delta x\to 0}\frac{\delta y}{\delta x} = \lim_{\delta x\to 0}\frac{1}{\cos(x+\delta x).cosx}.\lim_{\delta x\to 0}\left(\frac{sin\delta x}{\delta x}\right)$$

$\Rightarrow \frac{dy}{dx} = \frac{1}{cosx.cosx}(1) = \frac{1}{\cos^2 x} = \sec^2 x$

$\Rightarrow \frac{dy}{dx} = \sec^2 x$

$\Rightarrow$ Thus $\frac{d}{dx}Tanx = \sec^2 x$

4. $\frac{d}{dx}(\boldsymbol{secx}) = \boldsymbol{secxtanx}$

Proof:

$$let\ y = secx \to (i)$$

$$y + \delta x = \sec(x + \delta x) \to (ii)$$

$by\ (ii) - (i) \quad \Rightarrow y + \delta y - y = secx(x + \delta x) - secx$

$$\Rightarrow \delta y = \frac{1}{\cos(x+\delta x)} - \frac{1}{cosx} = \frac{cosx - \cos(x+\delta x)}{\cos(x+\delta x).cosx}$$

$$\left(\because\ cosA - cosB = -2\sin\left(\frac{A+B}{2}\right)\sin\left(\frac{A-B}{2}\right)\right)$$

$$= \frac{-2\sin\left(\frac{x+x+\delta x}{2}\right)\sin\left(\frac{x-x-\delta x}{2}\right)}{\cos(x+\delta x).\delta x}$$

$$= \frac{-2}{\cos(x+\delta x).cosx}\left[\sin\left(x+\frac{\delta x}{2}\right)\sin\left(-\frac{\delta x}{2}\right)\right]$$

$$= \frac{2}{\cos(x+\delta x).cosx}\left[\sin\left(x+\frac{\delta x}{2}\right)\sin\left(\frac{\delta x}{2}\right)\right]$$

$\because \sin(-\theta) - sin\theta$

Dividing by $\delta x\ and\ take\ limit\ \delta x \to 0$

$$\lim_{\delta x\to 0}\frac{\delta y}{\delta x} = \lim_{\delta x\to 0}\frac{2}{\cos(x+\delta x).cosx}\left[\lim_{\delta x\to 0}\sin\left(x+\frac{\delta x}{2}\right)\lim_{\delta x\to 0}\sin\frac{\left(\frac{\delta x}{2}\right)}{2\times\frac{\delta x}{2}}\right]$$

$\Rightarrow \frac{dy}{dx} = -\frac{1}{cosx.cosx}(sinx.1) = \frac{1}{\cos x}.\frac{sinx}{\cos x}$

$\Rightarrow \frac{dy}{dx} = secxtanx \qquad hence\ \frac{d}{dx}(secx) = secxtanx$

5. $\frac{d}{dx}(\boldsymbol{cosecx}) - \boldsymbol{cosecxcotx}$

Proof:

$$let\ y = cosecx \to (i)$$

$$y + \delta x = co\sec(x + \delta x) \to (ii)$$

$by\ (ii) - (i) \quad \Rightarrow y + \delta y - y = cosecx(x + \delta x) - cosecx$

$$\Rightarrow \delta y = \frac{1}{\sin(x+\delta x)} - \frac{1}{sinx} = \frac{sinx - \sin(x+\delta x)}{\sin(x+\delta x).sinx}$$

$$\left(\because\ sinA - sinB = 2\cos\left(\frac{A+B}{2}\right)\sin\left(\frac{A-B}{2}\right)\right)$$

$$= \frac{2\cos\left(\frac{x+x+\delta x}{2}\right)\sin\left(\frac{x-x-\delta x}{2}\right)}{\sin(x+\delta x).sinx}$$

$$= \frac{1}{\sin(x+\delta x).sinx}\left[2\cos\left(x+\frac{\delta x}{2}\right)\sin\left(-\frac{\delta x}{2}\right)\right]$$

$$= \frac{1}{\cos(x+\delta x).cosx}\left[\cos\left(x+\frac{\delta x}{2}\right)\sin\left(-\frac{\delta x}{2}\right)\right]$$

$\because \sin(-\theta) - sin\theta$

Dividing by $\delta x\ and\ take\ limit\ \delta x \to 0$

$$\lim_{\delta x\to 0}\frac{\delta y}{\delta x} = \lim_{\delta x\to 0}\frac{2}{\sin(x+\delta x).sinx}$$

$$\left[\lim_{\delta x\to 0}2\cos x\left(x+\frac{\delta x}{2}\right)\lim_{\delta x\to 0}\sin\frac{\left(\frac{\delta x}{2}\right)}{2\times\frac{\delta x}{2}}\right]$$

$\Rightarrow \frac{dy}{dx} = -\frac{1}{sinx.sinx}(cosx.1) = \frac{-1}{sinx}.\frac{cosx}{sinx}$

$\Rightarrow \frac{dy}{dx} = -cosecxcotx$

$\Rightarrow \quad hence\ \frac{d}{dx}(cosecx) = -cosecxcotx$

6. $\frac{d}{dx}(\boldsymbol{cotx}) = \boldsymbol{cosec^2x}$

Proof: *let* $y = cotx \to (i)$

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⇨ $y+\delta y = cot(x+\delta x) \rightarrow (ii)$

$by\ (ii)-(i) \Rightarrow y+\delta y - y = \cot(x+\delta x) - cotx$

⇨ $\delta y = \frac{\cos(x+\delta x)}{\sin(x+\delta x)} - \frac{cosx}{sinx}$

⇨ $\frac{\cos(x+\delta x-x)sinx - \sin(x+\delta x).cosx}{\sin(x+\delta x).sinx}$

$(\because sin\alpha cos\beta - cos\alpha sin\beta = \sin(\alpha-\beta)$

⇨ $\delta y = \frac{\sin(x-\delta s-x)}{\sin(x+\delta x).sinx} = \frac{\sin(-\delta x)}{\sin(x+\delta x).sinx}$

$dividing\ by\ \delta x\ and\ take\ limit\ \delta x \rightarrow 0$

$\lim_{\delta x\to 0}\frac{\delta y}{\delta x} = \lim_{\delta x\to 0}\frac{-1}{\sin(x+\delta x).sinx}.\lim_{\delta x\to 0}\left(\frac{sin\delta x}{\delta x}\right)$

⇨ $\frac{dy}{dx} = \frac{1}{sinx.sinx}(1) = \frac{1}{\sin^2 x} = -co\sec^2 x$

⇨ $\frac{dy}{dx} = -\text{cosec}^2\, x$

⇨ Thus $\frac{d}{dx}cotx = cosec^2 x$

Derivative of inverse trigonometric function:

1. $\frac{d}{dx}[sin^{-1}x] = \frac{1}{\sqrt{1-x^2}}$ $\quad x \in (-1,1)\ or\ -1 < x < 1$

Proof: $let\ y = sin^{-1}x$

⇨ $siny = x \Rightarrow \frac{d}{dx}(siny) = \frac{d}{dx}(x)$

⇨ $cosy.\frac{dy}{dx} = 1 \Rightarrow \frac{dy}{ds} = \frac{1}{cosy}$

⇨ $\frac{dy}{dx} = \frac{1}{\sqrt{1-sin^2 y}}$ $\quad \because cos^2\theta = 1 - sin^2\theta$

⇨ $cos\theta = \sqrt{1-sin^2\theta}$

⇨ $\frac{dy}{dx} = \frac{1}{\sqrt{1-x^2}}$ $\quad \because siny = x$

Hence $\frac{d}{dx}(sin^{-1}) = \frac{1}{\sqrt{1-x^2}}$

2. $\frac{d}{dx}(cox^{-1}x) = -\frac{1}{\sqrt{1-x^2}}$ $\quad x \in (-1,1)$

Proof: $let\ y = cos^{-1}x$

⇨ $cosy = x \Rightarrow \frac{d}{dx}(cosy) = \frac{d}{dx}(x)$

⇨ $-siny.\frac{dy}{dx} = 1 \Rightarrow \frac{dy}{ds} = \frac{-1}{siny}$

⇨ $\frac{dy}{dx} = \frac{-1}{\sqrt{1-cos^2 y}}$ $\quad \because sin^2\theta = 1 - cos^2\theta$

⇨ $sin\theta = \sqrt{1-cos^2\theta}$

⇨ $\frac{dy}{dx} = \frac{-1}{\sqrt{1-x^2}}$ $\quad \because cosy = x$

Hence $\frac{d}{dx}(cos^{-1}) = \frac{-1}{\sqrt{1-x^2}}$

3. $\frac{d}{dx}(tan^{-1}x) = \frac{1}{1+x^2}$ $\quad ,x \in R$

Proof: $lety = Tan^{-1}x$

$Tany = x \Rightarrow \frac{d}{dx}(tany) = \frac{d}{dx}(x)$

$\Rightarrow sec^2 y.\frac{dy}{dx} = 1 \Rightarrow \frac{dy}{dx} = \frac{1}{sec^2 y}$

⇨ $\frac{dy}{dx} = \frac{1}{1+tan^2 y}$ $\quad \because 1 + tan^2\theta = sec^2\theta$

⇨ $\frac{dy}{dx} = \frac{1}{1+x^2}$ $\quad \because tany = x$

⇨ Hence $\frac{d}{dx}(tan^{-1}x) = \frac{1}{1+x^2}$

4. $\frac{d}{dx}(cot^1 x) = \frac{-1}{1+x^2}$ $\quad ,x \in [-1,1]'$,

$[-1,1] = (-\infty,-1) \cup (1,+\infty)$

Proof: $lety = cot^{-1}x$

$coty = x \Rightarrow \frac{d}{dx}(coty) = \frac{d}{dx}(x)$

$\Rightarrow -cosec^2 y.\frac{dy}{dx} = 1 \Rightarrow \frac{dy}{dx} = \frac{1}{cosec^2 y}$

⇨ $\frac{dy}{dx} = \frac{-1}{1+cot^2 y}$ $\quad \because cosec^2\theta = 1 + cot^2$

⇨ $\frac{dy}{dx} = \frac{-1}{1+x^2}$ $\quad \because coty = x$

⇨ Hence $\frac{d}{dx}(cot^{-1}x) = \frac{-1}{1+x^2}$

5. $\frac{d}{dx}(sec^{-1}x) = \frac{1}{x\sqrt{x^2-1}}$ $\quad ,x \in [-1,1]$

Proof: $let\ y = sec^{-1}x \Rightarrow secy = x$

$\Rightarrow \frac{d}{dx}(secy) = \frac{d}{dx}(x) \Rightarrow secytany.\frac{dy}{dx} = 1$

$\Rightarrow \frac{dy}{dx} = \frac{1}{secytany} = \frac{1}{secy\sqrt{sec^2 y-1}}$ $\quad \because 1 + tan^2\theta = sec^2\theta$

$\Rightarrow tan^2\theta = sec^2 - 1$

$\Rightarrow tan\theta = \sqrt{sec^2\theta - 1}$

$\frac{dy}{dx} = \frac{1}{x\sqrt{x^2-1}}$ $\quad \because secy = x$

$hence\ \frac{d}{dx}(sec^{-1}x) = \frac{1}{x\sqrt{x^2-1}}$

6. $\frac{d}{dx}(cosec^{-1}x) = \frac{-1}{x\sqrt{x^2-1}}$ $\quad ,x \in R$

Proof: $let\ y = cosec^{-1}x \Rightarrow cosecy = x$

$\Rightarrow \frac{d}{dx}(cosecy) = \frac{d}{dx}(x) \Rightarrow cosecycoty.\frac{dy}{dx} = 1$

$\Rightarrow \frac{dy}{dx} = \frac{-1}{cosecycoty} = \frac{1}{cosecy\sqrt{cosec^2 y-1}}$

$\because 1 + cot^2\theta = cosec^2\theta$

$\Rightarrow cot^2\theta = cosec^2 - 1$

$\Rightarrow cot\theta = \sqrt{cosec^2\theta - 1}$

$\frac{dy}{dx} = \frac{-1}{x\sqrt{x^2-1}}$ $\quad \because cosecy = x$

$hence\ \frac{d}{dx}(cosec^{-1}x) = \frac{-1}{x\sqrt{x^2-1}}$

Exercise 2.5

QUESTION NO.1:

Differentiate the following trigonometric functions from the first principals.

i) $sin2x$

ii) Tan3x

iii) $sin2x + cos2x$

iv) $cosx^2$

v) $tan^2 x$

vi) $\sqrt{tanx}$

vii) $cos\sqrt{x}$

Solution:

(i)

Let $y = sin2x \rightarrow (1)$

$y + \delta y = sin2(x+\delta x) \rightarrow (2)$

$Eq\ 2 - eq1$

$y - \delta y - y = sin2(x+\delta x) - sin2x$

$\delta y = sin(2x + 2\delta x) - sin2x$

$using\ sinP - sinQ = 2cos\left(\frac{P+Q}{2}\right)sin\left(\frac{P-Q}{2}\right)$

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$$\delta y = 2cos\left(\frac{2x+2\delta x+2x}{2}\right)sin\left(\frac{2x+2\delta x-2x}{2}\right)$$
$$= 2cos\left(\frac{4x+2\delta x}{2}\right)sin\left(\frac{2\delta x}{2}\right)$$
$$\delta y = 2cos(2x+\delta x)sin\delta x$$

Dividing both side by δx

$$\frac{\delta y}{\delta x} = \frac{2cos(2x+\delta x)sin\delta x}{\delta x}$$
$$\lim_{\delta x\to 0}\frac{\delta y}{\delta x} = 2\lim_{\delta x\to 0} cos(2x+\delta x).\lim_{\delta x\to 0}\frac{sin\delta x}{\delta x}$$
$$\frac{dy}{dx} = 2cos(2x+0).1$$
$$\frac{dy}{dx} = 2cos2x$$

(ii) $\boldsymbol{let \quad y = tan3x \to (1)}$

$$y+\delta y = tan3(x+\delta x) \to (2)$$

Eq2-eq1

$$y+\delta y - y = tan(3x+3\delta x) - tan3x$$
$$\delta y = \frac{sin(3x+3\delta x)}{cos(3x+3\delta x)} - \frac{sin3x}{cos3x}$$
$$= \frac{sin(3x+3\delta x).cos3x - cos(3x+3\delta x).sin3x}{cos(3x+3\delta x).cos3x}$$
$$using\ sin\alpha co\beta - c0s\alpha sin\beta = sin(\alpha-\beta)$$
$$= \frac{sin(3x+3\delta x-3x)}{cos(3x+3\delta x).cos3x}$$
$$dividing\ both\ sides\ by\ \delta x$$
$$\frac{\delta y}{\delta x} = \frac{1}{cos(3x+3\delta x).cos3x}.\frac{sin3\delta x}{\delta x}$$
$$\lim_{\delta x\to 0}\frac{\delta y}{\delta x} = \lim_{\delta x\to 0}\frac{3}{cos(3x+3\delta x).cos3x}.\lim_{\delta x\to 0}\frac{sin3\delta x}{\delta x}$$
$$\frac{dy}{dx} = \frac{3}{cos(3x+0)cos3x}.1$$
$$= \frac{3}{cos3x.cos3x}$$
$$= \frac{3}{cos^2 3x}$$
$$\frac{dy}{dx} = 3sec^2 3x$$

(iii) Let $y = sin2x + cos2x \to (1)$

$$y+\delta y = sin2(x+\delta x) + cos2(x+\delta x) \to (2)$$
$$Eq2 - eq1$$
$$y+\delta y - y = sin(2x+2\delta x) + cos(2x+2\delta x) - sin2x - cos2x$$
$$\delta y = [sin(2x+2\delta x) - sin2x] + [cos(2x+2\delta x) - cos2x]$$
$$= 2cos\left(\frac{2x+2\delta x+2x}{2}\right).sin\left(\frac{2x+2\delta x-2x}{2}\right)$$
$$-2sin\left(\frac{2x+2\delta x+2x}{2}\right)sin\left(\frac{2x+2\delta x-2x}{2}\right)$$
$$= 2cos\left(\frac{4x+2\delta x}{2}\right).sin\delta x - 2sin\left(\frac{4x+2\delta x}{2}\right).sin\delta x$$
$$\delta y = [2cos(2x+2\delta x) - 2sin(2x+2\delta x)].sin\delta x$$
$$dividing\ both\ sides\ by\ \delta x$$
$$\frac{\delta y}{\delta x} = [2cos(2x+2\delta x) - 2sin(2x+2\delta x)].\frac{sin\delta c}{\delta x}$$
$$\lim_{\delta x\to 0}\frac{\delta y}{\delta x} = \lim_{\delta x\to 0}[2cos(2x+2\delta x) - 2sin(2x+2\delta x)]\lim_{\delta x\to 0}\frac{sin\delta x}{\delta x}$$
$$= \{2cos2x - 2sin2x\}.1$$
$$\frac{dy}{dx} = 2cos2x - 2sin2$$

(iv) $cosx^2$

Solution:

$$let\ y = cos$$
$$Let \quad y = cosx^2 \to (1)$$
$$y+\delta y = cos(x+\delta x)^2 \to (2)$$
$$Eq(2) - eq(1)$$
$$y+\delta y - y = cos(x+\delta x)^2 - cosx^2$$
$$\delta y = -2sin\left(\frac{x^2+\delta x^2+2x\delta x+x^2}{2}\right).sin\left(\frac{x^2+\delta x^2+2x\delta x-x^2}{2}\right)$$
$$\delta y = \frac{-2sin(2x^2+2x\delta x+\delta x^2).sin\left(\frac{\delta x(\delta x+2x)}{2}\right)}{\delta x}$$
$$\frac{\delta y}{\delta x} = -2sin(2x^2+2x\delta x+\delta x^2).\frac{sin\left(\delta x(\delta x+2x)\right)}{\delta x.\frac{\delta x+2x}{2}}.\frac{\delta x+2x}{2}$$
$$\lim_{\delta x\to 0}\frac{\delta y}{\delta x} = \lim_{\delta x\to 0} -2sin(2x^2+2x\delta x+\delta x^2).\frac{sin\left(\delta x(\delta x+2x)\right)}{\delta x.\frac{\delta x+2x}{2}}.\frac{\delta x+2x}{2}$$
$$\frac{dy}{dx} = -2sin\left(\frac{2x^2}{2}\right).1.\frac{2x}{2}$$
$$\frac{dy}{dx} = -2xsinx^2$$

(v)

$$let \quad y = tan^2 x \to (1)$$
$$y+\delta y = tan^2(x+\delta x) \to (2)$$
$$eq2 - eq1$$
$$y+\delta y - y = tan^2(x+\delta x) - tan^2 x$$
$$\delta y = \{tan(x+\delta x) - tanx\}.\{tan(x+\delta x) + tanx\}$$
$$\delta y = \{tan(x+\delta x) - tanx\}.\left[\frac{sin(x+\delta x)}{cos(x+\delta x)} - \frac{sinx}{cosx}\right]$$
$$\delta y = [tan(x+\delta x) - tanx].\left[\frac{sin(x+\delta x).cosx - cos(x+\delta x).sinx}{cos(x+\delta x).cosx}\right]$$
$$\delta y = [tan(x+\delta x) - tanx].\left[\frac{sin(x+\delta x-x)}{cos(x+\delta x).cosx}\right]$$
$$\delta y = \frac{[tan(x+\delta x) - tanx]}{cos(x+\delta x).cosx}.sin\delta x$$
$$dividing\ both\ sides\ by\ \delta x$$
$$\frac{\delta y}{\delta x} = \frac{[tan(x+\delta x) - tanx]}{cos(x+\delta x).cosx}.\frac{sin\delta x}{\delta x}$$
$$\lim_{\delta x\to 0}\frac{\delta y}{\delta x} = \lim_{\delta x\to 0}\frac{[tan(x+\delta x) - tanx]}{cos(x+\delta x).cosx}.\lim_{\delta x\to 0}\frac{sin\delta x}{\delta x}$$
$$\frac{dy}{dx} = \frac{tanx+tanx}{cosx.cosx}.1$$
$$\frac{dy}{dx} = \frac{2tanx}{cos^2 x}$$
$$\frac{dy}{dx} = 2tanxsec^2$$

(vi)

$$let \qquad y = \sqrt{tanx} - - - - - 1$$
$$y+\delta y = \sqrt{tan(x+\delta x)} - - - - - 2$$
$$eq2 - eq1$$
$$\delta y = \sqrt{tan(x+\delta x)} - \sqrt{tanx}$$

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$$\delta y = \sqrt{tan(x+\delta x)} - \sqrt{tanx} \times \frac{\sqrt{tan(x+\delta x)} + \sqrt{tanx}}{\sqrt{tan(x+\delta x)} + \sqrt{tanx}}$$

$$\delta y = \frac{\left(\sqrt{tan(x+\delta x)}\right)^2 - \left(\sqrt{tanx}\right)^2}{\sqrt{tan(x+\delta x)} + \sqrt{tanx}}$$

$$\delta y = \frac{tan(x+\delta x) - tanx}{\sqrt{tan(x+\delta x)} + \sqrt{tanx}}$$

$$\delta y = \frac{1}{\sqrt{tan(x+\delta x)} + \sqrt{tanx}} \left[\frac{sin(x+\delta x)}{cos(x+\delta x)} - \frac{sinx}{cosx}\right.$$

$$\delta y = \frac{1}{\sqrt{tan(x+\delta x)} + \sqrt{tanx}} \left[\frac{sin(x+\delta x)cosx - cos(x+\delta x).sins}{cos(x+\delta x)cosx}\right]$$

$$\delta y = \frac{sin(x+\delta x - x)}{\sqrt{tan(x+\delta x)} + \sqrt{tanx}.cos(x+\delta x)cosx}.$$

$$\delta y = \frac{1}{\sqrt{tan(x+\delta x)} + \sqrt{tanx}.cos(x+\delta x)cosx}.sin\delta x$$

dividing both sides by δx and take limit on both sides

$$\lim_{\delta x \to 0} \frac{\delta y}{\delta x}$$

$$= \lim_{\delta x \to 0} \frac{1}{\sqrt{tan(x+\delta x)} + \sqrt{tanx}.cos(x+\delta x)cosx}. \lim_{\delta x \to 0} sin\delta x$$

$$\frac{dy}{dx} = \frac{1}{\sqrt{tan(x)} + \sqrt{tanx}.cos(x)cosx}.1$$

$$\frac{dy}{dx} = \frac{1}{2\sqrt{tanx}cos^2x}$$

$$\frac{dy}{dx} = \frac{sec^2x}{2\sqrt{tanx}}$$

(vii)

$$cos\sqrt{x}$$

Solution:

$$Let\ y = cox\sqrt{x} \rightarrow (i)$$

$$\Rightarrow y + \delta y = cos\sqrt{x+\delta x} \rightarrow (ii)$$

$$by\ (ii) - (i) \Rightarrow y + \delta y - y = cos\sqrt{x+\delta x} - cos\sqrt{x}$$

$$\Rightarrow \delta y = cos\sqrt{x+\delta x} - cos\sqrt{x}$$

$$\because \cos A - cosB = -2\sin\left(\frac{A+B}{2}\right)\sin\left(\frac{A-B}{2}\right)$$

$$= -2\sin\left(\frac{\sqrt{x+\delta x}+\sqrt{x}}{2}\right)sin\left(\frac{\sqrt{x+\delta x}-\sqrt{x}}{2}\right)$$

dividing by δx

$$\Rightarrow \frac{\delta y}{\delta x} = \frac{-2sin\frac{\sqrt{x+\delta x}+\sqrt{x}}{2}sin\frac{\sqrt{x+\delta x}-\sqrt{x}}{2}}{\delta x}$$

$$\because \left(\sqrt{x+\delta x}+\sqrt{x}\right)\left(\sqrt{x+\delta x}-\sqrt{x}\right) = \delta x$$

$$\Rightarrow \frac{\delta y}{\delta x} = \frac{-2sin\frac{\sqrt{x+\delta x}+\sqrt{x}}{2}}{\left(\sqrt{x+\delta x}+\sqrt{x}\right)} \times \frac{sin\frac{\sqrt{x+\delta x}-\sqrt{x}}{2}}{2 \times \frac{\sqrt{x+\delta x}-\sqrt{x}}{2}}$$

$$Take\ limit\ \delta x \rightarrow 0$$

$$\Rightarrow \lim_{\delta x \to 0} \frac{\delta y}{\delta x} = \lim_{\delta x \to 0} \frac{-sin\frac{\sqrt{x+\delta x}+\sqrt{x}}{2}}{\left(\sqrt{x+\delta x}+\sqrt{x}\right)} \times \lim_{\delta x \to 0} \frac{sin\frac{\sqrt{x+\delta x}-\sqrt{x}}{2}}{\frac{\sqrt{x+\delta x}-\sqrt{x}}{2}}$$

$$\Rightarrow \frac{dy}{dx} = \frac{-sin\frac{\sqrt{x}+\sqrt{x}}{2}}{\sqrt{x}+\sqrt{x}}\ (1)$$

$$= \frac{-\sin\left(\frac{2\sqrt{x}}{2}\right)}{2\sqrt{x}}$$

$$\Rightarrow \frac{dy}{dx} = \frac{-sin\sqrt{x}}{2\sqrt{x}}$$

Question # 2.

Differentiate the following w.r.t the variable involved.

(i) x^2sec4x

Solution.

Let $y = x^2sec4x$

Differentiate w.r.t x

$$\frac{dy}{dx} = \frac{d}{dx}(x^2sec4x)$$

$$\frac{dy}{dx} = (x^2)\frac{d}{dx}(sec4x) + (sec4x)\frac{d}{dx}(x^2)$$

$$\frac{dy}{dx} = (x^2)(sec4x\ tan4x\ (4x)) + (sec4x)(2x)$$

$$\frac{dy}{dx} = 4x^2\ (sec4xTan4x) + 2xsec4x$$

Which is required.

(ii) $y = \tan^3\theta.\sec^2\theta$

Solution.

Let $y = \tan^3\theta\ .\sec^2\theta$

Differentiate w.r.t θ

$$\frac{dy}{d\theta} = \frac{d}{d\theta}(\tan^3\theta.\sec^2\theta)$$

$$\frac{dy}{d\theta} = (\sec^2\theta)\frac{d}{\theta}(\tan^3\theta) + (\tan^3\theta)\frac{d}{d\theta}(\sec^2\theta)$$

$$\frac{dy}{d\theta} = (\sec^2\theta)(3\tan^2\theta(\sec^2\theta)) + (\tan^3\theta)(2sec\,\theta\ (sec\theta\ tan\theta))$$

$$\frac{dy}{d\theta} = (3\sec^4\theta\tan^2\theta) + (2\tan^4\theta\sec^2\theta)$$

$$\frac{dy}{d\theta} = \sec^2\theta\tan^2\theta(3\sec^2\theta\tan^2\theta + 2\tan^2\theta\sec^2\theta)$$

(iii) $y = (sin2\theta - cos3\theta)^2$

Solution.

Let $y = (sin2\theta - cos3\theta)^2$

Differentiate w.r.t θ

$$\frac{dy}{d\theta} = \frac{d}{d\theta}(sin2\theta - cos3\theta)^2$$

$$\frac{dy}{d\theta} = 2(sin2\theta - cos3\theta)\left(\frac{d}{d\theta}\right)(sin2\theta - cos3\theta)$$

$$\frac{dy}{d\theta} = 2(sin2\theta - cos3\theta)\left[cos\theta\frac{d}{d\theta}(2\theta) - (-sin3\theta\frac{d}{d\theta}(3\theta)\right]$$

$$\frac{dy}{dx} = 2(sin2\theta - cos3\theta)(2cos2\theta + 3sin3\theta)$$

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Which is required.

(iv) $y = cos\sqrt{x} + \sqrt{sinx}$

Solution.

Let $y = cos\sqrt{x} + \sqrt{sinx}$

Differentiate w.r.t x

$$\frac{dy}{dx} = \frac{d}{dx}(cos\sqrt{x} + \sqrt{sinx})$$

$$\frac{dy}{dx} = \frac{-sin\sqrt{x}}{2\sqrt{x}} + \frac{1}{2\sqrt{sinx}}(cosx)$$

$$\frac{dy}{dx} = \frac{1}{2}\left(\frac{-sin\sqrt{x}}{\sqrt{x}}\frac{cosx}{\sqrt{sinx}}\right)$$

Which is required.

Question # 3. Find $\frac{dy}{dx}$ if:

(i) $y = x\,cosy$

Solution.

Since $y = x\,cosy$

Differentiate w.r.t x

$$\frac{dy}{dx} = \frac{d}{dx}(x\,cosy)$$

$$\frac{dy}{dx} = (cosy)\frac{d}{dx}(x) + (x)\frac{d}{dx}(cosy)$$

$$\frac{dy}{dx} = cosy(1) + x(-siny)\frac{dy}{dx}$$

$$\frac{dy}{dx} = cosy - x\,siny\,\frac{dy}{dx}$$

$$\frac{dy}{dx} + x\,siny\,\frac{dy}{dx} = cosy$$

$$\frac{dy}{dx}(1 + x\,siny) = cosy$$

$$\frac{dy}{dx} = \frac{cosy}{1 + x\,siny}$$

Which is required.

(ii) $x = y\,siny$

Solution.

Since $x = y\,siny$

Differentiate w.r.t x

$$\frac{dx}{dx} = \frac{d}{dx}(y\,siny)$$

$$1 = (siny)\frac{d}{dx}(y) + (y)\frac{d}{dx}(siny)$$

$$1 = (siny)\left(\frac{dx}{dx}\right) + (y)(cosy)\left(\frac{dx}{dx}\right)$$

$$1 = \frac{dx}{dx}(siny + y\,cosy)$$

$$\frac{dy}{dx} = \frac{1}{siny + y\,cosy}$$

Question # 4. Find derivative w.r.t x

(i) $y = cos\sqrt{\frac{1+x}{1+2x}}$

Solution.

Let $y = cos\sqrt{\frac{1+x}{1+2x}}$

Differentiate w.r.t x

$$\frac{dy}{dx} = \frac{d}{dx}\left(cos\sqrt{\frac{1+x}{1+2x}}\right)$$

$$\frac{dy}{dx} = \left(-sin\sqrt{\frac{1+x}{1+2x}}\right)\frac{d}{dx}\left(\sqrt{\frac{1+x}{1+2x}}\right)$$

$$\frac{dy}{dx} = \left(-sin\sqrt{\frac{1+x}{1+2x}}\right)\frac{1}{2}\left(\frac{1+x}{1+2x}\right)^{-\frac{1}{2}}\frac{d}{dx}\left(\frac{1+x}{1+2x}\right)$$

$$\frac{dy}{dx} = \left(-sin\sqrt{\frac{1+x}{1+2x}}\right)\frac{1}{2}\left(\frac{1+2x}{1+x}\right)^{\frac{1}{2}}\left(\frac{(1+2x)\frac{d}{dx}(1+x) - (1+x)\frac{d}{dx}(1+2x)}{(1+2x)^2}\right)$$

$$\frac{dy}{dx} = \left(-sin\sqrt{\frac{1+x}{1+2x}}\right)\frac{1}{2}\left(\frac{1+2x}{1+x}\right)^{\frac{1}{2}}\left(\frac{(1+2x)(1) - (1+x)(2)}{(1+2x)^2}\right)$$

$$\frac{dy}{dx} = \left(-sin\sqrt{\frac{1+x}{1+2x}}\right)\frac{1}{2}\left(\frac{1+2x}{1+x}\right)^{\frac{1}{2}}\left(\frac{1+2x-2-2x}{(1+2x)^2}\right)$$

$$\frac{dy}{dx} = \left(-sin\sqrt{\frac{1+x}{1+2x}}\right)\frac{1}{2}\left(\frac{1}{1+x}\right)^{\frac{1}{2}}\left(\frac{-1}{(1+2x)^{2-\frac{1}{2}}}\right)$$

$$\frac{dy}{dx} = \left(\frac{sin\sqrt{\frac{1+x}{1+2x}}}{2(1+2x)^{\frac{3}{2}}\sqrt{1+x}}\right)$$

Which is required.

(i) $y = sin\sqrt{\frac{1+2x}{1+x}}$

Solution.

Let $y = sin\sqrt{\frac{1+2x}{1+x}}$

Differentiate w.r.t x

$$\frac{dy}{dx} = \frac{d}{dx}\left(sin\sqrt{\frac{1+2x}{1+x}}\right)$$

$$\frac{dy}{dx} = \left(cos\sqrt{\frac{1+2x}{1+x}}\right)\frac{d}{dx}\left(\sqrt{\frac{1+2x}{1+x}}\right)$$

$$\frac{dy}{dx} = \left(cos\sqrt{\frac{1+2x}{1+x}}\right)\frac{1}{2}\left(\frac{1+2x}{1+x}\right)^{-\frac{1}{2}}\frac{d}{dx}\left(\frac{1+2x}{1+x}\right)$$

$$\frac{dy}{dx} = \left(cos\sqrt{\frac{1+2x}{1+x}}\right)\frac{1}{2}\left(\frac{1+2x}{1+x}\right)^{\frac{1}{2}}$$

$$\left(\frac{(1+x)\frac{d}{dx}(1+2x) - (1+2x)\frac{d}{dx}(1+x)}{(1+x)^2}\right)$$

$$\frac{dy}{dx}$$

$$= \left(cos\sqrt{\frac{1+2x}{1+x}}\right)\frac{1}{2}\left(\frac{1+x}{1+2x}\right)^{\frac{1}{2}}\left(\frac{(1+x)(2) - (1+2x)(1)}{(1+x)^2}\right)$$

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$$\frac{dy}{dx}=\left(cos\sqrt{\frac{1+2x}{1+x}}\right)\frac{1}{2}\left(\frac{1+x}{1+2x}\right)^{\frac{1}{2}}\left(\frac{2+2x-1-2x}{(1+2x)^2}\right)$$

$$\frac{dy}{dx}=\left(cos\sqrt{\frac{1+2x}{1+x}}\right)\frac{1}{2}\left(\frac{1}{1+2x}\right)^{\frac{1}{2}}\left(\frac{1}{(1+x)^{2-\frac{1}{2}}}\right)$$

$$\frac{dy}{dx}=\left(\frac{cos\sqrt{\frac{1+2x}{1+x}}}{2(1+x)^{\frac{3}{2}}\sqrt{1+2x}}\right)$$

Which is required.

Question # 5 Differentiate

(i) $\boldsymbol{sinx\ w.r.t.cotx}$

Solution.

Let $y=$ sinx and $u=cotx$

we find $\frac{dy}{du}$

Now $y=$ sinx

Differentiate w.r.t x

$$\frac{dy}{dx}=\frac{d}{dx}(\text{sinx})$$

$$\frac{dy}{dx}=cosx$$

Now $u=$ cotx

Differentiate w.r.t x

$$\frac{du}{dx}=-Cosec^2x$$

$$\frac{dx}{du}=-\frac{1}{Cose^2x}$$

$$\frac{dx}{du}=-Sin^2x$$

Now using chain rule

$$\frac{dy}{du}=\frac{dy}{dx}.\frac{dx}{du}$$

$$\frac{dy}{du}=(Cosx)(-Sin^2x)=-Sin^2x\ Cosx.$$

(ii). $\boldsymbol{Sin^2x\ w.r.t\ \ Cos^4x}$

Solution.

Let $y=\ Sin^2x$ and $u=Cos^4x$

we find $\frac{dy}{du}$

Now $y=\ Sin^2x$

Differentiate w.r.t x

$$\frac{dy}{dx}=\frac{d}{dx}(Sin^2x)$$

$$\frac{dy}{dx}=2Sinx\ cosx$$

Now $u=Cos^4x$

Differentiate w.r.t x

$$\frac{du}{dx}=4Cos^3x(-Sinx)=-4SinxCos^3x$$

$$\frac{dx}{du}=-\frac{1}{4SinxCos^3x}$$

Now using chain rule

$$\frac{dy}{du}=\frac{dy}{dx}.\frac{dx}{du}$$

$$\frac{dy}{du}=(2Sinx\ cosx)\left(-\frac{1}{4SinxCos^3x}\right)$$

$$\frac{dy}{du}=-\frac{1}{2Cos^2x}$$

$$\frac{dy}{du}=-\frac{1}{2}Sec^2x$$

Question.6.

If $tany\ (1+tanx)=1-tanx$, *show that* $\frac{dy}{dx}=-1$.

Solution.

Since

$$tany\ (1+tanx)=1-tanx$$

$$tany=\frac{1-tanx}{1+tanx}$$

$$tany=\frac{\tan\frac{\pi}{4}-tanx}{\tan\frac{\pi}{4}+tanx}$$

$$tany=\tan\left(\frac{\pi}{4}-x\right)$$

$$y=\frac{\pi}{4}-x$$

Differentiate w.r.t "x" , we have

$$\frac{dy}{dx}=0-1$$

$$\frac{dy}{dx}=-1$$

Hence proved.

Question.7.

If $\boldsymbol{y=\sqrt{tanx+\sqrt{tanx+\sqrt{tanx+\cdots\infty}}}}$

, $\boldsymbol{Prove\ that\ (2y-1)\frac{dy}{dx}=Sec^2x}$

Solution.

Since

$$y=\sqrt{tanx+\sqrt{tanx+\sqrt{tanx+\cdots\infty}}}---(i)$$

Squarring on both sides, we have

$$y^2=tanx+\sqrt{tanx+\sqrt{tanx+\cdots\infty}}$$

$$y^2=tanx+\sqrt{tanx+\sqrt{tanx+\sqrt{tanx+\cdots\infty}}}$$

$$y^2=tanx+y$$

Differentiate w.r.t "x" , we have

$$2y\frac{dy}{dx}=Sec^2x+\frac{dy}{dx}$$

$$2y\frac{dy}{dx}-\frac{dy}{dx}=Sec^2x$$

$$(2y-1)\frac{dy}{dx}=Sec^2x$$

Hence Proved.

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Question.8

If $x = aCos^3\theta$, $y = b\,Sin^3\theta$, Show that $a\frac{dy}{dx} + b\,tanx = 0$

Solution

$$x = aCos^3\theta \quad , y = b\,Sin^3\theta$$

Diff. $''x''$ w r.t.$''\theta''$, *we have*

$$\frac{dx}{d\theta} = \frac{d}{d\theta}(aCos^3\theta)$$
$$\frac{dx}{d\theta} = a.3Cos^2\frac{d}{d\theta}(Cos\theta)$$
$$\frac{dx}{d\theta} = a.3Cos^2(-Sin\theta)$$
$$\frac{dx}{d\theta} = -3aSin\theta Cos^2$$
$$\frac{d\theta}{dx} = \frac{-1}{3aSin\theta Cos^2}$$

Diff. $''y''$ w r.t.$''\theta''$, *we have*

$$\frac{dy}{d\theta} = \frac{d}{d\theta}(b\,Sin^3\theta)$$
$$\frac{dy}{d\theta} = b.3Sin^2\frac{d}{d\theta}(Sin\theta)$$
$$\frac{dy}{d\theta} = b.3Sin^2Cos\theta$$
$$\frac{dy}{d\theta} = 3bSin^2Cos\theta$$

Now by chain rule

$$\frac{dy}{dx} = \frac{dy}{d\theta}.\frac{d\theta}{dx}$$
$$\frac{dy}{dx} = 3bSin^2Cos\theta.\frac{-1}{3aSin\theta Cos^2}$$
$$\frac{dy}{dx} = -\frac{b}{a}\,tan\theta$$
$$a\frac{dy}{dx} = -b\,tan\theta$$
$$a\frac{dy}{dx} + b\,tan\theta = 0$$

Hence proved.

Question.10. Differentiate $''x''$, *we have*

(i) $\boldsymbol{Cos^{-1}\frac{x}{a}}$

Solution.

Let $y = Cos^{-1}\frac{x}{a}$

Diff.w.r.t $''x''$, *we have*

$$\frac{dy}{dx} = \frac{d}{dx}\left(Cos^{-1}\frac{x}{a}\right)$$
$$\frac{dy}{dx} = -\frac{1}{\sqrt{1-\left(\frac{x}{a}\right)^2}}.\frac{d}{dx}\left(\frac{x}{a}\right)$$
$$\frac{dy}{dx} = -\frac{1}{\sqrt{1-\frac{x^2}{a^2}}}.\left(\frac{1}{a}\right)$$
$$\frac{dy}{dx} = -\frac{1}{\sqrt{\frac{a^2-x^2}{a^2}}}.\left(\frac{1}{a}\right)$$
$$\frac{dy}{dx} = -\frac{1}{\frac{\sqrt{a^2-x^2}}{a}}.\left(\frac{1}{a}\right)$$
$$\frac{dy}{dx} = -\frac{a}{\sqrt{a^2-x^2}}.\left(\frac{1}{a}\right)$$
$$\frac{dy}{dx} = -\frac{1}{\sqrt{a^2-x^2}}.$$

Which is required.

(ii) $\boldsymbol{Cot^{-1}\frac{x}{a}}$

Solution.

Let $y = Cot^{-1}\frac{x}{a}$

Diff.w.r.t $''x''$, *we have*

$$\frac{dy}{dx} = \frac{d}{dx}\left(Cot^{-1}\frac{x}{a}\right)$$
$$\frac{dy}{dx} = -\frac{1}{1+\left(\frac{x}{a}\right)^2}.\frac{d}{dx}\left(\frac{x}{a}\right)$$
$$\frac{dy}{dx} = -\frac{1}{1+\frac{x^2}{a^2}}.\left(\frac{1}{a}\right)$$
$$\frac{dy}{dx} = -\frac{1}{\frac{a^2+x^2}{a^2}}.\left(\frac{1}{a}\right)$$
$$\frac{dy}{dx} = -\frac{1}{\frac{a^2+x^2}{a^2}}.\left(\frac{1}{a}\right)$$
$$\frac{dy}{dx} = -\frac{a^2}{a^2+x^2}.\left(\frac{1}{a}\right)$$
$$\frac{dy}{dx} = -\frac{a}{a^2+x^2}.$$

Which is required.

(iii) $\frac{1}{a}Sin^{-1}\frac{a}{x}$

Solution.

Let $y = \frac{1}{a}Sin^{-1}\frac{a}{x}$

Diff.w.r.t $''x''$, *we have*

$$\frac{dy}{dx} = \frac{d}{dx}\left(\frac{1}{a}Sin^{-1}\frac{a}{x}\right)$$
$$\frac{dy}{dx} = \frac{1}{a}.\frac{1}{\sqrt{1-\left(\frac{a}{x}\right)^2}}.\frac{d}{dx}\left(\frac{a}{x}\right)$$
$$\frac{dy}{dx} = \frac{1}{a}.\frac{1}{\sqrt{1-\frac{a^2}{x^2}}}.\left(\frac{-a}{x^2}\right)$$
$$\frac{dy}{dx} = \frac{1}{a}.\frac{1}{\sqrt{\frac{x^2-a^2}{x^2}}}.\left(\frac{-a}{x^2}\right)$$
$$\frac{dy}{dx} = \frac{1}{a}.\frac{1}{\frac{\sqrt{x^2-a^2}}{x}}.\left(\frac{-a}{x^2}\right)$$
$$\frac{dy}{dx} = \frac{x}{\sqrt{x^2-a^2}}.\left(\frac{-1}{x^2}\right)$$
$$\frac{dy}{dx} = \frac{-1}{x\sqrt{x^2-a^2}}$$

Which is required.

(iv) $Sin^{-1}\sqrt{1-x^2}$

Solution.

Let $y = Sin^{-1}\sqrt{1-x^2}$

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Diff.w.r.t $''x'', we\ have$

$$\frac{dy}{dx}=\frac{d}{dx}\left(Sin^{-1}\sqrt{1-x^2}\right)$$

$$\frac{dy}{dx}=\frac{1}{\sqrt{1-\left(\sqrt{1-x^2}\right)^2}}.\frac{d}{dx}\left(\sqrt{1-x^2}\right)$$

$$\frac{dy}{dx}=\frac{1}{\sqrt{1-(1-x^2)}}.\left(\frac{1}{2\sqrt{1-x^2}}\frac{d}{dx}(1-x^2)\right)$$

$$\frac{dy}{dx}=\frac{1}{\sqrt{1-1+x^2}}.\left(\frac{1}{2\sqrt{1-x^2}}(-2x)\right)$$

$$\frac{dy}{dx}=\frac{1}{\sqrt{x^2}}.\left(\frac{-x}{\sqrt{1-x^2}}\right)$$

$$\frac{dy}{dx}=\frac{1}{x}.\left(\frac{-x}{\sqrt{1-x^2}}\right)$$

$$\frac{dy}{dx}=\frac{-1}{\sqrt{1-x^2}}$$

Which is required.

(v) $Sec^{-1}\left(\frac{x^2-1}{x^2+1}\right)$

Solution.

Let $y=\ Sec^{-1}\left(\frac{x^2+1}{x^2-1}\right)$

Diff.w.r.t $''x'', we\ have$

$$\frac{dy}{dx}=\frac{d}{dx}\left(Sec^{-1}\left(\frac{x^2+1}{x^2-1}\right)\right)$$

$$\frac{dy}{dx}=\frac{1}{\left(\frac{x^2+1}{x^2-1}\right)\sqrt{\left(\frac{x^2+1}{x^2-1}\right)^2-1}}.\frac{d}{dx}\left(\frac{x^2+1}{x^2-1}\right)$$

$$\frac{dy}{dx}=\frac{1}{\left(\frac{x^2+1}{x^2-1}\right)\frac{\sqrt{(x^2+1)^2-(x^2-1)^2}}{(x^2-1)}}.$$

$$\left(\frac{(x^2-1)\frac{d}{dx}(x^2+1)-(x^2+1)\frac{d}{dx}(x^2-1)}{(x^2-1)^2}\right)$$

$$\frac{dy}{dx}=\frac{1}{(x^2+1)\frac{\sqrt{4.x^2.1}}{(x^2-1)^2}}.\left(\frac{(x^2-1)(2x)-(x^2+1)(2x)}{(x^2-1)^2}\right)$$

$$\frac{dy}{dx}=\frac{1}{(x^2+1).2x}.(2x(x^2-1-x^2-1)$$

$$\frac{dy}{dx}=\frac{1}{(x^2+1)}.(-2)$$

$$\frac{dy}{dx}=\frac{-2}{x^2+1}$$

Which is required.

(vi) $\boldsymbol{Cot^{-1}\left(\frac{2x}{1-x^2}\right)}$

Solution.

Let $y=Cot^{-1}\left(\frac{2x}{1-x^2}\right)$

Diff.w.r.t $''x'', we\ have$

$$\frac{dy}{dx}=\frac{d}{dx}\left(Cot^{-1}\left(\frac{2x}{1-x^2}\right)\right)$$

$$\frac{dy}{dx}=-\frac{1}{1+\left(\frac{2x}{1-x^2}\right)^2}.\frac{d}{dx}\left(\frac{2x}{1-x^2}\right)$$

$$\frac{dy}{dx}=-\frac{1}{1+\frac{4x^2}{(1-x^2)^2}}.\left(2.\frac{(1-x^2)\frac{d}{dx}(x)-x\frac{d}{dx}(1-x^2)}{(1-x^2)^2}\right)$$

$$\frac{dy}{dx}=-\frac{1}{\frac{(1-x^2)^2+4x^2}{(1-x^2)^2}}.\left(2.\frac{(1-x^2).1-x(-2x)}{(1-x^2)^2}\right)$$

$$\frac{dy}{dx}=-\frac{1}{1+x^4-2x^2+4x^2}.(2(1-x^2+2x^2))$$

$$\frac{dy}{dx}=-\frac{1}{1+x^4+2x^2}.(2(1+x^2))$$

$$\frac{dy}{dx}=-\frac{1}{(1+x^2)^2}.(2(1+x^2))$$

$$\frac{dy}{dx}=\frac{-2}{1+x^2}$$

Which is required.

(vii) $Cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)$

Solution.

Let $y=Cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)$

Diff.w.r.t $''x'', we\ have$

$$\frac{dy}{dx}=\frac{d}{dx}\left(Cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)\right)$$

$$\frac{dy}{dx}=-\frac{x^2-1}{x^2-1\sqrt{1-\left(\frac{1-x^2}{1+x^2}\right)^2}}.\frac{d}{dx}\left(\frac{1-x^2}{1+x^2}\right)$$

$$\frac{dy}{dx}$$

$$=-\frac{x^2-1}{x^2-1\sqrt{1-\frac{(1-x^2)^2}{(1+x^2)^2}}}.\left(\frac{(1+x^2)\frac{d}{dx}(1-x^2)-(1-x^2)\frac{d}{dx}(1+x^2)}{(x^2+1)^2}\right)$$

$$\frac{dy}{dx}$$

$$=-\frac{x^2-1}{x^2-1\sqrt{\frac{(1+x^2)^2-(1-x^2)^2}{(1+x^2)^2}}}.\left(\frac{(x^2+1)(-2x)-(1-x^2)(2x)}{(x^2+1)^2}\right)$$

$$=\frac{x^2-1}{\frac{(x^2-1)\sqrt{x^4+2x^2+1-(x^4+1-2x)}}{x^2-1}}.\frac{2x(x^2-1-x^2-1)}{(x^2-1)^2}$$

$$=\frac{(x^2-1)^2}{(x^2+1)\sqrt{2x^2+2x^2}}.\frac{2x(-2)}{(x^2-1)^2}$$

$$\frac{-4x}{(x^2\mp1)\sqrt{4x^2}}=\frac{-4x}{(x^2-1).2x}=\frac{-2x}{(x^2+1)}$$

$$\Rightarrow\frac{dy}{dx}=-\frac{2}{(x^2+1)}$$

Which is required.

Question.11.

Show that $\boldsymbol{\frac{dy}{dx}=\frac{y}{x}\ \ if\ \frac{y}{x}=\tan^{-1}\frac{x}{y}}$

Solution.

Since

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$$\frac{y}{x} = \tan^{-1}\frac{x}{y}$$
$$y = x\tan^{-1}\frac{x}{y}$$

Diff. w.r.t $''x'', we\ have$

$$\frac{dy}{dx} = x\frac{d}{dx}\tan^{-1}\frac{x}{y} + \tan^{-1}\frac{x}{y}\frac{d}{dx}(x)$$
$$\frac{dy}{dx} = x\frac{1}{1+\left(\frac{x}{y}\right)^2}\frac{d}{dx}\left(\frac{x}{y}\right) + \tan^{-1}\frac{x}{y}.1$$
$$\frac{dy}{dx} = x\frac{1}{1+\frac{x^2}{y^2}}\left(\frac{y\frac{d}{dx}x - x\frac{dy}{dx}}{y^2}\right) + \frac{y}{x}$$
$$\frac{dy}{dx} = x\frac{1}{\frac{y^2+x^2}{y^2}}\left(\frac{y - x\frac{dy}{dx}}{y^2}\right) + \frac{y}{x}$$
$$\frac{dy}{dx} = x\frac{1}{y^2+x^2}\left(y - x\frac{dy}{dx}\right) + \frac{y}{x}$$
$$\frac{dy}{dx} = \frac{x}{x^2+y^2}\left(y - x\frac{dy}{dx}\right) + \frac{y}{x}$$
$$\frac{dy}{dx} = \frac{xy}{x^2+y^2} - \frac{x^2}{x^2+y^2}\frac{dy}{dx} + \frac{y}{x}$$
$$\frac{dy}{dx} + \frac{x^2}{x^2+y^2}\frac{dy}{dx} = \frac{xy}{x^2+y^2} + \frac{y}{x}$$
$$\left(1+\frac{x^2}{x^2+y^2}\right)\frac{dy}{dx} = \frac{y}{x}\left(\frac{x^2}{x^2+y^2}+1\right)$$
$$\left(1+\frac{x^2}{x^2+y^2}\right)\frac{dy}{dx} = \frac{y}{x}\left(1+\frac{x^2}{x^2+y^2}\right)$$
$$\frac{dy}{dx} = \frac{y}{x}$$

Hence Proved.

Question.12.

If $y = \tan(p\tan^{-1}x)$, *Show that* $(1+x^2)y_1 - p(1+y^2)$

Solution.

Since

$$y = \tan(p\tan^{-1}x)$$
$$\tan^{-1}y = p\tan^{-1}x$$

Differentiate with respect to $''x'', we\ have$

$$\frac{1}{1+y^2}\frac{dy}{dx} = p\frac{1}{1+x^2}$$

Since $\frac{dy}{dx} = y_1$

$$\frac{1}{1+y^2}y_1 = p\frac{1}{1+x^2}$$
$$(1+x^2)y_1 = p(1+y^2)$$
$$(1+x^2)y_1 - p(1+y^2) = 0$$

Hence Proved.

Derivative of exponential Functions:

if $f(x) = a^x$ *wher* $a > 0$ *then* $f(x)$*is called general expontial function. here the base a is constant and exponent x is variable.*

if $f(x) = e^x$ *wher e* $\cong 2.71$ *then* $f(x)$*is called natural expontial function here base e is constant and expontent x is variable.*

1. $\frac{d}{dx}(e^x) = e^x$

Proof:

Let $y = e^x \to (i)$

$\Rightarrow y + \delta y = e^{x+\delta x} \to (ii)$

by $(ii) - (i) \Rightarrow y + \delta y - y = e^{x+\delta x} - e^x$

$\Rightarrow \delta y = e^x.e^{\delta x} - e^x = e^x(e^{\delta x} - 1)$

Dividing by δx *and take limit* $\delta x \to 0$

$$\lim_{\delta x\to 0}\frac{\delta y}{\delta x} = \lim_{\delta x\to 0}\left[e^x\left(\frac{e^{\delta x}-1}{\delta x}\right)\right]$$
$$\frac{dy}{dx} = e^x.\lim_{\delta x\to 0}\frac{e^{\delta x}-1}{\delta x}$$
$$\frac{dy}{dx} = e^x.lne \qquad \because \lim_{\delta x\to 0}\frac{a^x-1}{x} = lna$$
$$\frac{dy}{dx} = e^x \qquad and\ \lim_{\delta x\to 0}\frac{e^x-1}{x} = lne = 1$$
$$\frac{d}{dx}(e^x) = e^x$$

2. $\frac{d}{dx}(a^x) = a^x lna$

Proof:

$$Let\ y = a^x \to (i)$$
$$\Rightarrow y + \delta y = a^{x+\delta x} \to (ii)$$
$$by\ (ii) - (i) \Rightarrow y + \delta y - y = a^{x+\delta x} - a^x$$
$$\Rightarrow \delta y = a^x.a^{\delta x} - a^x = a^x(a^x - 1)$$

Divivding by δx *and take limit* $\delta x \to 0$

$$\lim_{\delta x\to 0}\frac{\delta y}{\delta x} = \lim_{\delta x\to 0}\left[a^x\frac{a^{\delta x}-1}{\delta x}\right]$$
$$\frac{dy}{dx} = a^x\lim_{\delta x\to 0}\frac{a^x-1}{\delta x} = a^x lna$$
$$Hence\frac{d}{dx}(a^x) = a^x lna \qquad (\because \lim_{\delta x\to 0}\frac{a^x-1}{x} = lna)$$

Note: as $\boldsymbol{a^x = a^x \Rightarrow lna^x = lna^x}$

$$\boldsymbol{\Rightarrow lna^x = xlna}$$
$$\boldsymbol{= xlna.lne \qquad \because lna^b = blna}$$
$$\boldsymbol{= xlna \qquad \because lne = 1}$$
$$\boldsymbol{ln^x = lne^{xlna}}$$

Derivative of the logarithmic function:

logarithmic function: *if* $a > 0\ a \neq 1$ *and* $x = a^y$ *the the functioon defined by* $y = log_a x.\ (x > 0)$ *is called the logarthmic of x to the base a.*

The logarithmic functions $log_e x$ *and* $log_{10}x$ *are called natural and common logarthmic resp.* $y = log_e x$ *is written as* $y = lnx$

1. $\frac{d}{dx}(lnx) = \frac{1}{x}$

Proof:

Let $y = lnx \to (i)$

$$\Rightarrow y + \delta y = \ln(x+\delta x) - (ii)$$
$$\Rightarrow (ii) - (i) \Rightarrow y + \delta y - y = \ln(x+\delta x) - lnx$$

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$$\delta y = \ln\frac{(x+\delta x)}{x} \qquad (\because lnx - lny = \ln\frac{x}{y})$$

$$\delta y = \ln\left(\frac{x}{x} + \frac{\delta x}{x}\right) = \ln\left(1 + \frac{\delta x}{x}\right)$$

$$Dividing\ by\ \delta x\ and\ take\ limit\ \delta x \to o$$

$$\lim_{\delta x\to 0}\frac{\delta y}{\delta x} = \lim_{\delta x\to 0}\left[\frac{1}{\delta x}\ln\left(1+\frac{\delta x}{x}\right)\right]$$

$$\frac{dy}{dx} = \frac{1}{x}\lim_{\delta x\to 0}\ln\left(1+\frac{\delta x}{x}\right)^{\frac{x}{\delta x}} \qquad \left(\because \lim_{\delta x\to 0}\left(1+\frac{1}{x}\right)^{n} = e\right)$$

$$\frac{dy}{dx} = \frac{1}{x}lne = \frac{1}{x}$$

$$Hence\ \frac{d}{dx}(lnx) = \frac{1}{x}$$

2. $\frac{d}{dx}[\log_a x] = \frac{1}{x}\frac{1}{lna}$

Proof:

$$Let\ u = \log_a x \to (i)$$

$$\Rightarrow y + \delta y = \log_a(x+\delta x) - (ii)$$

$$\Rightarrow (ii) - (i) \Rightarrow y + \delta y - y = \log_a(x+\delta x) - \log_a x$$

$$\delta y = \log_a\frac{(x+\delta x)}{x} \qquad (\because \log_a x - lny = \log_a\frac{x}{y})$$

$$\delta y = \log_a\left(\frac{x}{x} + \frac{\delta x}{x}\right) = \log_a\left(1+\frac{\delta x}{x}\right)$$

$$Dividing\ by\ \delta x\ and\ take\ limit\ \delta x \to o$$

$$\lim_{\delta x\to 0}\frac{\delta y}{\delta x} = \lim_{\delta x\to 0}\left[\frac{1}{\delta x}\log_a\left(1+\frac{\delta x}{x}\right)\right]$$

$$\frac{dy}{dx} = \frac{1}{x}\lim_{\delta x\to 0}\log_a\left(1+\frac{\delta x}{x}\right)^{\frac{x}{\delta x}} \qquad \left(\because \lim_{\delta x\to 0}\left(1+\frac{1}{x}\right)^{n} = e\right)$$

$$\frac{dy}{dx} = \frac{1}{x}\log_a e = \frac{1}{x}.\frac{1}{\log_e a} = \frac{1}{x}.\frac{1}{lna}$$

$$Hence\ \frac{d}{dx}(\log_a x) = \frac{1}{xlna}$$

Logarithmic differentiation:

Derivative of hyperbolic function:

1. $\frac{d}{dx}(sinhx) = coshx$

Proof:

$$we\ know\ thta\ sinhx = \frac{e^x - e^{-x}}{2}$$

$$so\ \frac{d}{dx}(sinhx) = \frac{d}{dx}\left(\frac{e^x - e^{-x}}{2}\right)$$

$$\frac{1}{2}\frac{d}{dx}\left(\frac{e^x - e^{-x}}{2}\right) = \frac{1}{2}\left\{\frac{d}{dx}(e^x) - \frac{d}{dx}(e^{-x})\right\}$$

$$\frac{1}{2}(e^x - e^{-x})\frac{d}{dx}(-x) = \frac{1}{2}\left(e^x - e^{-x}(-1)\right)$$

$$\frac{1}{2}(e^x + e^{-x}) = \frac{e^x + e^{-x}}{2} = coshx$$

$$\frac{d}{dx}(sinhx) = coshx \qquad \because\ coshx = \frac{e^x + e^{-x}}{2}$$

2. $\frac{d}{dx}(coshx) = sinhx$

Proof:

$$we\ know\ thta\ coshx = \frac{e^x + e^{-x}}{2}$$

$$so\ \frac{d}{dx}(coshx) = \frac{d}{dx}\left(\frac{e^x + e^{-x}}{2}\right)$$

$$\frac{1}{2}\frac{d}{dx}\left(\frac{e^x + e^{-x}}{2}\right) = \frac{1}{2}\left\{\frac{d}{dx}(e^x) + \frac{d}{dx}(e^{-x})\right\}$$

$$\frac{1}{2}(e^x + e^{-x})\frac{d}{dx}(-x) = \frac{1}{2}\left(e^x + e^{-x}(-1)\right)$$

$$\frac{1}{2}(e^x - e^{-x}) = \frac{e^x - e^{-x}}{2} = sinhx$$

$$\frac{d}{dx}(coshx) = sinhx \qquad \because\ sinhx = \frac{e^x - e^{-x}}{2}$$

3. $\frac{d}{dx}(tanhx) = \mathbf{sech^2}\,x$

Proof:

We know that $tanhx = \frac{e^x - e^x}{e^x + e^x}$

So $\frac{d}{dx}(tanhx) = \frac{d}{dx}\left(\frac{e^x - e^x}{e^x + e^x}\right)$

$$= \frac{(e^x + e^{-x})\frac{d}{dx}(e^x - e^{-x}) - (e^x - e^{-x})\frac{d}{dx}(e^x + e^{-x})}{(e^x + e^{-x})^2}$$

$$= \frac{(e^x + e^{-x})(e^x - e^{-x}(-1)) - (e^x - e^{-x})(e^x + e^{-x}(-1))}{(e^x + e^{-x})^2}$$

$$= \frac{(e^x + e^{-x})(e^x + e^{-x}) - (e^x - e^{-x})(e^x - e^{-x})}{(e^x + e^{-x})^2}$$

$$= \frac{(e^x + e^{-x})^2 - (e^x - e^{-x})^2}{(e^x + e^{-x})^2}$$

$$= \frac{e^{2x} + e^{-2x} + 2e^x e^{-x} - (e^{2x} + e^{-2x} - 2e^x e^{-x})}{(e^x + e^{-x})^2}$$

$$= \frac{e^{2x} + e^{-2x} + 2 - e^{2x} - e^{-2x} + 2}{(e^x + e^{-x})^2} = \frac{4}{(e^x + e^{-x})^2}$$

$$\frac{d}{dx}(tanhx) = \left(\frac{2}{e^x + e^{-x}}\right)^2 = (sechx)^2 = \mathrm{sech}^2\,x$$

$$\Rightarrow \frac{d}{dx}(tanhx) = \mathrm{sech}^2\,x \quad \because sechx = \frac{2}{e^x + e^{-x}}$$

4.

$$\frac{d}{dx}(cosechx) = -cosechxcothx$$

Proof:

we know that $cosechx = \frac{1}{sinhx}$

$$so\ \frac{d}{dx}(cosechx) = \frac{d}{dx}\left(\frac{1}{sinhx}\right)$$

$$= \frac{sinhx\frac{d}{dx}(1) - (1)\frac{d}{dx}(sinhx)}{\sinh^2 x}$$

$$= \frac{sinhx(0) - coshx}{\sinh^2 x}$$

$$\frac{d}{dx}(cosechx) = \frac{-coshx}{sinhx.sinhx} = -cothxcosechx$$

$$\frac{d}{dx}(cosechx) = -cosechxcothx$$

5.

$$\frac{d}{dx}(sechx) = -sechxtanhx$$

Proof:

$$we\ know\ that\ \ sechx = \frac{1}{coshx}$$

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$\Rightarrow \frac{d}{dx}(sechx) = \frac{d}{dx}\left(\frac{1}{coshx}\right)$

$\Rightarrow = \frac{coshx\frac{d}{dx}(1)-(1)\frac{d}{dx}(coshx)}{\cosh^2 x}$

$= \frac{coshx(0)-(1)(sinhx)}{\cosh^2 x}$

$= \frac{-sinhx}{coshx.coshx} = -tanhx.sechx$

$\frac{d}{dx}(sechx) = -sechxtanhx$

6.

$$\frac{d}{dx}(cothx) = -cosech^2 x$$

Proof:

$$we\ konw\ that\ cothx = \frac{1}{tanhx}$$

$\Rightarrow \frac{d}{dx}(cothx) = \frac{d}{dx}\left(\frac{1}{tanhx}\right)$

$= \frac{tanhx\frac{d}{dx}(1)-(1)\frac{d}{dx}(tanhx)}{\tanh^2 x}$

$= \frac{tanhx(o)-(1)(\text{sech}^2 x)}{\tanh^2 x}$

$= \frac{-\text{sech}^2 x}{\frac{\sinh^2 x}{\cosh^2 x}} = \frac{-\text{sech}^2 x}{\sinh^2 x}\cosh^2 x$

$= \frac{-\text{sech}^2 x}{\sinh^2 x} \times \frac{1}{\text{sech}^2 x} = -\frac{1}{\sinh^2 x}$

Hence $\frac{d}{dx}(cothx) = -cosech^2 x$

Derivative of inverse hyperbolic function:

The inverse hyperbolic functions are

1. $y = \sinh^{-1} x \quad iff \quad x = sinhy \;\; ; \;\; x, y \in R$
2. $y = \cosh^{-1} x \quad iff\ x = coshy; \;\; x \in [1,\infty), y \in [0,\infty)$
3. $y = \tanh^{-1} x \quad iff\ x = tanhy \;; x \in (-1,1), y \in R$
4. $y = \coth^{-1} x \quad iff\ x = cothy \;\; ; \;\; x \in [-1,1]', y \in R - \{0\}$
5. $y = \text{sech}^{-1} x \;\; iff\ x = sechy; \;\; x \in (0,1], y \in [0,\infty)$
6. $y = cosech^{-1} x \;\; iff \;\; x = cosechy\,; x \in R - \{0\}, y \in R - \{0\}$

Prove that $\sinh^{-1} x = \ln\left(x + \sqrt{x^2+1}\right)$

Proof:

$$Let\ y = \sinh^{-1} x \Rightarrow x = sinhy$$

$\Rightarrow x = \frac{e^y - e^{-y}}{2} \qquad \because sinhx = \frac{e^x - e^{-x}}{2}$

$\Rightarrow 2x = e^y - e^{-y} = e^y - \frac{1}{e^y} = \frac{(e^y)^2 - 1}{e^y}$

$\Rightarrow 2xe^y = e^{2y} - 1 \Rightarrow e^{2y} - 2xe^y - 1 = 0$

$\Rightarrow (e^y)^2 - 2xe^y - 1 = 0$

$\Rightarrow e^y = \frac{-(-2x) \pm \sqrt{(-2x)^2 - 4(1)(-1)}}{2(1)} \;\; using\ x$

$= \frac{-b \pm \sqrt{b^2-4ac}}{2a}$

$= \frac{2x \pm \sqrt{4x^2+4}}{2} = \frac{2 \pm \sqrt{4(x^2+1)}}{2}$

$= \frac{2x \pm 2\sqrt{x^2+1}}{2} = 2\left(\frac{x \pm \sqrt{x^2+1}}{2}\right)$

$e^y = x \pm \sqrt{x^2+1}$

$\Rightarrow e^y = x + \sqrt{x^2+1} \quad , e^y = x - \sqrt{x^2+1}$

$As\ e^y\ is + ve\ for\ y \in R, so\ (rejected\ being - ve\ for\ x \in R)$

$\Rightarrow e^y = x + \sqrt{x^2+1}\ ,$

$\Rightarrow y = \ln(x + \sqrt{x^2+1})$

Hence $\mathbf{\sinh^{-1} x = \ln\left(x + \sqrt{x^2+1}\right)}$

Note:
$\because e^y = x$
$lne^y = lnx$
$ylne = lnx$
$y(1) = lnx$
$y = lnx$

Prove that $\cosh^{-1} x = \ln\left(x + \sqrt{x^2-1}\right)$

Proof:

$$Let\ y = \cosh^{-1} x \Rightarrow x = coshy$$

$\Rightarrow x = \frac{e^y + e^{-y}}{2} \qquad \because coshx = \frac{e^x + e^{-x}}{2}$

$\Rightarrow 2x = e^y + e^{-y} = e^y + \frac{1}{e^y} = \frac{(e^y)^2 + 1}{e^y}$

$\Rightarrow 2xe^y = e^{2y} + 1 \Rightarrow e^{2y} - 2xe^y + 1 = 0$

$\Rightarrow (e^y)^2 - 2xe^y + 1 = 0$

$\Rightarrow e^y = \frac{-(-2x) \pm \sqrt{(-2x)^2 - 4(1)(1)}}{2(1)} \;\; using\ x$

$= \frac{-b \pm \sqrt{b^2-4ac}}{2a}$

$= \frac{2x \pm \sqrt{4x^2-4}}{2} = \frac{2 \pm \sqrt{4(x^2-1)}}{2}$

$= \frac{2x \pm 2\sqrt{x^2-1}}{2} = 2\left(\frac{x \pm \sqrt{x^2-1}}{2}\right)$

$e^y = x \pm \sqrt{x^2-1}$

$\Rightarrow e^y = x + \sqrt{x^2-1} \quad , e^y = x - \sqrt{x^2-1}$

$As\ e^y\ is + ve\ for\ y \in R, so\ (rejected\ being - ve\ for x > 1$

$\Rightarrow e^y = x + \sqrt{x^2-1}\ ,$

$\Rightarrow y = \ln(x + \sqrt{x^2-1})$

Hence $\mathbf{\cosh^{-1} x = \ln\left(x + \sqrt{x^2-1}\right)}$

Note:
$\because e^y = x$
$lne^y = lnx$
$ylne = lnx$
$y(1) = lnx$
$y = lnx$

1. $\frac{d}{dx}(\sinh^{-1} x) = \frac{1}{\sqrt{1+x^2}} \qquad x, y \in R$

Proof:

$$Let\ y = \sinh^{-1} x$$

$\Rightarrow \frac{d}{dx}(x) = \frac{d}{dx}(sinhy) = coshy\frac{dy}{dx}$

$\Rightarrow \cosh\frac{dy}{dx} = 1 \Rightarrow \frac{dy}{dx} = \frac{1}{coshy}$

$\frac{dy}{dx} = \frac{1}{\sqrt{coh^2 y}} = \frac{1}{\sqrt{1+\sinh^2 y}}$

$\because \cosh^2 x - \sinh^2 x = 1$

$\frac{dy}{dx} = \frac{1}{\sqrt{\cosh^2 y}} = \frac{1}{\sqrt{1+\sin^2 y}}$

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$$(\because sinhy = x)$$

2. $\frac{d}{dx}(\cosh^{-1} x) = \frac{1}{\sqrt{x^2-1}}$ $\quad x \in [1,\infty), y \in [0,\infty)$

Proof:

$$Let\ y = \cosh^{-1} x$$
$$\Rightarrow \frac{d}{dx}(x) = \frac{d}{dx}(coshy) = sinhy\frac{dy}{dx}$$
$$\Rightarrow \sinh\frac{dy}{dx} = 1 \Rightarrow \frac{dy}{dx} = \frac{1}{sinhy}$$
$$\frac{dy}{dx} = \frac{1}{\sqrt{sinh^2 y}} = \frac{1}{\sqrt{cosh^2 y - 1}}$$
$$\because \cosh^2 x - \sinh^2 x = 1$$
$$\frac{dy}{dx} = \frac{1}{\sqrt{\cosh^2 y - 1}} = \frac{1}{\sqrt{x^2-1}}$$

3.

$$\frac{d}{dx}(\tanh^{-1} x) = \frac{1}{1-x^2}$$

Proof:

$$Let\ y = \tanh^{-1} x\ ; x \in (-1,1), y \in R$$
$$\Rightarrow tanhy = x$$
$$\Rightarrow \frac{d}{dx}(tany) = \frac{d}{dx}(x)$$
$$\Rightarrow \operatorname{sech}^2 y\frac{dy}{dx} = 1$$
$$\frac{dy}{dx} = \frac{1}{\operatorname{sech}^2 y} = \frac{1}{1-\tanh^2 y}$$
$$\frac{dy}{dx} = \frac{1}{1-x^2} \quad (\because \operatorname{sech}^2 x + \tanh^2 x = 1)$$

4.

$$\frac{d}{dx}(\coth^{-1} x) = \frac{1}{1-x^2}$$

Proof:

$$Let\ y = \coth^{-1} x\ ;\ |x| > 1$$
$$\Rightarrow cothy = x$$
$$\Rightarrow \frac{d}{dx}(tany) = \frac{d}{dx}(x)$$
$$\Rightarrow -cosech^2 y\frac{dy}{dx} = 1$$
$$\frac{dy}{dx} = \frac{1}{co\operatorname{sech}^2 y} = \frac{1}{1-\coth^2 y}$$
$$\frac{dy}{dx} = \frac{1}{1-x^2} \quad (\because -co\operatorname{sech}^2 x + \coth^2 x = 1)$$

5. $\frac{d}{dx}(\operatorname{sech}^{-1} x) = -\frac{1}{x\sqrt{1-x^2}}$ $\quad ,0 < x < 1$

Proof:

$$Let\ y = \operatorname{sech}^{-1} x$$
$$\Rightarrow sechy = x \Rightarrow \frac{d}{dx}(sechy) = \frac{d}{dx}(x)$$
$$\Rightarrow -sechytanhy\frac{dy}{dx} = 1$$
$$\Rightarrow \frac{dy}{dx} = \frac{-1}{sechytanhy} = \frac{-1}{sechy\sqrt{\tanh^2 y}}$$
$$\frac{dy}{dx} = \frac{-1}{sechy\sqrt{1-\operatorname{sech}^2 y}} (\because \operatorname{sech}^2 y + \tanh^2 y = 1)$$

$$\frac{d}{dx}(\operatorname{sech}^{-1} x) = -\frac{1}{x\sqrt{1-x^2}}$$

6. $\frac{d}{dx}(\operatorname{cosech}^{-1} x) = -\frac{1}{x\sqrt{1+x^2}}$ $\quad ,x \in R - \{0\}$

Proof:

$$Let\ \ y = co\operatorname{sech}^{-1} x$$
$$\Rightarrow cosechy = x \Rightarrow \frac{d}{dx}(cosechy) = \frac{d}{dx}(x)$$
$$\Rightarrow -cosechycothy\frac{dy}{dx} = 1$$
$$\Rightarrow \frac{dy}{dx} = \frac{-1}{cosechycothy} = \frac{-1}{cosechy\sqrt{\coth^2 y}}$$

$$\begin{pmatrix}\because \coth^2 x - cosec^2 x = 1\\ \coth^2 x = 1 + cosech^2 x\end{pmatrix}$$
$$\frac{dy}{dx} = -\frac{1}{cosechy\sqrt{1+cosech^2 x}} = \frac{-1}{x\sqrt{1+x^2}}$$

Exercise 2.6

Question # 1) Find $f'(x)$ if

i) $\quad f(x) = e^{\sqrt{x}-1}$

Solution:

Let $f(x) = e^{\sqrt{x}-1}$

Differentiate w.r.t x

$$\frac{d(f(x))}{dx} = \frac{d}{dx}\left(e^{\sqrt{x}-1}\right)$$
$$f'(x) = e^{\sqrt{x}-1}\frac{d}{dx}(\sqrt{x}-1)$$
$$\frac{dy}{dx} = \frac{e^{\sqrt{x}-1}}{2\sqrt{x}}$$

Which is required.

ii) $\quad f(x) = x^3 e^{\frac{1}{x}}$

Solution:

Let $f(x) = x^3 e^{\frac{1}{x}}$

Differentiate w.r.t x

$$\frac{d(f(x))}{dx} = \frac{d}{dx}\left(x^3 e^{\frac{1}{x}}\right)$$
$$f'(x) = e^{\frac{1}{x}}\frac{d}{dx}(x^3) + x^3\frac{d}{dx}\left(e^{\frac{1}{x}}\right)$$
$$f'(x) = e^{\frac{1}{x}}(3x^2) + x^3\left(e^{\frac{1}{x}}\right)\frac{d}{dx}\left(\frac{1}{x}\right)$$
$$f'(x) = e^{\frac{1}{x}}(3x^2) + x^3\left(e^{\frac{1}{x}}\right)\frac{d}{dx}\left(\frac{1}{x}\right)$$
$$f'(x) = e^{\frac{1}{x}}(3x^2) + x^3\left(e^{\frac{1}{x}}\right)\left(\frac{-1}{x^2}\right)$$
$$f'(x) = xe^{\frac{1}{x}}(3x-1)$$

Which is required.

iii) $\quad f(x) = e^x(1 + lnx)$

Solution:

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Let $f(x) = e^x(1 + lnx)$

Differentiate w.r.t x

$$\frac{d(f(x))}{dx} = \frac{d}{dx}(e^x(1 + lnx))$$

$$f'(x) = e^x \frac{d}{dx}(1 + lnx) + (1 + lnx)\frac{d}{dx}(e^x)$$

$$f'(x) = e^x(0 + \frac{1}{x}) + (1 + lnx)(e^x)$$

$$f'(x) = \frac{e^x + x(1 + lnx)(e^x)}{x}$$

$$f'(x) = \frac{(1 + x(1 + lnx))e^x}{x}$$

Which is required.

iv) $\boldsymbol{f(x) = \frac{e^x}{e^{-x}+1}}$

Solution.

Let $f(x) = \frac{e^x}{e^{-x}+1}$

Differentiate w.r.t x

$$\frac{d(f(x))}{dx} = \frac{d}{dx}\left(\frac{e^x}{e^{-x}+1}\right)$$

$$f'(x) = \frac{(e^{-x}+1)\frac{d}{dx}(e^x) - (e^x)\frac{d}{dx}(e^{-x}+1)}{(e^{-x}+1)^2}$$

$$f'(x) = \frac{(e^{-x}+1)(e^x) - (e^x)(e^{-x}(-1))}{(e^{-x}+1)^2}$$

$$f'(x) = \frac{(1 + e^x) - (-1)}{(e^{-x}+1)^2}$$

$$f'(x) = \frac{2 + e^x}{(e^{-x}+1)^2}$$

Which is required.

v) $\boldsymbol{f(x) = ln(e^x + e^{-x})}$

Solution:

Let $f(x) = ln(e^x + e^{-x})$

Differentiate w.r.t x

$$\frac{d(f(x))}{dx} = \frac{d}{dx}(ln(e^x + e^{-x}))$$

$$f'(x) = \frac{1}{e^x + e^{-x}}\frac{d}{dx}(e^x + e^{-x})$$

$$f'(x) = \frac{1}{e^x + e^{-x}}(e^x + e^{-x}(-1))$$

$$f'(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$

$$f'(x) = \frac{e^x - \frac{1}{e^x}}{e^x + \frac{1}{e^x}}$$

$$f'(x) = \frac{e^{2x} - 1}{e^{2x} + 1}$$

Which is required.

vi) $\boldsymbol{f(x) = \frac{e^{ax}-e^{-ax}}{e^{ax}+e^{-ax}}}$

Solution.

Let $f(x) = \frac{e^{ax}-e^{-ax}}{e^{ax}+e^{-ax}}$

Differentiate w.r.t x

$$\frac{d(f(x))}{dx} = \frac{d}{dx}\left(\frac{e^{ax} - e^{-ax}}{e^{ax} + e^{-ax}}\right)$$

$$f'(x) = \frac{(e^{ax} + e^{-ax})\frac{d}{dx}(e^{ax} - e^{-ax}) - (e^{ax} - e^{-ax})\frac{d}{dx}(e^{ax} + e^{-ax})}{(e^{ax} + e^{-ax})^2}$$

$$f'(x) = \frac{(e^{ax} + e^{-ax})(e^{ax}(a) - e^{-ax}(-a)) - (e^{ax} - e^{-ax})(e^{ax}(a) + e^{-ax}(-))}{(e^{ax} + e^{-ax})^2}$$

$$f'(x) = \frac{a[(e^{ax} + e^{-ax})^2 - (e^{ax} - e^{-ax})^2]}{(e^{ax} + e^{-ax})^2}$$

$$f'(x) = \frac{a[(e^{ax})^2 + (e^{-ax})^2 + 2(e^{ax})(e^{-ax}) - (e^{ax})^2 - (e^{-ax})^2 - 2(e^{ax})(e^{-ax})]}{(e^{ax} + e^{-ax})^2}$$

$$f'(x) = \frac{a[4e^0]}{(e^{ax} + e^{-ax})^2}$$

$$f'(x) = \frac{4a}{(e^{ax} + e^{-ax})^2}$$

Which is required.

vii) $\boldsymbol{f(x) = \sqrt{ln(e^{2x} + e^{-2x})}}$

Solution:

Let $f(x) = \sqrt{ln(e^{2x} + e^{-2x})}$

Differentiate w.r.t x

$$\frac{d(f(x))}{dx} = \frac{d}{dx}\left(\sqrt{ln(e^{2x} + e^{-2x})}\right)$$

$$f'(x) = \frac{1}{2\sqrt{ln(e^{2x} + e^{-2x})}}\frac{d}{dx}(ln(e^{2x} + e^{-2x}))$$

$$f'(x) = \frac{1}{2\sqrt{ln(e^{2x} + e^{-2x})}}[\frac{1}{e^{2x} + e^{-2x}}\frac{d}{dx}(e^{2x} + e^{-2x})]$$

$$f'(x) = \frac{2(e^{2x} - e^{-2x})}{2\sqrt{ln(e^x + e^{-x})}(e^{2x} + e^{-2x})}$$

$$f'(x) = \frac{(e^{2x} - e^{-2x})}{\sqrt{ln(e^x + e^{-x})}(e^{2x} + e^{-2x})}$$

Which is required.

viii) $\boldsymbol{f(x) = ln\sqrt{(e^{2x} + e^{-2x})}}$

Solution:

Let $f(x) = ln\sqrt{(e^{2x} + e^{-2x})} = \frac{1}{2}(\ln(e^{2x} + e^{-2x}))$

Differentiate w.r.t x

$$\frac{d(f(x))}{dx} = \frac{1}{2}\frac{d}{dx}(\ln(e^{2x} + e^{-2x}))$$

$$f'(x) = \frac{2e^{2x} - 2e^{-2x}}{2(e^{2x} + e^{-2x})}$$

$$f'(x) = \frac{e^{2x} - e^{-2x}}{e^{2x} + e^{-2x}}]$$

$$f'(x) = tanh2x$$

Question # 2. Find $\frac{dy}{dx}$ if

Solution.

i) $y = \mathrm{x}^2 \ln\sqrt{\mathrm{x}}$

Let $y = \mathrm{x}^2 \ln\sqrt{\mathrm{x}} = \frac{1}{2}x^2 lnx$

Differentiate w.r.t x

$$\frac{dy}{dx} = \frac{d}{dx}(\frac{1}{2}x^2 lnx)$$

$$\frac{dy}{dx} = \frac{1}{2}\left[(x^2)\frac{d}{dx}(lnx) + (lnx)\frac{d}{dx}(x^2)\right]$$

$$\frac{dy}{dx} = \frac{1}{2}\left[x^2\left(\frac{1}{x}\right) + lnx\,(2x)\right]$$

ii) $\boldsymbol{y = x\sqrt{lnx}}$

Solution.

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Let $y = x\sqrt{lnx}$

Differentiate w.r.t x

$$\frac{dy}{dx} = \frac{d}{dx}(x\sqrt{lnx})$$

$$\frac{dy}{dx} = (x)\frac{d}{dx}(\sqrt{lnx}) + (\sqrt{lnx})\frac{d}{dx}(x)$$

$$\frac{dy}{dx} = x\frac{1}{2\sqrt{lnx}}\frac{d}{dx}(lnx) + \sqrt{lnx}\,(1)$$

$$\frac{dy}{dx} = x\frac{1}{2\sqrt{lnx}}\left(\frac{1}{x}\right) + \sqrt{lnx}\,(1)$$

$$\frac{dy}{dx} = \frac{1 + 2\left(\sqrt{lnx}\right)^2}{2\sqrt{lnx}}$$

$$\frac{dy}{dx} = \frac{1 + 2lnx}{2\sqrt{lnx}}$$

Which is required.

iii) $\boldsymbol{y = \frac{x}{lnx}}$

Solution.

Let $y = \frac{x}{lnx}$

Differentiate w.r.t x

$$\frac{dy}{dx} = \frac{d}{dx}\left(\frac{x}{lnx}\right)$$

$$\frac{dy}{dx} = \left(\frac{(lnx)\frac{d}{dx}(x) - (x)\frac{d}{dx}(lnx)}{(lnx)^2}\right)$$

$$\frac{dy}{dx} = \left(\frac{(lnx)(1) - (x)\left(\frac{1}{x}\right)}{(lnx)^2}\right)$$

$$\frac{dy}{dx} = \frac{lnx - 1}{(lnx)^2}$$

iv) $\boldsymbol{y = x^2 \ln\left(\frac{1}{x}\right)}$

Solution.

Let $y = x^2 \ln\left(\frac{1}{x}\right) = x^2 \ln(x^{-1}) = -x^2 \ln(x)$

Differentiate w.r.t x

$$\frac{dy}{dx} = \frac{d}{dx}(-x^2 \ln(x))$$

$$\frac{dy}{dx} = -\left[(x^2)\frac{d}{dx}(lnx) + (lnx)\frac{d}{dx}(x^2)\right]$$

$$\frac{dy}{dx} = -\left[(x^2)\left(\frac{1}{x}\right) + (lnx)(2x)\right]$$

$$\frac{dy}{dx} = -[x + 2xlnx]$$

$$\frac{dy}{dx} = -x(1 + 2lnx)$$

Which is required.

v) $\boldsymbol{y = ln\sqrt{\frac{x^2-1}{x^2-1}}}$

Solution.

Let $y = ln\sqrt{\frac{x^2-1}{x^2+1}} = \frac{1}{2}\ln\left(\frac{x^2-1}{x^2+1}\right)$

Differentiate w.r.t x

$$\frac{dy}{dx} = \frac{d}{dx}\left(\frac{1}{2}\ln\left(\frac{x^2-1}{x^2+1}\right)\right)$$

$$\frac{dy}{dx} = \frac{1}{2}\left[\frac{1}{\frac{x^2-1}{x^2+1}}\frac{d}{dx}\left(\frac{x^2-1}{x^2+1}\right)\right]$$

$$\frac{dy}{dx} = \frac{1(x^2+1)}{2(x^2-1)}\left(\frac{(x^2+1)\frac{d}{dx}(x^2-1) - (x^2-1)\frac{d}{dx}(x^2+1)}{(x^2+1)^2}\right)$$

$$\frac{dy}{dx} = \left(\frac{(x^2+1)(2x) - (x^2-1)(2x)}{2(x^2-1)(x^2+1)}\right)$$

$$\frac{dy}{dx} = \left(\frac{2x(x^2+1-x^2+1)}{2(x^2-1)(x^2+1)}\right)$$

$$\frac{dy}{dx} = \left(\frac{2x}{(x^4-1)}\right)$$

Which is required.

vi) $\boldsymbol{y = \ln(x + \sqrt{x^2+1})}$

Solution.

Let $y = \ln(x + \sqrt{x^2+1})$

Differentiate w.r.t x

$$\frac{dy}{dx} = \frac{1}{x + \sqrt{x^2+1}}\left(\frac{\sqrt{x^2+1} + x}{\sqrt{x^2+1}}\right)$$

$$\frac{dy}{dx} = \frac{1}{\sqrt{x^2+1}}$$

Which is required.

vii) $\boldsymbol{y = \ln(9 - x^2)}$

Solution.

Let $y = \ln(9 - x^2)$

Differentiate w.r.t x

$$\frac{dy}{dx} = \frac{d}{dx}(\ln(9 - x^2)$$

$$\frac{dy}{dx} = \frac{1}{9 - x^2}\frac{d}{dx}(9 - x^2)$$

$$\frac{dy}{dx} = \frac{1}{9 - x^2}(-2x)$$

$$\frac{dy}{dx} = \frac{-2x}{9 - x^2}$$

Which is required.

viii) $\boldsymbol{y = e^{-2x}.sin2x}$

Solution.

Let $y = e^{-2x}.sin2x$

Differentiate w.r.t x

$$\frac{dy}{dx} = \frac{d}{dx}(e^{-2x}.sin2x)$$

$$\frac{dy}{dx} = (e^{-2x})(cos2x\,(2)) + (sin2x)(e^{-2x}(-2))$$

$$\frac{dy}{dx} = 2e^{-2x}(cos2x - sin2x)$$

Which is required.

ix) $\boldsymbol{y = e^{-x}(x^3 + 2x^2 + 1)\,.}$

Solution.

Since $y = e^{-x}(x^3 + 2x^2 + 1)$

Differentiating w.r.t "x"

$$\frac{dy}{dx} = (x^3 + 2x^2 + 1)\frac{d}{dx}(e^{-x}) + e^{-x}\frac{d}{dx}(x^3 + 2x^2 + 1)$$

$$\frac{dy}{dx} = (x^3 + 2x^2 + 1)(e^{-x}(-1)) + e^{-x}(3x^2 + 4x)$$

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$\frac{dy}{dx} = e^{-x}(-x^3 - 2x^2 - 1 + 3x^2 + 4x)$

$\frac{dy}{dx} = -e^{-x}(x^3 - x^2 - 4x + 1)$

x) $\boldsymbol{y = xe^{sinx}}$

Solution.

Since $y = xe^{sinx}$

Differentiate w.r.t.''x''

$\frac{dy}{dx} = \frac{d}{dx}(xe^{sinx})$

$\frac{dy}{dx} = (e^{sinx})\frac{d}{dx}(x) + (x)\frac{d}{dx}(e^{sinx})$

$\frac{dy}{dx} = (e^{sinx})(1) + (x)(e^{sinx}(cosx))$

$\frac{dy}{dx} = e^{sinx}(1 + xcosx)$

Which is required.

xi) $\boldsymbol{y = 5e^{3x-4}}$

Solution.

Let $y = 5e^{3x-4}$

Differentiate w.r.t x

$\frac{dy}{dx} = \frac{d}{dx}(5e^{3x-4})$

$\frac{dy}{dx} = 5e^{3x-4}\ \frac{d}{dx}(3x - 4)$

$\frac{dy}{dx} = 5e^{3x-4}\ (3)$

$\frac{dy}{dx} = 15e^{3x-4}$

Which is required.

xii) $\boldsymbol{y = (x + 1)^x}$

Solution.

Let $y = (x + 1)^x$

Differentiate w.r.t x

Taking ln on both sides

$lny = ln(x + 1)^x$

$lny = xln(x + 1)$

$\frac{1}{y}\frac{dy}{dx} = (\ln(x + 1))\frac{d}{dx}(x) + x\frac{d}{dx}(ln(x + 1))$

$\frac{1}{y}\frac{dy}{dx} = (\ln(x + 1))(1) + (x)\frac{1}{(x+1)}\left(\frac{1}{x}\right)$

$\frac{dy}{dx} = y\left(\frac{\ln(x+1)+1}{x+1}\right)$

$\frac{dy}{dx} = (x + 1)^x\left(\frac{\ln(x+1)+1}{x+1}\right)$

$\frac{dy}{dx} = (x + 1)^{x-1}(\ln(x + 1) + 1)$

Which is required.

xiii) $\boldsymbol{y = (lnx)^{lnx}}$

Solution.

Let $y = (lnx)^{lnx}$

Differentiate w.r.t x

Taking ln on both sides

$lny = ln(lnx)^{lnx}$

$lny = lnx.ln(lnx)$

$\frac{1}{y}\frac{dy}{dx} = (\ln(lnx))\frac{d}{dx}(lnx) + (lnx)\frac{d}{dx}(ln(lnx))$

$\frac{1}{y}\frac{dy}{dx} = (\ln(lnx))\left(\frac{1}{x}\right) + (lnx)\frac{1}{(lnx)}\left(\frac{1}{x}\right)$

$\frac{dy}{dx} = y\left(\frac{\ln(lnx)+1}{x}\right)$

$\frac{dy}{dx} = (lnx)^x\left(\frac{\ln(x+1)+1}{x}\right)$

$\frac{dy}{dx} = \frac{(x+1)^x}{x}(\ln(lnx) + 1)$

Q#3) Find $\frac{dy}{dx}$ if:

(i) $\boldsymbol{y = \cosh 2x}$

Solution.

Let $y = \cosh 2x$

Differentiate w.r.t x

$\frac{dy}{dx} = \frac{d}{dx}(\cosh 2x)$

$\frac{dy}{dx} = sinh2x\ \frac{d}{dx}(2x)$

$\frac{dy}{dx} = sinh2x\ (2)$

$\frac{dy}{dx} = 2sinh2x$

Which is required.

(ii) $\boldsymbol{y = \sinh 3x}$

Let $y = \cosh 2x$

Differentiate w.r.t x

$\frac{dy}{dx} = \frac{d}{dx}(\sinh 3x)$

$\frac{dy}{dx} = cosh2x\ \frac{d}{dx}(3x)$

$\frac{dy}{dx} = cosh2x\ (3)$

$\frac{dy}{dx} = 3cosh3x$

(iii) $\boldsymbol{y = \tanh^{-1}(sinx), -\frac{\pi}{2} < x < \frac{\pi}{2}}$

Sol:

Let $y = \tanh^{-1}(sinx)$

Differentiate w.r.t x

$\frac{dy}{dx} = \frac{d}{dx}(\tanh^{-1}(sinx),)$

$\frac{dy}{dx} = \frac{1}{1-(sinx)^2}\frac{d}{dx}(sinx)$

$\frac{dy}{dx} = \frac{1}{1-\sin^2 x}(\cos x)$

$\frac{dy}{dx} = \frac{cosx}{\cos^2 x} = secx$

(iv) $\boldsymbol{y = \sinh^{-1}(x^3)}$

Let $y = \sinh^{-1}(x^3)$

Differentiate w.r.t x

$\frac{dy}{dx} = \frac{d}{dx}(\sinh^{-1}(x^3))$

$\frac{dy}{dx} = \frac{1}{\sqrt{1-(x^3)^2}}\frac{d}{dx}(x^3)$

$\frac{dy}{dx} = \frac{1}{\sqrt{1-x^6}}(3x^2)$

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$\frac{dy}{dx} = \frac{3x^2}{\sqrt{1-x^6}}$

(v) $y = \ln(\tanh x)$

Let $y = \ln(\tanh x)$

Differentiate w.r.t x

$\frac{dy}{dx} = \frac{d}{dx}(\ln(\tanh x))$

$\frac{dy}{dx} = \frac{1}{tanhx}\frac{d}{dx}(tanhx)$

$\frac{dy}{dx} = \frac{coshx}{sinhx}(\operatorname{sech}^2 x)$

$\frac{dy}{dx} = \frac{coshx}{sinhx\cosh^2 x}$

$\frac{dy}{dx} = \frac{1}{sinhx\ coshx}$

(vi) $y = \sinh^{-1}\left(\frac{x}{2}\right)$

Sol:

Let $y = \sinh^{-1}\left(\frac{x}{2}\right)$

Differentiate w.r.t x

$\frac{dy}{dx} = \frac{d}{dx}(\sinh^{-1}\left(\frac{x}{2}\right))$

$\frac{dy}{dx} = \frac{1}{1-(sinx)^2}\frac{d}{dx}(sinx)$

$\frac{dy}{dx} = \frac{1}{\sqrt{1+\left(\frac{x}{2}\right)^2}}\frac{d}{dx}\left(\frac{x}{2}\right)$

$\frac{dy}{dx} = \frac{1}{\sqrt{1+\frac{x^2}{4}}}\left(\frac{1}{2}\right)$

$\frac{dy}{dx} = \frac{1}{\sqrt{\frac{4+x^2}{4}}}\left(\frac{1}{2}\right)$

$\frac{dy}{dx} = \frac{2}{\sqrt{4+x^2}}\left(\frac{1}{2}\right)$

$\frac{dy}{dx} = \frac{1}{\sqrt{4+x^2}}$

Successive differentiation (or higher derivative)

Let $y = f(x)$ *be a function, then its successive or higher are given below.*

1st derivative	*2nd derivative*	*3rd derivative*
y'	y''	y'''
$\frac{dy}{dx}$	$\frac{d^2y}{dx^2}$	$\frac{d^3y}{dx^3}$
y_1	y_2	y_3
D_y	D_y^2	D_y^3
$f'(x)$	$f''(x)$	$f'''(x)$
$\frac{df}{dx}$	$\frac{d^2y}{d^2x}$	$\frac{d^3y}{dx^3}$

Exercise 2.7

Question # 1

Find y_2 if:

(i) $y = 2x^5 - 3x^4 + 4x^3 + x - 2$

Solution.

Let $y = 2x^5 - 3x^4 + 4x^3 + x - 2$

Differentiate w.r.t x

$\frac{dy}{dx} = \frac{d}{dx}(2x^5 - 3x^4 + 4x^3 + x - 2)$

$y_1 = 10x^4 - 12x^3 + 12x^2 + 1$

Again differentiate w.r.t x

$y_2 = \frac{d}{dx}(10x^4 - 12x^3 + 12x^2 + 1)$

$y_2 = 40x^3 - 36x^2 + 24x$

Which is required.

(ii) $y = (2x+5)^{\frac{3}{2}}$

Solution.

$\frac{dy}{dx} = \frac{d}{dx}(2x+5)^{\frac{3}{2}}$

$y_1 = \frac{3}{2}(2x+5)^{\frac{1}{2}}(2)$

$y_1 = 3(2x+5)^{\frac{1}{2}} = 3\sqrt{2x+5}$

Again differentiate w.r.t x

$y_2 = 3\frac{d}{dx}\left(\sqrt{2x+5}\right)$

$y_2 = \frac{3}{2\sqrt{2x+5}}(2)$

$y_2 = \frac{3}{\sqrt{2x+5}}$

Which is required.

(iii) $y = \sqrt{x} - \frac{1}{\sqrt{x}}$

Sol: Since $y = \sqrt{x} - \frac{1}{\sqrt{x}}$

$y = x^{\frac{1}{2}} - x^{-\frac{1}{2}}$

Differentiating w.r.t "x"

$y_1 = \frac{1}{2}x^{-\frac{1}{2}} + \frac{1}{2}x^{-\frac{3}{2}}$

Again differentiate w.r.t x

$y_2 = \frac{d}{dx}\left(\frac{1}{2}x^{-\frac{1}{2}} + \frac{1}{2}x^{-\frac{3}{2}}\right)$

$y_2 = \frac{-1}{4}x^{-\frac{3}{2}} + \frac{-3}{4}x^{-\frac{5}{2}}$

$y_2 = \frac{-1}{4}x^{-\frac{3}{2}} + \frac{-3}{4}x^{-\frac{5}{2}}$

Which is required.

Question # 2 Find y_2 if:

(i) $y = x^2.e^{-x}$

Solution.

Let $y = x^2.e^{-x}$

Differentiate w.r.t x

$\frac{dy}{dx} = \frac{d}{dx}(x^2.e^{-x})$

$y_1 = (x^2)(e^{-x}(-1)) + (e^{-x})(2x)$

$y_1 = e^{-x}(2x - x^2)$

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Again differentiate w.r.t x

$$y_2 = \frac{d}{dx}(e^{-x}(2x - x^2))$$
$$y_2 = ((2x - x^2))(e^{-x}(-1)) + (e^{-x})(2 - 2x)$$
$$y_2 = e^{-x}(-2x + x^2 + 2 - 2x)$$
$$y_2 = e^{-x}(x^2 - 4x + 2)$$

Which is required.

(ii) $y = \ln\left(\frac{2x+3}{3x+2}\right)$

$$\Rightarrow \frac{dy}{dx} = y_1 = \frac{d}{dx}.\ln\left(\frac{2x+3}{3x+2}\right)$$
$$= \frac{d}{dx}[\ln(2x+3) - \ln(3x+2)]$$
$$y_1 = 2(2x+3)^{-1} - 3(3x+2)^{-1}$$
$$\Rightarrow \frac{d^2y}{dx^2} = y_2 = 2\frac{d}{dx}(2x+3)^{-1} - 3\frac{d}{dx}(3x+2)^{-1}$$
$$= 2(-1)(2x+3)^{-1}\frac{d}{dx}(2x+3) - 3(-1)(3x+2)^{-2}\frac{d}{dx}(3x+2)$$
$$= -2(2x+3)^{-2}(2) + 3(3x+2)^{-2})(3)$$
$$y_2 = -\frac{4}{(2x+5)^2} + \frac{9}{(3x+2)^2}$$
$$= \frac{-4(3x+2)^2 + 9(2x+3)^2}{(2x+5)^2(3x+2)^2}$$
$$= \frac{-4(9x^2+4+12x) + 9(4x^2+3+12x)}{(2x+5)^2(3x+2)^2}$$
$$= \frac{-36x^2 - 16 - 48x + 36x62 + 81 + 108x}{(2x+5)^2(3x+2)^2}$$
$$y_2 = \frac{60x + 65}{(2x+5)^2(3x+2)^2}$$

Q#3) Find y_2 if:

(i) $\boldsymbol{x^2 + y^2 = a^2}$

Sol:

Differentiate w.r.t x

$$\frac{d}{dx}(x^2 + y^2) = \frac{d}{dx}(a^2)$$
$$2x + 2yy_1 = 0$$
$$2yy_1 = -2x$$
$$y_1 = -\frac{x}{y}$$

Differentiating again w.r.t "x"

$$\frac{dy_1}{dx} = -\frac{d}{dx}\left(\frac{x}{y}\right)$$
$$\frac{dy_1}{dx} = -\frac{(y)\frac{d}{dx}(x) - (x)\frac{d}{dx}(y)}{(y)^2}$$
$$y_2 = -\left(\frac{(y)(1) - (x)(y_1)}{(y)^2}\right)$$

Put value of "y_1"

$$y_2 = -\left(\frac{(y) - (x)\left(-\frac{x}{y}\right)}{(y)^2}\right)$$
$$y_2 = -\left(\frac{\left(\frac{y^2 + x^2}{y}\right)}{(y)^2}\right)$$
$$y_2 = -\left(\frac{y^2 + x^2}{y^3}\right)$$
$$y_2 = \frac{-a^2}{y^3}$$

(ii) $\boldsymbol{x^3 - y^3 = a^3}$

Sol:

Differentiate w.r.t x

$$\frac{d}{dx}(x^3 - y^3) = \frac{d}{dx}(a^3)$$
$$3x^2 - 3y^2y_1 = 0$$
$$3y^2y_1 = 3x^2$$
$$y_1 = \frac{x^2}{y^2}$$

Differentiating again w.r.t "x"

$$\frac{dy_1}{dx} = \frac{d}{dx}\left(\frac{x^2}{y^2}\right)$$
$$\frac{dy_1}{dx} = \frac{(y^2)\frac{d}{dx}(x^2) - (x^2)\frac{d}{dx}(y^2)}{(y^2)^2}$$
$$y_2 = \frac{(y^2)(2x) - (x^2)(2yy_1)}{y^4}$$

Put value of "y_1"

$$y_2 = \frac{(y^2)(2x) - (x^2)\left(2y\left(\frac{x^2}{y^2}\right)\right)}{y^4}$$
$$y_2 = \frac{(2xy^2) - \left(\frac{2x^4}{y}\right)}{y^4}$$
$$y_2 = \frac{\left(\frac{2y^2x^2 - 2x^4}{y}\right)}{y^4}$$
$$y_2 = \frac{2x(y^3 - x^3)}{y^5}$$
$$= \frac{-2x}{y^5}(x^3 - y^3)$$
$$\frac{d^2y}{dx^2} = -\frac{2x}{y^5}(a^3) \qquad \because x^3 - y^3 = a^3$$
$$y_2 = -\frac{2xa^3}{y^5}$$

(iii) $\boldsymbol{x = acos\theta\ y = asin\theta}$

Sol:

$$x = acos\theta\ , y = asin\theta$$
$$\frac{dx}{d\theta} = \frac{d}{d\theta}(acos\theta) = a(-sib\theta) = -asin\theta$$
$$\frac{dy}{d\theta} = \frac{d}{d\theta}(asin\theta) = acos\theta$$

By chain rule, we have

$$\frac{dy}{dx} = \frac{d}{d\theta}.\frac{d\theta}{dx} = (acos\theta)\left(\frac{1}{-asin\theta}\right)$$
$$\frac{dy}{dx} = -\frac{cox\theta}{sin\theta} = -cot\theta \Rightarrow y_1 = -cot\theta$$

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$$\Rightarrow \frac{d^2y}{dx^2} = \frac{d}{dx}(-cot\theta) = -(-cosec^2\theta)\frac{d\theta}{dx}$$
$$= cosec^2\theta\left(\frac{1}{-asin\theta}\right) = \frac{1}{sin^2\theta}\left(\frac{1}{-asin\theta}\right)$$
$$\frac{d^2y}{dx^2} = \frac{-1}{asin^3\theta}$$

$$y_2 = \frac{-1}{a\ \sin^3\theta}$$

Which is required

(iv) $x = at^2 \quad y = bt^4$

Sol:

$$x = at^2 \Rightarrow t^2 = \frac{x}{a}$$
$$y = bt^4 \Rightarrow y = b\left(\frac{x}{a}\right)^2$$
$$y = \frac{bx^2}{a^2}$$

Differentiate w.r.t x

$$\frac{dy}{dx} = \frac{b}{a^2}\frac{d}{dx}(x^2)$$
$$y_1 = \frac{b}{a^2}(2x)$$

Again differentiate w.r.t x

$$y_2 = \frac{2b}{a^2}\frac{d}{dx}(x)$$
$$y_2 = \frac{2b}{a^2}$$

(v) $x^2 + y^2 + 2gx + 2fy + c = 0$

Sol: differentiate w.r.t "x"

$$\frac{d}{dx}(x^2 + y^2 + 2gx + 2fy + c) = \frac{d}{dx}(0)$$
$$2x + 2yy_1 + 2g + 2fy_1 = 0$$
$$yy_1 + fy_1 = -x - g$$
$$y_1(y + f) = -(x + g)$$
$$y_1 = -\frac{x + g}{y + f}$$

Differentiating again w.r.t "x"

$$\frac{dy_1}{dx} = -\frac{d}{dx}\left(\frac{x + g}{y + 1}\right)$$
$$\frac{dy_1}{dx} = -\frac{(y + f)\frac{d}{dx}(x + g) - (x + g)\frac{d}{dx}(y + f)}{(y + f)^2}$$
$$y_2 = -\left(\frac{(y + f)(1) - (x + g)(y_1)}{(y + f)^2}\right)$$

Put value of "y_1"

$$y_2 = -\left(\frac{(y + f)(1) - (x + g)\left(-\frac{x + g}{y + 1}\right)}{(y + f)^2}\right)$$
$$y_2 = -\left(\frac{(y + f)^2 + (x + g)^2}{(y + f)^3}\right)$$
$$y_2 = -\left(\frac{y^2 + f^2 + 2fy + x^2 + g^2 + 2gx}{(y + f)^3}\right)$$
$$y_2 = -\left(\frac{(x^2 + y^2 + 2gx + 2fy + c) + f^2 + g^2 - c}{(y + f)^3}\right)$$
$$y_2 = -\left(\frac{(0) + f^2 + g^2 - c}{(y + f)^3}\right)$$
$$y_2 = \frac{-f^2 - g^2 + c}{(y + f)^3}$$
$$\frac{d^2y}{dx^2} = \frac{-f^2 - g^2 + c}{(y + f)^3}$$
$$\frac{d^2y}{dx^2} = \frac{c - f^2 - g^2}{(y + f)^3}$$

Q#4 Find y_4 if:

(i) $y = sin3x$

Sol: Differentiate w.r.t x

$$\frac{dy}{dx} = \frac{d}{dx}(sin3x) = cos3x\frac{d}{dx}(3x)$$
$$y_1 = cos3x\ (3)$$
$$y_1 = 3cos3x$$

Again differentiate w.r.t x

$$\frac{d^2y}{dx^2} = y_2 = \frac{d}{dx}(3cos3x) = 3(-sin3x)\frac{d}{dx}(3x)$$
$$3\ (-sin3x\ (3))$$
$$y_2 = -9sin3x$$

Again differentiate w.r.t x

$$\frac{d^3y}{dx^3} = y_3 = \frac{d}{dx}(-9sin3x) = -9(cos3x)\frac{d}{dx}(3x)$$
$$= -9\ cos3x\ (3)$$
$$y_3 = -27cos3x$$

Again differentiate w.r.t x

$$\frac{d^4y}{dx^4} = y_4 = \frac{d}{dx}(-27cos3x) = -27(-sin3x)\frac{d}{dx}(3x)$$
$$= \ -27\ (-sin3x\ (3))$$
$$y_4 = 81sin3x$$

Which is required

(ii) $y = \cos^3 x$

Sol: $\begin{pmatrix} \because cos3x = 4\cos^3 x - 3cosx \\ \Rightarrow cos3x + 3cosx = 4\cos^3 x \\ \Rightarrow \frac{1}{4}(cos3x + 3cosx = \cos^2 x \end{pmatrix}$

Thus

$$y = \frac{1}{4}[cos3x + 3cosx]$$
$$\frac{dy}{dx} = y_1 = \frac{1}{4}\left[\frac{d}{dx}(cos3x) + \frac{d}{dx}(3cosx)\right]$$
$$= \frac{1}{4}[(-sin3x(3) + 3(-sinx)]$$
$$y_1 = -\frac{3}{4}(sin3x + sinx)$$
$$\Rightarrow \frac{d^2y}{dx^2} = -\frac{3}{4}\left(\frac{d}{dx}(sin3x) + \frac{d}{dx}(sinx)\right)$$
$$= -\frac{3}{4}(cos3x\ (3) + cosx)$$
$$\Rightarrow \frac{d^2y}{dx^2} = y_2 = -\frac{3}{4}[(3cos3x + cosx)]$$
$$\Rightarrow \frac{d^3y}{dx^3} = y_3 = -\frac{3}{4}\left[3(\frac{d}{dx}(cos3x) + \frac{d}{dx}(cosx)\right]$$

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$$= -\frac{3}{4}[3(-sin3x)(3) + (-sinx)]$$
$$= -\frac{3}{4}(-9sin3x - sinx)$$
$$y_3 = \frac{3}{4}(9sin3x + sinx)$$
$$\Rightarrow \frac{d^4y}{dx^4} = y_4 = \frac{3}{4}\left[9\frac{d}{dx}(sin3x) + \frac{d}{dx}(sinx)\right]$$
$$= \frac{3}{4}[9cos3x(3) + cosx]$$
$$= \frac{3}{4}[27cos3x + cosx]$$
$$= \frac{3}{4}(27(4\cos^3 x - 3cosx) + cosx)$$
$$= \frac{3\times 27\times 4}{4}\cos^3 x - \frac{3\times 27\times 3}{4}cosx + \frac{3}{4}cosx$$
$$81\cos^3 x - \frac{243}{4}cosx + \frac{3}{4}cosx$$
$$y_4 = 81\cos^3 + \left(\frac{3-243}{4}\right)cosx$$
$$\Rightarrow y_4 = 81\cos^3 x - \frac{240}{4}cosx$$

$y_4 = -60cosx + 81\cos^3 x$

$y_4 = 81\cos^3 x - 60cosx$

(iii) $y = \ln(x^2 - 9) = \ln(x-3)(x+3)$

$y = \ln(x-3) + \ln(x+3)$

$\Rightarrow \frac{dy}{dx} = y_1 = \frac{d}{dx}[\ln(x-3) + \ln(x+3)]$

$y_1 = \frac{1}{x-3} + \frac{1}{x+3} = (x-3)^{-1} + (x+3)^{-1}$

$\frac{d^2y}{d^2x} = y_2 = \frac{d}{dx}(x-3)^{-1} + \frac{d}{dx}(x+3)^{-1}$

$y_2 = (-1)(x-3)^{-2} + (-1)(x+3)^{-2}$

$\frac{d^3y}{dx^3} = y_3 = \frac{d}{dx}(-1)(x-3)^{-2} + (-1)(x+3)^{-2}$

$y_3 = (-1)(-2)(x-3)^{-3} + (-1)(-2)(x+3)^{-3}$

$\frac{d^4y}{dx^4} = y_4 = \frac{d}{dx}((-1)(-2)(x-3)^{-3} + \frac{d}{dx}(-1)(-2)(x+3)^{-3}$

$= (-1)(-2)(-3)(x-3)^{-4} + (-1)(-2)(-3)(x+3)^{-4}$

$$y_4 = -\frac{6}{(x-3)^4} - \frac{6}{(x+3)^4}$$
$$= -6\left[\frac{1}{(x-3)^4} + 1/(x+3)^4\right]$$

Q#5)

$if\ x = sinm\theta, y = cos\theta$, show that $(1-x^2)y_2 - xy_1 + m^2y = 0$

Sol: Since $x = sinm\theta \Rightarrow \theta = \frac{1}{m}\sin^{-1}(x)$

And $y = cos\theta \Rightarrow \theta = \cos^{-1}(y)$

$\Rightarrow \frac{1}{m}\sin^{-1}(x) = \cos^{-1}(y)$

Differentiate w.r.t x

$\frac{1}{m}\frac{d}{dx}(\sin^{-1}(x)) = \frac{d}{dx}(\cos^{-1}(y)$

$\frac{1}{m}\frac{1}{\sqrt{1-x^2}} = \frac{-1}{\sqrt{1-y^2}}(y_1)$

$\Rightarrow m\sqrt{1-y^2} = -\sqrt{1-x^2}y_1$

On squaring

$(1-x^2)y_1^2 = m^2(1-y^2)$

Differentiating again w.r.t x

$(1-x^2)(2y_1y_2) + (-2x)(y_1^2) = m^2(-2yy_1)$

Taking common $2y_1$

$2y_1[(1-x^2)y_2 - xy_1] = -2m^2yy_1$

$(1-x^2)y_2 - xy_1 = -m^2y$

$$(1-x^2)y_2 - xy_1 + m^2y = 0$$

Hence proved.

Q#6) $If\ y = e^x sinx$, show that $\frac{d^2y}{dx^2} - \frac{2dy}{dx} + 2y = 0$

Sol: $y = e^x sinx$

Differentiate w.r.t x

$\frac{dy}{dx} = \frac{d}{dx}(e^x sinx)$

$\frac{dy}{dx} = (e^x)(cosx) + (sinx)(e^x)$

$\frac{dy}{dx} = e^x(cosx + sinx)$

Differentiating again w.r.t x

$\frac{d^2y}{dx^2} = (e^x)(-sinx + cosx) + (cosx + sinx)(e^x)$

$\frac{d^2y}{dx^2} = e^x(-sinx + cosx + cosx + sinx)$

$\frac{d^2y}{dx^2} = e^x(2cosx)$

Now consider

$$\frac{d^2y}{dx^2} - \frac{2dy}{dx} + 2y$$
$$= \left(e^x(2cosx)\right) - 2\left(e^x(cosx + sinx)\right)$$
$$+ 2(e^x sinx)$$

$= e^x(2cosx - 2sinx - 2cosx + sinx)$

$= e^x(0)$

$\frac{d^2y}{dx^2} - \frac{2dy}{dx} + 2y = 0$

Hence proved.

Q#7) $If\ y = e^{ax}sinbx$, show that $\frac{d^2y}{dx^2} - 2a\frac{dy}{dx} + (a^2 + b^2)y = 0$

Sol: $y = e^{ax}sinbx$

$differentiate\ w.r.t\ \ x$

$\frac{dy}{dx} = \frac{d}{dx}(e^{ax}sinbx)$

$\frac{dy}{dx} = (e^{ax})(cosbx\ (b)) + (sinbx)(e^{ax}(a))$

$\frac{dy}{dx} = e^{ax}(bcosbx + asinbx)$

Differentiating again w.r.t x

$\frac{d^2y}{dx^2} = (e^{ax})(-bsinbx\ (b) + a\ cosbx\ (a)) + (bcosbx$

$+ asinbx)(e^{ax}(a))$

$\frac{d^2y}{dx^2} = e^{ax}(-b^2sinbx + abcosbx + abcosbx$

$+ a^2sinbx)$

$\frac{d^2y}{dx^2} = e^{ax}(-b^2sinbx + 2abcosbx + a^2sinbx)$

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Now consider

$$\frac{d^2y}{dx^2} - 2a\frac{dy}{dx} + (a^2+b^2)y$$
$$= \left(e^{ax}(-b^2 sinbx + 2ab\, cosbx + a^2 sinbx)\right) - 2a(e^{ax}(bcosbx + asinbx)) + a^2 + b^2(e^{ax}sinbx)$$
$$= e^{ax}(-b^2 sinbx + 2ab\, cosbx + a^2 sinbx - 2ab\, cosx - 2a^2 sinbx + a^2\, sinbx + b^2 sinbx)$$
$$= 0$$
$$\frac{d^2y}{dx^2} - 2a\frac{dy}{dx} + (a^2+b^2)y = 0$$

Hence proved.

Q#8) $\boldsymbol{If\ y = (cos^{-1}x)^2}$**,**

prove that $\boldsymbol{(1-x^2)y_2 - xy_1 - 2 = 0}$

Sol:

Let $y = (\cos^{-1}x)^2$

Differentiate w.r.t x

$$\frac{dy}{dx} = \frac{d}{dx}((\cos^{-1}x)^2)$$
$$y_1 = 2\cos^{-1}x\left(\frac{-1}{\sqrt{1-x^2}}\right)$$

On squaring

$$y_1^2 = 4(\cos^{-1}x)^2\left(\frac{1}{1-x^2}\right)$$
$$(1-x^2)y_1^2 = 4(\cos^{-1}x)^2$$
$$(1-x^2)y_1^2 = 4y$$

Differentiating again w.r.t x

$$(1-x^2)(2y_1y_2) + (-2x)(y_1^2) = 4y_1$$

Taking common $2y_1$

$$2y_1[(1-x^2)y_2 - xy_1] = 4y_1$$
$$(1-x^2)y_2 - xy_1 = 2$$
$$(1-x^2)y_2 - xy_1 - 2 = 0$$

Hence proved.

Q#9) If $y = a\cos(lnx) + b\sin(lnx)$, prove that $x^2\frac{d^2y}{dx^2} + x\frac{dy}{dx} + y = 0$

Solution.

Let $y = a\cos(lnx) + b\sin(lnx)$

Differentiate w.r.t x

$$\frac{dy}{dx} = \frac{d}{dx}(a\cos(lnx) + b\sin(lnx))$$
$$\frac{dy}{dx} = a\left(-\sin(lnx)\left(\frac{1}{x}\right)\right) + b\left(\cos(lnx)\left(\frac{1}{x}\right)\right)$$

Multiply by x

$$x\frac{dy}{dx} = -a\sin(lnx) + b\cos(lnx)$$

Differentiating again w.r.t x

$$(x)\left(\frac{d^2y}{dx^2}\right) + (1)\left(\frac{dy}{dx}\right) = -a\left(\cos(lnx)\left(\frac{1}{x}\right)\right) + b\left(-\sin(lnx)\left(\frac{1}{x}\right)\right)$$
$$x\frac{d^2y}{dx^2} + \frac{dy}{dx} = -\left(\frac{1}{x}\right)(a\cos(lnx) + b\sin(lnx))$$

Multiply by x

$$x^2\frac{d^2y}{dx^2} + x\frac{dy}{dx} = -y$$
$$x^2\frac{d^2y}{dx^2} + x\frac{dy}{dx} + y = 0$$

Hence proved

McLaurin Series: *The series*

$f(x) = f(0) + xf'(o) + \frac{x^2}{2!}f''(0) + \cdots + \frac{x^n}{n!}f^n(x)$ is known as "McLaurin Series expansion**.**

Proof:

We know that power series of $\boldsymbol{f(x)in\ x}$

$$f(x) = a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 + \cdots + a_nx^n \rightarrow (i)$$
$$\Rightarrow f(o) = a_0 \Rightarrow = f(o)$$
$$f'(x) = a_1 + 2a_2x + 3a_3x^2 + 4a_4x^3 + \cdots + na_nx^{n-1}$$
$$\Rightarrow f'(0) = a_1 \Longrightarrow a_1 = f'(0)$$
$$f''(x) = 2a_2 + 6a_3x + 12a_4x^2 + \cdots + n(n-1)a_nx^{n-2}$$
$$f''(x) = 2a_2 \Rightarrow a_2 = \frac{f''(0)}{2} = \frac{f''(0)}{3!}$$
$$f'''(x) = 6a_3 + 24a_4x + \cdots + n(n-1)(n-2)a_nx^{n-3}$$
$$\Rightarrow f'''(0) = 6a_3 \Rightarrow a_3 = \frac{f'''(0)}{6} = \frac{f'''(0)}{3!}$$

Similarly, $\Rightarrow a_n = \frac{f^n(0)}{n!}$

Putting valves of $a_1, a_2, a_3, .., a_n\ in\ (1)$

$$f(x) = f(0) + xf'(0) + \frac{x^2}{2}f''(0) + \frac{x^3}{3!}f'''(0) + \cdots + \frac{x^n}{n!}f^n(0$$

NOTE:

A function f can be expanded in Machianrium serie

If the function is defined in the interval containing 0 and its derivative exist at $\boldsymbol{x = 0}$ **the expansion is only valid if it is converge.**

The above expansion is also named as "McLaren's theorem"

Note:

McLaren's series for $\frac{1}{x+1}$ is the geometric series with first term 1 and common ratio x

$$\because S = \frac{a}{1-r}\ we\ have$$
$$1 - x + x^2 - x^3 + \cdots + \frac{1}{1-(-x)} = \frac{1}{1+x}$$

Taylor Series:

The series

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$$f(x+h)=f(x)+f'(x)h+\frac{f''(x)}{2!}h^2+\frac{f'''(x)}{3!}h^3+\cdots+\frac{f^n(x)}{n!}h^n$$

Is known as **"Taylor series expansion"**

Proof:

Let the power series of $f(x) in\ (x-a)$

$$f(x)=a_0+a_1(x-a)+a_2(x-a)^2+a_3(x-a)^3 + a_4(x-a)^4+\cdots+a_n(x-a)^n \rightarrow (1)$$

$$\Rightarrow f(a)=a_0+a_1(a-a)+a_2(a-a)^2+a_3(a-a) +\cdots+a_n(a-a)^n$$

$$\Rightarrow f(a)=a_0 \Rightarrow a_o=f(a)$$

$$f'(x)=a_1+2a_2(x-a)+3a_3(x-a)^2+4a(x-a)^3 +\cdots+na_n(x-a)^{n-1}$$

$$\Rightarrow \boldsymbol{f'(a)=a_1 \Rightarrow a_1=f'(a)}$$

$$\boldsymbol{f''(x)=2a_2+6a_3(x-a)2a_4(x-a)^2+\cdots + n(n-1)a_n(x-a)^{n-2}}$$

$$\Rightarrow \boldsymbol{f''(a)=6a_3 \Rightarrow a_3=\frac{f'''(a)}{3!}}$$

Similarly, $\boldsymbol{a_n=\frac{f^n(a)}{n!}}$

Putting valves of $\boldsymbol{a_1,a_2,a_3,\ldots,a_n\ in\ (1)}$

$$\boldsymbol{f(x)=f(a)+f'(a)(x-a)+\frac{f''(x)}{2!}(x-a)^2+ \frac{f'''(a)}{3!}(x-a)^3+\cdots+\frac{f^n(a)}{n!}(x-a)^n \rightarrow (2)}$$

This expansion is the Taylor series $\boldsymbol{for\ f\ at\ x=a.}$

Replace $\boldsymbol{x\ by\ x+h\ and\ a\ by\ x\ we\ get}$

$$\boldsymbol{f(x+h)=f(x)+f'(x)(x+h-x) +\frac{f''(x)}{2!}(a+h-x)^2 +\frac{f'''(x)}{3!}(x+h-x)^3+\cdots +\frac{f^n(x)}{n!}(x+h-h)^n+\cdots}$$

$$\boldsymbol{f(a+h)=f(x)+f'(x)(h)+\frac{f''(x)}{2!}(h)^2 +\frac{f'''(x)}{3!}(h)^3+\cdots+\frac{f^n(x)}{n!}(h)^n+\cdots}$$

Note:

$$if\ we\ put\ a=0\ in(2)then\ f(x) =f(0)+f'(0)x+\frac{f'''(o)}{2!}x^2+\cdots +\frac{f^n(0)}{n!}x^n$$

Which is Machlaurin's series.

The above expansion is also named as Taylor theorem.

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these notes are Witten by Amir Shehzad

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Exercise 2.8

Question # 1. Apply the Maclaurin series expansion to prove that:

(i) $\ln(1+x)=x-\frac{x^2}{2}+\frac{x^3}{4}-\frac{x^4}{4}+\ \ldots\ldots\ldots$

Solution. Let $f(x)=\ln(x+1)$

$$f(0)=\ln(0+1)=ln1=0$$
$$f'(x)=\frac{1}{x+1}$$
$$f'(0)=\frac{1}{0+1}=1$$
$$f''(x)=\frac{-1}{(x+1)^2}$$
$$f''(0)=\frac{-1}{(0+1)^2}=-1$$
$$f'''(x)=\frac{2}{(x+1)^3}$$
$$f'''(0)=\frac{2}{(0+1)^3}=2$$
$$f^{(4)}(x)=\frac{-6}{(x+1)^4}$$
$$f^{(4)}(0)=\frac{-6}{(0+1)^4}=-6$$

By Maclaurin Series

$$f(x)=f(0)+xf'(0)+\frac{x^2}{2!}f''(0)+\frac{x^3}{3!}f'''(0) +\frac{x^4}{4!}f^4(0)+\ \ldots\ldots\ldots$$

$$\ln(x+1)=0+x(1)+\frac{x^2}{2}(-1)+\frac{x^3}{6}(2)+\frac{x^4}{24}(-6) +\ \ldots\ldots\ldots$$

$$\ln(x+1)=x-\frac{x^2}{2}+\frac{x^3}{3}-\frac{x^4}{4}+\ \ldots\ldots\ldots$$

Hence Proved.

(ii) $\boldsymbol{\mathrm{Cosx}=1-\frac{x^2}{2!}+\frac{x^4}{4!}-\frac{x^6}{6!}+\ \ldots\ldots\ldots}$

Solution. Let $f(x)=\mathrm{Cosx}$

$$f(0)=Cos0=0$$
$$f'(x)=-Sinx$$
$$f'(0)=-Sin0=0$$
$$f''(x)=-Cosx$$
$$f''(0)=-Cos0=-1$$
$$f'''(x)=Sinx$$
$$f'''(0)=Sin0=0$$
$$f^{(4)}(x)=Cosx$$
$$f^{(4)}(0)=Cos0=1$$
$$f^{(5)}(x)=-Sinx$$
$$f^{(5)}(0)=-Sin0=0$$
$$f^{(6)}(x)=-Cosx$$
$$f^{(6)}(0)=-Cos0=-1$$

By Maclaurin Series

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$$f(x) = f(0) + xf'(0) + \frac{x^2}{2!}f''(0) + \frac{x^3}{3!}f'''(0) + \frac{x^4}{4!}f^4(0) + \frac{x^5}{5!}f^{(5)}(0) + \frac{x^6}{6!}f^{(6)}(0) \ \dots\dots\dots$$

$$Cosx = 1 + x(0) + \frac{x^2}{2!}(-1) + \frac{x^3}{3!}(0) + \frac{x^4}{4!}(1) + \frac{x^5}{5!}(0) + \frac{x^6}{6!}(-1) \ \dots\dots\dots$$

$$Cosx = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} \ \dots\dots\dots$$

Hence Proved.

(iii) $\boldsymbol{\sqrt{1+x} = 1 + \frac{x}{2} - \frac{x^2}{8} + \frac{x^3}{16} + \ \dots\dots\dots}$

Solution. Let $f(x) = \sqrt{1+x}$

$$f(0) = \sqrt{1+0} = 1$$
$$f'(x) = \frac{1}{2}(1+x)^{-\frac{1}{2}}$$
$$f'(0) = \frac{1}{2}(1+0)^{-\frac{1}{2}} = \frac{1}{2}$$
$$f''(x) = \frac{1}{2}.\frac{-1}{2}(1+x)^{-\frac{3}{2}}$$
$$f''(0) = \frac{1}{2}.\frac{-1}{2}(1+0)^{-\frac{3}{2}} = -\frac{1}{4}$$
$$f'''(x) = \frac{1}{2}.\frac{-1}{2}.\frac{-3}{2}(1+x)^{-\frac{5}{2}}$$
$$f'''(0) = \frac{1}{2}.\frac{-1}{2}.\frac{-3}{2}(1+0)^{-\frac{5}{2}} = \frac{3}{8}$$

By Maclaurin Series

$$f(x) = f(0) + xf'(0) + \frac{x^2}{2!}f''(0) + \frac{x^3}{3!}f'''(0) + \ \dots\dots\dots$$

$$\sqrt{1+x} = 1 + x\left(\frac{1}{2}\right) + \frac{x^2}{2!}\left(-\frac{1}{4}\right) + \frac{x^3}{3!}\left(\frac{1}{8}\right) + \ \dots\dots\dots$$

$$Cosx = 1 + \frac{x}{2} - \frac{x^2}{8} + \frac{x^3}{16} \ \dots\dots\dots$$

Hence Proved.

(iv) $\boldsymbol{e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \ \dots\dots\dots}$

Solution. Let $f(x) = e^x$

$$f(0) = e^0 = 1$$
$$f'(x) = e^x$$
$$f'(0) = e^0 = 1$$
$$f''(x) = e^x$$
$$f''(0) = e^0 = 1$$
$$f'''(x) = e^x$$
$$f'''(0) = e^0 = 1$$

By Maclaurin Series

$$f(x) = f(0) + xf'(0) + \frac{x^2}{2!}f''(0) + \frac{x^3}{3!}f'''(0) + \ \dots\dots\dots$$

$$e^x = 1 + x(1) + \frac{x^2}{2!}(1) + \frac{x^3}{3!}(1) + \ \dots\dots\dots$$

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} \ \dots\dots\dots$$

Hence Proved.

(v) $e^{2x} = 1 + 2x + \frac{4x^2}{2!} + \frac{8x^3}{3!} + \ \dots\dots\dots$

Solution. Let $f(x) = e^{2x}$

$$f(0) = e^{2(0)} = 1$$
$$f'(x) = 2e^{2x}$$
$$f'(0) = 2e^0 = 2$$
$$f''(x) = 4e^{2x}$$
$$f''(0) = 4e^0 = 4$$
$$f'''(x) = 8e^{2x}$$
$$f'''(0) = 8e^0 = 8$$

By Maclaurin Series

$$f(x) = f(0) + xf'(0) + \frac{x^2}{2!}f''(0) + \frac{x^3}{3!}f'''(0) + \ \dots\dots\dots$$

$$e^{2x} = 1 + x(2) + \frac{x^2}{2!}(4) + \frac{x^3}{3!}(8) + \ \dots\dots\dots$$

$$e^{2x} = 1 + 2x + \frac{4x^2}{2!} + \frac{8x^3}{3!} \ \dots\dots\dots$$

Hence Proved.

Question # 2 Show that $\boldsymbol{Cos(x+h) = Cosx - hSinx - \frac{h^2}{2!}Cosx + \frac{h^3}{3!}Sinx + \ \dots\dots\dots}$

And evaluate$\boldsymbol{Cos61^0}$.

Solution.

Let $f(x+h) = Cos(x+h)$

$$f(x) = \text{Cosx}$$
$$f'(x) = -Sinx$$
$$f''(x) = -Cosx$$
$$f'''(x) = Sinx$$

By Taylor Series

$$f(x+h) = f(x) + hf'(x) + \frac{h^2}{2!}f''(x) + \frac{h^3}{3!}f'''(x) + \ \dots\dots\dots$$

$$Cos(x+h) = Cosx + h(-Sinx) + \frac{h^2}{2!}(-Cosx) + \frac{h^3}{3!}(Sinx) + \ \dots\dots\dots$$

$$Cos(x+h) = Cosx - hSinx - \frac{h^2}{2!}Cosx + \frac{h^3}{3!}Sinx + \ \dots\dots\dots$$

Hence Proved.

$$Now\ , Put\ x = 60^0\ and\ h = 1^0 = \frac{\pi}{180}\ rad$$
$$= 0.01745\ rad$$

$$Cos(60^0 + 1^0) = Cos60^0 - (0.0174\ 5)Sin60^0 - \frac{(0.01745)^2}{2!}Cos60^0 + \frac{(0.01745)^3}{3!}Sin60^0 + \ \dots\dots\dots$$

$$Cos(61^0) = 0.5 - (0.01745)(0.866) - \frac{(0.01745)^2}{2!}(0.5) + \frac{(0.01745)^3}{3!}(0.866) + \ \dots\dots\dots$$

$$Cos(61^0) = 0.5 - 0.0151117 - 0.000076125 + 0.000000072 + \ \dots\dots\dots$$

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$$Cos(61^0) = 0.484812247 \approx 0.4848$$

Which is required.

Question # 3 Show that $2^{x+h} = 2^x\left[1 + (ln2)h + (ln2)^2\frac{h^2}{2!} + (ln2)^3\frac{h^3}{3!} + \dots\dots\dots\right]$

Solution.

Let $f(x+h) = 2^{x+h}$

$$f(x) = 2^x$$
$$f'(x) = 2^x.lna$$
$$f''(x) = -Cosx$$
$$f'''(x) = Sinx$$

By Taylor Series

$$f(x+h) = f(x) + hf'(x) + \frac{h^2}{2!}f''(x) + \frac{h^3}{3!}f'''(x) + \dots\dots\dots$$

$$Cos(x+h) = Cosx + h(-Sinx) + \frac{h^2}{2!}(-Cosx) + \frac{h^3}{3!}(Sinx) + \dots\dots\dots$$

$$Cos(x+h) = Cosx - hSinx - \frac{h^2}{2!}Cosx + \frac{h^3}{3!}Sinx + \dots\dots\dots$$

$$Hence\ Proved$$

Geometrical meaning of a derivative:

Let us draw a curve of the function $y = f(x)$ take two neighboring points $P(x,y)$ and $Q(x+\delta x, y+\delta y)$ on the curve. Draw a tangent line to the curve at pt. $p(x,y)$ Such that it makes an angle φ with $x-$ axis .

Also Draw a secant line to the curve passing through the pts. $p(x,y) and\ Q(x+\delta x, y+\delta y). the\ secent\ line\ makes\ angle\ \emptyset\ with\ x-axis.$

Draw $\perp\ PM\ and\ QN\ from\ P\ and\ Q\ on\ x-axis. Also\ draw\ perpendicular\ PR\ on\ QN. in\ \triangle PQR$

$$tan\emptyset = \frac{|QR|}{|PR|} \quad |QR| = \delta y \quad |PR| = \delta x$$

$$\Rightarrow Tan\emptyset = \frac{\delta y}{\delta x}$$

When $Q \to P\ \ then\ \delta x \to 0 \quad \delta y \to 0$

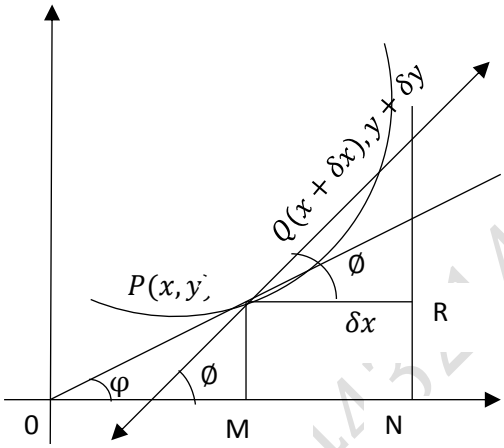
$\emptyset \to \varphi\ it\ means\ secent\ line\ becomes\ tangent\ line.$

Now,

$$\lim_{\emptyset\to\varphi} Tan\emptyset = \lim_{\emptyset\to\varphi}\frac{\delta y}{\delta x} \Rightarrow Tan\varphi = \frac{dy}{dx}$$

But $Tan\varphi$ is slope of tangent line. Thus geometrically derivative of

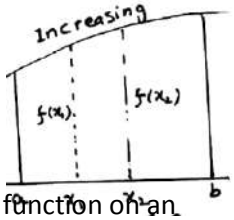
$y\ at\ x\ represent\ the\ slope\ of\ tangent\ line\ at\ p(x,y)$



Increasing and decreasing function:

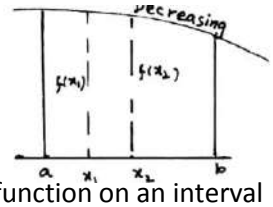
Increasing functions:

A function f is said to be increasing function on an interval (a,b) if for every $x_1, x_2 \in (a,b)\ then$ $f(x_1) < f(x_2)\ whenever\ x_1 < x_2$



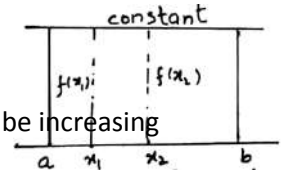
Decreasing function:

A function f is said to be decreasing function on an interval (a,b) if for every $x_1, x_2 \in (a,b)\ then$ $f(x_1) > f(x_2)\ whenever\ x_1 < x_2$



Constant function:

A function f is said to be constant function on an interval (a,b) if for every $x_1, x_2 \in (a,b)\ then$ $f(x_1) = f(x_2)\ whenever\ x_1 < x_2$



Alternative definitions:

Increasing functions:

A differentiable function f is said to be increasing function $if\ f'(x) > 0\ \forall x \in (a,b)$

Decreasing function:

A differentiable function f is said to be decreasing function $if\ f'(x) < 0\ \forall x \in (a,b)$

Constant function:

A differentiable function f is said to be constant function $if\ f'(x) = 0\ \forall x \in (a,b)$

Stationary point:

Any point where f is neither increasing nor decreasing is called stationary point.

Provided that $f'(x) = 0\ at\ that\ point.$

Relative Maxima:

A function $f(x)$ has relative maxima $f(c)$ at $x = c \in (a,b) if$

$(i) f(c-\delta x) > 0 \quad (ii) f'(c) = 0 \quad (iii) f'(c+\delta x) < 0$

$where\ \delta x\ is\ small + ve\ is\ small\ number.$

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Relative Minima:

A function $f(x)$ has relative minima $f(c)$ at $x = c \in (a,b)$ if

$(i) f(c-\delta x) < 0 \quad (ii) f'(c) = 0 \quad (iii) f'(c+\delta x) > 0$

$where\ \delta x\ is\ small + ve\ is\ small + ve\ number.$

Alternative definitions.

Relative maxima:

A function $f(x)$ has relative maxima $f(c)$ if $f'(x)$ $chnages\ sign\ from + ve\ to - ve\ as\ x\ change\ through\ C.$

Relative minima:

A function $f(x)$ has relative minima $f(c)$ if $f'(x)$ $chnages\ sign\ from - ve\ to + ve\ as\ x\ change\ through\ C$

Relative Extrema:

Both relative maxima and relative minima are called in general "relative extreme"

Critical Valves and critical points:

$if\ c \in D_f\ and\ f'(x) = 0\ or\ f'(c) does\ not\ exist,$

Them the number $c\ is\ called\ a\ critical\ value\ for\ f$ $while\ the\ point\ (c, f(x)) on\ the\ graph\ of\ f\ is$ Named as a critical as a critical point.

(i)

Important note:

There are functions which have extrema (maxima or minima) at the points where there derivatives do not exist. For example

$$f(x) = |x| \quad and \quad \emptyset(x) = \begin{cases} 2-x & , x > 0 \\ x+2, & x \le 0 \end{cases}$$

$do\ not\ exist\ at\ (0,0)\ and\ (0,20)\ resp.$

But

$f\ has\ minima\ at\ (0,0) and\ \emptyset\ has\ relative\ maxima\ at\ (0,2)$

First derivative:

Let $f\ be\ diffrentiable\ in\ neighbourhood\ of\ C\ where\ f'(c) = 0$

1. A function $f(x) has\ relative\ maxima\ at\ x = c \in (a,b)\ if\ f'(x) has\ realtive\ maxima\ at\ x = f'(c-\delta x) < 0$
2. A function $f(x) has\ relative\ minima\ at\ x = c \in (a,b)\ if\ f'(c-\delta x) < 0, f'(c) = 0\ and\ f'(c+\delta x) > 0$

Second derivative:

Let $f\ be\ diffrentiable\ in\ neighbourhood\ of\ C\ where\ f'(c) = 0$

Then,

1. $f\ has\ relative\ maxima\ at\ c\ if\ f''(c) < 0$
2. $f\ has\ relative\ minima\ at\ c\ if\ f''(c) > 0$

Turning point:

A stationary point is called a turning point or a minimum point.

Point of inflexion:

A point at which the function has neither maximum nor minimum value is called point of inflexion.

Procedure for finding maxima and minima of a function:

$Let\ y = f(x) be\ given\ function$

Step1. $Find \frac{dy}{dx}\ or\ f'(x)$

Step11: put $\frac{dy}{dx} = 0\ f'(x) = 0$

$stepIII \quad find \frac{d^2y}{dx^2}\ or\ f"(x)\ at\ the\ roots.$

$StepIV \quad check\ the\ sign\ of\ f''(x) at\ the\ roots$

*if $f''(x) > 0\ then\ f(x) has\ minimum\ value$

*if $f''(x) < 0\ then\ f(x)\ has\ maximum\ value.$

*$if\ f''(x) = 0$

$then\ no\ information\ about\ the\ function\ then\ use\ first\ derivative\ rule$"

Exercise 2.9

Question No.1

Determine the intervals in which *f* is increasing or decreasing for the domain mentioned in each case

$\boldsymbol{f(x) = Sinx \quad ; x \in [-\pi, \pi]}$

Solution.

Given that

$$f(x) = Sinx \quad ; x \in [-\pi, \pi]$$

$$f'(x) = Cosx$$

$$Cosx = 0$$

$$x = -\frac{\pi}{2}, \frac{\pi}{2}$$

So we have Sub-intervals $\left(-\pi, -\frac{\pi}{2}\right), \left(-\frac{\pi}{2}, \frac{\pi}{2}\right), \left(\frac{\pi}{2}, \pi\right)$

$f'(x) = Cosx < 0\ whenever\ x \in \left(-\pi, -\frac{\pi}{2}\right)$

$So\ f(x) is\ decreasing\ in\ the\ interval\ \left(-\pi, -\frac{\pi}{2}\right)$

$f'(x) = Cosx > 0\ whenever\ x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

$So\ f(x) is\ increasing\ in\ the\ interval\ \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

$f'(x) = Cosx < 0\ \ whenever\ x \in \left(\frac{\pi}{2}, \pi\right)$

$So\ f(x) is\ decreasing\ in\ the\ interval\ \left(\frac{\pi}{2}, \pi\right)$

(ii) $\boldsymbol{f(x) = Cosx \quad ; x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)}$

Solution.

Given that

$$f(x) = Cosx \quad ; x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$$

$$f'(x) = -Sinx$$

$$Put\ f'(x) = 0$$

$$-Sinx = 0$$

$$x = 0$$

So we have Sub-intervals $\left(-\frac{\pi}{2}, 0\right), \left(0, \frac{\pi}{2}\right)$

$f'(x) = -Sinx > 0\ whenever\ x \in \left(-\frac{\pi}{2}, 0\right)$

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$So\ f(x) is\ increasing\ in\ the\ interval\ \left(-\frac{\pi}{2}\ ,0\right)$

$f'(x) = -Sinx < 0\ whenever\ x \in \left(0\ ,\frac{\pi}{2}\right)$

$So\ f(x) is\ decreasing\ in\ the\ interval\ \left(0\ ,\frac{\pi}{2}\right).$

(iii) $\boldsymbol{f(x) = 4 - x^2 \quad ; x \in [-2, 2]}$

Solution.

Given that

$$f(x) = 4 - x^2 \quad ; x \in [-2,2]$$

$$f'(x) = -2x$$

$$Put\ f'(x) = 0$$

$$-2x = 0$$

$$x = 0$$

So we have Sub-intervals $(-2\ ,0\ ), (0\ ,2)$

$f'(x) = -2x > 0\ whenever\ x \in (-2\ ,0\ )$

$So\ f(x) is\ increasing\ in\ the\ interval\ (-2\ ,0\ )$

$f'(x) = -2x < 0\ whenever\ x \in (0\ ,2)$

$So\ f(x) is\ decreasing\ in\ the\ interval\ (0\ ,2).$

(iv) $\boldsymbol{f(x) = x^2 + 3x + 2 \quad ; x \in [-4, 1]}$

Solution.

$$\Rightarrow f'(x) = 2x + 3$$

$Put\ f'(x) = 0 \Rightarrow 2x + 3 = 0$

$$\Rightarrow x = -\frac{3}{2}$$

$$\because 2x + 3\ is + ve\ \ in \left(-\frac{3}{2}, 1\right)$$

So $f(x) is\ increasing\ on\ the\ interval\ \left(-\frac{3}{2}, 1\right)$

$\because 2x + 3\ is - ve\ in\ \left(-4, -\frac{3}{2}\right)\ so\ f(x) is\ decreasing$

On the interval $\left(-4, -\frac{3}{2}\right)$

Question. 2.

Find the extreme values of the following functions defined as

(i) $\boldsymbol{f(x) = 1 - x^3}$

Solution.

Given that

$$f(x) = 1 - x^3$$

Differentiate w.r.t $''x''$

$$f'(x) = -3x^2 - - - (i)$$

For stationary points. Put $f'(x) = 0$

$$-3x^2 = 0$$

$$x = 0$$

Differentiate (i) w.r.t $''x'', we\ have$

$$f''(x) = -6x - - - (ii)$$

Now put $x = 0\ in\ (ii)$

$$f''(0) = -6(0) = 0$$

So second derivative test fails to determine the extreme points

Put $x = 0 - \varepsilon\ in\ (i)$

$$f'(x) = -3(-\varepsilon)^2 = -3\varepsilon^2 < 0$$

Put $x = 0 + \varepsilon\ in\ (i)$

$$f'(x) = -3(-\varepsilon)^2 = -3\varepsilon^2 < 0$$

As $f'(x)$ does not change its sign before and after $x = 0$. Since $at\ x = 0\ , f(x) = 1$ therefore $(0,1)$ is the point inflection.

(ii) $\boldsymbol{f(x) = x^2 - x - 2}$

Solution.

Given that

$$f(x) = x^2 - x - 2$$

Differentiate w.r.t $''x''$

$$f'(x) = 2x - 1 \rightarrow (i)$$

For stationary points , Put $f'(x) = 0$

$$2x - 1 = 0$$

$$x = \frac{1}{2}$$

Differentiate (i) w.r.t $''x'', we\ have$

$$f''(x) = 2 - - - (ii)$$

Now put $x = \frac{1}{2}\ in\ (ii)$

$$f''\left(\frac{1}{2}\right) = 2$$

So second derivative test $f(x) is\ minimum\ at\ x = \frac{1}{2}$.

Now $f\left(\frac{1}{2}\right) = \left(\frac{1}{2}\right)^2 - \frac{1}{2} - 2$

$$f\left(\frac{1}{2}\right) = \frac{1}{4} - \frac{1}{2} - 2 = -\frac{9}{4}.$$

(iii) $\boldsymbol{f(x) = 5x^2 - 6x + 2}$

Solution.

Given that

$$f(x) = 5x^2 - 6x + 2$$

Differentiate w.r.t $''x''$

$$f'(x) = 10x - 6 - - - (i)$$

For stationary points. Put $f'(x) = 0$

$$10x - 6 = 0$$

$$x = \frac{3}{5}$$

Differentiate (i) w.r.t $''x'', we\ have$

$$f''(x) = 10 - - - (ii)$$

Now put $x = \frac{3}{5}\ in\ (ii)$

$$f''\left(\frac{3}{5}\right) = 10$$

So second derivative test $f(x) is\ minimum\ at\ x = \frac{3}{5}$.

Now

$$f\left(\frac{3}{5}\right) = 5\left(\frac{3}{5}\right)^2 - 6\left(\frac{3}{5}\right) + 2$$

$$f\left(\frac{3}{5}\right) = \frac{9}{5} - \frac{18}{5} + 2 = \frac{1}{5}.$$

(iv) $\boldsymbol{f(x) = 3x^2}$

Solution.

Given that

$$f(x) = 3x^2$$

Differentiate w.r.t $''x''$

$$f'(x) = 6x - - - (i)$$

For stationary points. Put $f'(x) = 0$

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$$6x = 0$$
$$x = 0$$

Differentiate (i) w.r.t $''x''$, *we have*

$$f''(x) = 6 - - - (ii)$$

Now put $x = 0$ *in* (ii)

$$f''(0) = 6$$

So second derivative test $f(x)$*is minimum at* $x = 0$.

Now

$$f(0) = 3(0)^2$$
$$f(0) = 0.$$

(v) $\boldsymbol{f(x) = 3x^2 - 4x + 5}$

Solution.

$$\Rightarrow f'(x) = 6x - 4 \Rightarrow f''(x) = 6 \rightarrow (i)$$
$$Put\ f'(x) = 0 \Rightarrow 6x = 0 \Rightarrow x = 0\ put\ in (i)$$
$$\Rightarrow f''(o) = 6 > 0\ so\ f(x) hax\ minimum\ value\ \ at$$
$$x = 0\ and\ f(0) = 3(0)^2 = 0$$

(vi) $\boldsymbol{f(x) = 2x^3 - 2x^2 - 36x + 3}$

Solution.

Given that

$$f(x) = 2x^3 - 2x^2 - 36x + 3$$

Differentiate w.r.t $''x''$

$$f'(x) = 6x^2 - 4x - 36 - - - (i)$$

For stationary points . Put $f'(x) = 0$

$$6x^2 - 4x - 36 = 0$$
$$3x^2 - 2x - 18 = 0$$
$$x = \frac{2 \pm \sqrt{(2)^2 - 4(3)(-18)}}{2(3)}$$
$$x = \frac{2 \pm \sqrt{4 + 216}}{6}$$
$$x = \frac{2 \pm \sqrt{220}}{6}$$
$$x = \frac{2 \pm 2\sqrt{55}}{6}$$
$$x = \frac{1 \pm \sqrt{55}}{3}$$

Differentiate (i) w.r.t $''x''$, *we have*

$$f''(x) = 12x - 4 - - - (ii)$$

Now put $x = \frac{1+\sqrt{55}}{3}$ *in* (ii)

$$f''\left(\frac{1+\sqrt{55}}{3}\right) = 12\left(\frac{1+\sqrt{55}}{3}\right) - 4$$

$$f''\left(\frac{1+\sqrt{55}}{3}\right) = 4(1+\sqrt{55}) - 4$$

$$f''\left(\frac{1+\sqrt{55}}{3}\right) = 4 + 4\sqrt{55} - 4$$

$$f''\left(\frac{1+\sqrt{55}}{3}\right) = 4\sqrt{55} > 0$$

So second derivative test $f(x)$*is minimum at* $x = \frac{1+\sqrt{55}}{3}$.

Now

$$f\left(\frac{1+\sqrt{55}}{3}\right) = 2\left(\frac{1+\sqrt{55}}{3}\right)^3 - 2\left(\frac{1+\sqrt{55}}{3}\right)^2 - 36\left(\frac{1+\sqrt{55}}{3}\right) + 3$$

$$f\left(\frac{1+\sqrt{55}}{3}\right) = \frac{2}{27}(1+\sqrt{55})^3 - \frac{2}{9}(1+\sqrt{55})^2 - \frac{36}{3}(1+\sqrt{55}) + 3$$

$$f\left(\frac{1+\sqrt{55}}{3}\right) = \frac{2}{27}(1 + 3\sqrt{55} + 3.55 + 55\sqrt{55}) - \frac{2}{9}(1 + 2\sqrt{55} + 55) - 12(1+\sqrt{55}) + 3$$

$$f\left(\frac{1+\sqrt{55}}{3}\right) = \frac{2}{27}(166 + 58\sqrt{55}) - \frac{2}{9}(56 + 2\sqrt{55}) - 12(1+\sqrt{55}) + 3$$

$$f\left(\frac{1+\sqrt{55}}{3}\right) = \frac{332}{27} + \frac{116}{27}\sqrt{55} - \frac{112}{9} - \frac{4}{9}\sqrt{55} - 12 - 12\sqrt{55} + 3$$

$$f\left(\frac{1+\sqrt{55}}{3}\right) = -\frac{1}{27}(247 + 220\sqrt{55})$$

Now put $x = \frac{1-\sqrt{55}}{3}$ *in* (ii)

$$f''\left(\frac{1-\sqrt{55}}{3}\right) = 12\left(\frac{1-\sqrt{55}}{3}\right) - 4$$

$$f''\left(\frac{1-\sqrt{55}}{3}\right) = 4(1-\sqrt{55}) - 4$$

$$f''\left(\frac{1\pm\sqrt{55}}{3}\right) = 4 - 4\sqrt{55} - 4$$

$$f''\left(\frac{1-\sqrt{55}}{3}\right) = -4\sqrt{55} < 0$$

So second derivative test $f(x)$*is maximum at* $x = \frac{1-\sqrt{55}}{3}$.

Now

$$f\left(\frac{1-\sqrt{55}}{3}\right) = 2\left(\frac{1-\sqrt{55}}{3}\right)^3 - 2\left(\frac{1-\sqrt{55}}{3}\right)^2 - 36\left(\frac{1-\sqrt{55}}{3}\right) + 3$$

$$f\left(\frac{1-\sqrt{55}}{3}\right) = \frac{2}{27}(1-\sqrt{55})^3 - \frac{2}{9}(1-\sqrt{55})^2 - \frac{36}{3}(1-\sqrt{55}) + 3$$

$$f\left(\frac{1-\sqrt{55}}{3}\right) = \frac{2}{27}(1 - 3\sqrt{55} + 3.55 - 5\sqrt{55}) - \frac{2}{9}(1 - 2\sqrt{55} + 55) - 12(1-\sqrt{55}) + 3$$

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$$f\left(\frac{1+\sqrt{55}}{3}\right)=\frac{2}{27}(166-58\sqrt{55})-\frac{2}{9}(56-2\sqrt{55})$$
$$-12(1-\sqrt{55})+3$$
$$f\left(\frac{1+\sqrt{55}}{3}\right)=\frac{332}{27}-\frac{116}{27}\sqrt{55}-\frac{112}{9}+\frac{4}{9}\sqrt{55}-12$$
$$+12\sqrt{55}+3$$
$$f\left(\frac{1+\sqrt{55}}{3}\right)=-\frac{1}{27}(247-220\sqrt{55})$$

(vii) $\boldsymbol{f(x)=x^4-4x^2}$

Solution:

$\Rightarrow \boldsymbol{f'(x)}=4x^3-8x \Rightarrow f''(x)=12x^2-8\rightarrow(i)$

$Put\ f'(x)=0\ so\ 4x^3-8x=0$

$\Rightarrow 4x(x^2-8x)=0$

$\Rightarrow 4x(x^2-2)=$

$\Rightarrow\ 4x=0 \qquad x^2-2=0$

$\Rightarrow\ x=0 \qquad x=\pm\sqrt{2}$

$\Rightarrow\ thus\ x=0 \quad ,x=\sqrt{2} \quad ,x=-\sqrt{2}$

$Put\ x=0\ in\ (1)$

$\Rightarrow\ f''(0)=12(0)^2-8=-8<0$

$\Rightarrow$ So $f(x)has\ maximam\ value\ at\ x=o\ and$

$\Rightarrow\ f(0)=(0)^4-4(0)^2=0$

$Put\ x=\sqrt{2}\ in\ (1)$

$\Rightarrow\ f''(\sqrt{2})=12\left(\sqrt{2}\right)^2-8=12(2)-8=16>0$

$\Rightarrow$ So $f(x)has\ minimum\ value\ at\ x=\sqrt{2}\ and$

$$f(\sqrt{2})=12\left(\sqrt{2}\right)^2$$

$$=\left[\left(\sqrt{2}\right)^2\right]^2-4(2)=(2)^2-8=4-8=-4<0$$

Put $x=-\sqrt{2}\ in\ (1)$

$$f''(-\sqrt{2})=12\left(\sqrt{2}\right)^2-8=12(2)-8=24-8$$
$$=16>0$$

$so\ f(x)has\ minmum\ value\ at$

$$x=-\sqrt{2}\ ans\ f(-\sqrt{2})=\left(-\sqrt{2}\right)^4-4\left(-\sqrt{2}\right)^2$$
$$\left[\left(-\sqrt{2}\right)^2\right]^2-4(2)$$
$$=(2)^2-8=4-8=4$$

(viii) $\boldsymbol{f(x)=x^4-4x^2}$

Solution.

$\Rightarrow\ \boldsymbol{f'(x)}=\frac{d}{dx}\{(x-2)^2(x-1)\}$

$$=(x-2)\frac{d}{dx}(x-1)+(x-1)\frac{d}{dx}(x-2)^2$$
$$=(x-2)^2(1(x-1).2(x-2)\frac{d}{dx}(x-2)$$
$$=(x-2)^2+2(x-1)(x-2)(1)$$
$$f'(x)=(x-2)+2(x-1)(x-2)$$
$$=(x-2)\{x-2+2x-2\}$$
$$f'(x)=(x-2)(3x-4)$$
$$f''(x)=\frac{d}{dx}\{(x-2)(3x-4)\}$$
$$=(x-2)\frac{d}{dx}(3x-4)+(3x-4)\frac{d}{dx}(x-2)$$
$$(x-2)(3)+(3x-4)(1)$$
$$=3x-6+3x-4$$
$$=6x-10$$
$$Put\ f'(x)=0\ ,\qquad so\ (x-2)(3x-4)=0$$
$$x-2=0 \qquad 3x-4=0$$
$$x=2 \qquad x=\frac{4}{3}$$

$Put\ x=2\ in\ (1) f''(2)=6(2)-10=12-10=2>0$

$sp\ f(x)has\ minimum\ value\ at\ x=2\ and$

$$f(2)=(2-2)^2(2-1)=0$$

$put\ x=\frac{4}{3}\ in\ (1)\ f''\left(\frac{4}{3}\right)=6\left(\frac{4}{3}\right)-10=8-10=-2<0$

$so\ f(x)has\ maximum\ value\ at\ x=\frac{4}{3}\ and$

$$f\left(\frac{4}{5}\right)=\left(\frac{4}{3}-2\right)^2\left(\frac{4}{3}-1\right)=\left(\frac{4-6}{3}\right)^2\left(\frac{4-3}{3}\right)$$
$$=\left(-\frac{2}{3}\right)^2\left(\frac{1}{3}\right)=\frac{4}{27}$$

(ix) $\boldsymbol{f(x)=5+3x-x^3}$

$\Rightarrow\ f'(x)=3-3x^2$

$\Rightarrow\ f''(x)=-6x\rightarrow(1)$

$put\ f'(x)=0 \quad ,3-3x^2=0$

$\Rightarrow\ 3(1-x^2)=0 \quad 3\neq0\ so\ 1-x^2=0$

$\Rightarrow\ 1=x^2$

$\Rightarrow\ x=\pm1$

$Put\ x=1\ in\ (i)has\ maximum\ value\ at\ x$

$=1\ and\ f(1)=5+3(1)-(1)^3$

$=5+3-1=7$

$put\ x=-1\ in\ (i)f''(-1)=-6(-1)=6>0$

So f(x) $hax\ minimum\ value\ at\ x=1$

$and\ f(-1)=5+3(-1)-(-1)^3=(5-3-(-1)$

$f(-1)=2+1=3$

Question.3.

Find the maximum and minimum values of the function defined by the following equation occurring in the interval [0,2π]

$$\boldsymbol{f(x)=Sinx+Cos\ x}$$

Solution. Given function

$$f(x)=Sinx+Cos\ x \qquad ;x\in[0,2\pi]$$

Differentiate with respect to $''x''$.

$$f'(x)=Cosx-Sinx---(i)$$

For Stationary points , $Put\ f'(x)=0$

$$Cosx-Sinx=0$$
$$-Sinx=-Cosx$$
$$\frac{Sinx}{Cosx}=1$$
$$tanx=1$$
$$x=\tan^{-1}1=\frac{\pi}{4}\ ,\pi+\frac{\pi}{4}$$
$$x=\frac{\pi}{4}\ ,\frac{5\pi}{4} \qquad when\ x\epsilon[0,2\pi]$$

Now diff. $(i)\ w.r.t\ ''x''$

$$f''(x)=-Sinx-Cosx$$

Now put $x=\frac{\pi}{4}\ in\ (ii)$

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$$f''\left(\frac{\pi}{4}\right) = -Sin\left(\frac{\pi}{4}\right) - Cos\left(\frac{\pi}{4}\right) = -\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}} = -\frac{2}{\sqrt{2}}$$
$$< 0$$

So second derivative test $f(x) is\ MAximum\ at\ x = \frac{\pi}{4}$.

Now

$$f\left(\frac{\pi}{4}\right) = Sin\left(\frac{\pi}{4}\right) + Cos\left(\frac{\pi}{4}\right)$$
$$f\left(\frac{\pi}{4}\right) = \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} = \frac{2}{\sqrt{2}} = \sqrt{2}.$$

Now put $x = \frac{5\pi}{4}\ in\ (ii)$

$$f''\left(\frac{5\pi}{4}\right) = -Sin\left(\frac{5\pi}{4}\right) - Cos\left(\frac{5\pi}{4}\right)$$
$$= -\left(-\frac{1}{\sqrt{2}}\right) - \left(-\frac{1}{\sqrt{2}}\right)$$

$$f''\left(\frac{5\pi}{4}\right) = \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} = \frac{2}{\sqrt{2}} = \sqrt{2} > 0$$

So second derivative test $f(x) is\ Minimum\ at\ x = \frac{5\pi}{4}$.

Now

$$f\left(\frac{5\pi}{4}\right) = Sin\left(\frac{5\pi}{4}\right) + Cos\left(\frac{5\pi}{4}\right)$$
$$f\left(\frac{\pi}{4}\right) = -\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}} = -\frac{2}{\sqrt{2}} = -\sqrt{2}.$$

Question.4.

Show that $y = \frac{lnx}{x}$ has maximum value at $x = e$.

Solution. Given that

$$y = \frac{lnx}{x}$$

Diff. w r .t $''x'', we\ have$

$$\frac{dy}{dx} = \frac{d}{dx}\left(\frac{lnx}{x}\right)$$
$$\frac{dy}{dx} = \frac{x.\frac{1}{x} - lnx.1}{x^2}$$
$$\frac{dy}{dx} = \frac{1 - lnx}{x^2} - - - (i)$$

For critical points. $Put \frac{dy}{dx} = 0$

$$\frac{1 - lnx}{x^2} = 0$$
$$1 - lnx = 0$$
$$lnx = 1$$
$$lnx = lne$$
$$x = e$$

Now diff. (i) with respect to $''x'', we\ have$

$$\frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{1 - lnx}{x^2}\right)$$
$$\frac{d^2y}{dx^2} = \frac{x^2.\left(-\frac{1}{x}\right) - (1 - lnx)(2x)}{x^4}$$
$$\frac{d^2y}{dx^2} = \frac{-x - 2x + 2xlnx}{x^4}$$
$$\frac{d^2y}{dx^2} = \frac{-3x + 2xlnx}{x^4}$$

$At\ x = e$, we have

$$\frac{d^2y}{dx^2}|_{x=e} = \frac{-3(e) + 2(e)\ln(e)}{(e)^4}$$
$$\frac{d^2y}{dx^2}|_{x=e} = \frac{-3e + 2e}{e^4}$$
$$\frac{d^2y}{dx^2}|_{x=e} = \frac{-e}{e^4} = -\frac{1}{e^3} < 0$$

Hence $y\ has\ a\ maximum\ value\ at\ x = e$.

Question.5.

Show that $y = x^x$ has maximum value at $x = \frac{1}{e}$.

Solution. Given that

$$y = x^x$$

Taking Log on both sides

$$lny = lnx^x = xlnx$$

Diff. w r .t $''x'', we\ have$

$$\frac{d}{dx}(lny) = \frac{d}{dx}(xlnx)$$
$$\frac{1}{y}.\frac{dy}{dx} = x\left(\frac{1}{x}\right) + lnx.1$$
$$\frac{dy}{dx} = y(1 + lnx)$$
$$\frac{dy}{dx} = x^x(1 + lnx) - - - (i).$$

For critical points , $Put \frac{dy}{dx} = 0$

$$x^x(1 + lnx) = 0$$
$$As\ x^x \neq 0\ then\ ,\ 1 + lnx = 0$$
$$1 + lnx = 0$$
$$lnx = -1$$
$$lnx = -lne$$
$$lnx = lne^{-1}$$
$$x = e^{-1}$$
$$x = \frac{1}{e}$$

Now diff. (i) with respect to $''x'', we\ have$

$$\frac{d^2y}{dx^2} = \frac{d}{dx}\left(x^x(1 + lnx)\right)$$
$$\frac{d^2y}{dx^2} = (1 + lnx)\frac{d}{dx}x^x + x^x\frac{d}{dx}(1 + lnx)$$
$$\frac{d^2y}{dx^2} = (1 + lnx)x^x(1 + lnx) + x^x\left(\frac{1}{x}\right)$$
$$\frac{d^2y}{dx^2} = x^x\left[(1 + lnx)^2 + \frac{1}{x}\right]$$
$$\frac{d^2y}{dx^2} = x^x\left[\frac{x(1 + lnx)^2 + 1}{x}\right]$$

$At\ x = \frac{1}{e}$, we have

$$\frac{d^2y}{dx^2}|_{x=\frac{1}{e}} = \left(\frac{1}{e}\right)^{\frac{1}{e}}\left[\frac{\frac{1}{e}\left(1 + \ln\left(\frac{1}{e}\right)\right)^2 + 1}{\frac{1}{e}}\right]$$
$$\frac{d^2y}{dx^2} > 0$$

Hence $y\ has\ a\ minimum\ value\ at\ x = \frac{1}{e}$.

Application of maxima and minima:

1. If we first from the relation of the form $y = f(x)$ from the given information.

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2. Find the maximum or minimum value of f as required by using 1st or 2nd derivative value.

*mostly we will uses 2nd derivative rule.

Exercise 2.10

Question No1.find two integers whose sum is 20 and their product will be maximum.

Solution:

Let x and y be req. integer's product $f(x) = xy$

By given condition $x + y = 30$

$\Rightarrow \quad y = 30 - x \rightarrow (i)$

So, $f(x) = x(30 - x) = 30x - x^2$

$\Rightarrow f'(x) = 30 - 2x$

$\Rightarrow f''(x) = -2 \rightarrow (ii)$

Put $f'(x) = 0 \qquad 30 - 2x = 0$

$\Rightarrow \quad 2x = 30 \Rightarrow x = 15$ put in (ii)

$\Rightarrow \quad f''(15) = -2 < 0$ thus $f(x)$has maximum value

At $x = 15$ and put in $(i) \Rightarrow y = 30 - 15 = 15$

So $req.integer$ are 15 and 15.

Question No: 2 divide 20 into two parts so that the sum of their squares will be minimum.

Solution:

$let\ x\ and\ y\ be\ req.two\ parts\ of\ 20.so x + y = 20$

$\Rightarrow y = 20 - x \rightarrow (i)$

$by\ given\ condition\ f(x) = x^2 + y^2$

$\Rightarrow \quad f(x) = x^2 + (20 - x)^2 = x^2 + 400 + x^2 - 40x$

$\Rightarrow \quad f'(x) = 4x - 40 \Rightarrow f''(x) = 4 \rightarrow (ii)$

$\Rightarrow \quad Put\ f'(x) = 0 \quad 4x - 40 = 0$

$x - 10 = 0 \Rightarrow x = 10\ put\ in\ (i)$

$f''(10) = 4 > 0\ so\ f(x)hax\ minimum\ value\ at\ x = 10$

$and\ put\ in(i) y = 20 - 10 = 10$

So req. two parts of 20 are 10 and 10.

Q3. Find two positive integers whose sum is 12 and the product of one with the square of the other will be maximum.

Solution:

Let x and y be $req.integers\ by\ given\ condition;$

$x + y = 12\ y = 12 - x \rightarrow (i)$

Product $= f(x) = x^2 y = x^2\ (12 - x)$

$\Rightarrow \quad f(x) = 12x^2 - x^3$

$\Rightarrow \quad f'(x) = 24x - 3x^2 \qquad \Rightarrow f''(x) = 24 - 6x \rightarrow (ii)$

$put\ f'(x) = 0 \Rightarrow 24x - 30x^2 = 0 \Rightarrow 2(8 - x) = 0$

$\Rightarrow x = 0\ and\ x = 8$

$Put\ x = 0\ in\ (ii) \quad \Rightarrow f''(o) = 24 > 0$

So $f(x)has\ minimum\ value\ at\ x =$ $o\ which\ is\ not\ required$

$So\ Put\ x = 8\ in\ (ii) f''(8) = 24 - 6(8) = 24 - 48$

$f''(8) = -24 < 0$

$So\ f(x)hax\ maximum\ value\ at\ x = 8\ and\ put\ in\ (i)$

$y = 12 - 8 = 4\ \ so\ required\ are\ 8\ and\ 4.$

Question No.4 The perimeter of a triangle is 16cm. if one side of the other sides for maximum area of the triangle?

Solution:

Let $x\ and\ y\ be\ other\ sides.$

$\because\ perimeter = x + y + 6(sum\ of\ all\ sides)$

$\Rightarrow \quad 16 = x + y + 6$

$\Rightarrow \quad 16 - 6 = x + y$

$\Rightarrow \quad x + y = 10$

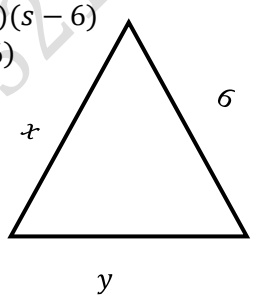
$\Rightarrow \quad$ And $y = 10 - x \rightarrow (i)$

$\because S = \dfrac{x + y + 6}{2} = \dfrac{16}{2} \quad or\ S = 8$

Also area $A^2 = f(x) = s(s - x)(s - y)(s - 6)$

$\Rightarrow \quad f(x) = 8(8 - x)(8 - y)(8 - 6)$

$\Rightarrow \quad f(x) = 16(8 - x)(8 - y)$



$= 16(8 - x)(8 - (10 - x))$

$f(x) = (8 - x)(x - 2)$

$f'(x) = 16\dfrac{d}{dx}\{(8 - x)(x - 2)\}$

$= 16\big((8 - x)(1) + (x - 2)(-1)\big)$

$= 16(8 - x - x + 2) = 16(10 - 2x)$

$f'(x) = 160 - 32x = 0$

$f''(X) = -32 \rightarrow (ii)$

Put $f'(x) = 0 \Rightarrow 160 - 32x = 0$

$\Rightarrow \quad 32x = 160 \Rightarrow x = 5$

$put\ x = 5\ in\ (ii) \ \Rightarrow f''(5) = -32 < 0$

Thus $f(x)has\ maximum\ value\ at\ x = 5\ and\ put\ in\ (i)$

$y = 10 - 5 = 5$

Hence req. sides are 5cm and 5cm.

Question No.5 find the dimension of a rectangular of largest area having perimeter 12cm.

Solution:

$Let\ x\ and\ y\ be\ required\ dimensions.$

$Perimeter = x + y + x + y(som\ of\ all\ sides)$

$\Rightarrow 120 = 2x + 2y$

$\Rightarrow x + y = 60$

And $y = 60 - x \rightarrow (i)$

$\because\ Area\ of\ f(x) = xy\ rectangle)$

$\Rightarrow f(x) = x(60 - x)$

$f(x) = 60x - x^2$

$\Rightarrow f'(x) = 60 - 2x$

$\Rightarrow f''(x) = -2 \rightarrow (ii)$

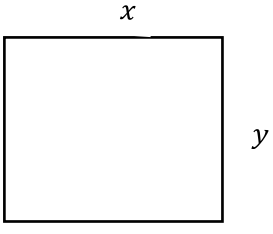
$Put\ f'(x) = 0 \ \ so\ 60 - 2x = 0$

$f''(x) = -2 \rightarrow (ii)$

$Put\ f'(x) = 0\ so\ 60 - 2x = 0$

$2x = 60 \Rightarrow x = \dfrac{60}{2} = 30\ put\ in\ (ii)$

$\Rightarrow f''(30) = -2 < 0$



$so\ f(x)hax\ maximum\ (largest)value\ at\ x = 30\ and\ put\ in\ (i) y = 60 - 30 = 30$

Thus required dimensions are $30cm^2$ where its perimter ?? minimum.

Question N0.6 find the length of sides of a variable rectangular having area $36cm^2$ where its permeter is minimum.

Solution:

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$let\ x\ and\ y\ be\ required\ length$

$$\boldsymbol{Perimeter = x + y + x + y}$$

$\Rightarrow\ f(x) = 2x + 2y$

$\Rightarrow\ f(x) = 2(x + y) \rightarrow (\boldsymbol{i})$

$\boldsymbol{area\ of\ reactangle = xy}$

$\Rightarrow\ \boldsymbol{36 = xy \Rightarrow y = \frac{36}{x}}$

$so(i) \Rightarrow f(x) = 2\left(x + \frac{36}{y}\right)$

$f(x) = 2x + 72^{-1}$

$f'(x) = 2 - \frac{72}{x^2}, \qquad f''(x) = -72(-2)x^{-3}$

$f''(x) = \frac{144}{x^3} \rightarrow (iii)$

$Put\ f'(x) = 0 \Rightarrow 2 - \frac{72}{x^2} = 0$

$2 = \frac{72}{x^2} \Rightarrow x^2 - 36 \Rightarrow x = \pm 6n$

$\Rightarrow\ x = 6 \qquad x = -6(rejected\ legth\ is\ always\ is + ve)$

$\Rightarrow$ Put $x = 6\ in\ (iii)\ f''(6) = \frac{144}{(6)^3} = \frac{144}{216} > 0$

So $f(x)$ has minimum value at $x = 6\ and\ put\ in (ii)$

$$\Rightarrow y = \frac{36}{6} = 6$$

So sides of rectangle are 6cm and 6cm.

Question No.7 A box with a square base and open top is top is to have a volume of 4 cube dm. find the dimensions of the box which will require the least material.

Solution:

Let length, width, and hieght of box are x,x and y repectively. So

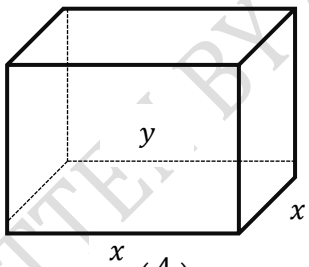
Volume=$x \times x \times y = x^2y \rightarrow (i)$

$\Rightarrow\ 4 = x^2y \Rightarrow y = \frac{4}{x^2}$

$\because\ surface\ area\ of\ the\ box =$
$(area\ of\ the\ square + area\ of\ 4\ verticle\ width)$

$\Rightarrow\ f(x) = x \times x \times 4(x \times y)$

$\Rightarrow\ f(x) = x^2 = 4xy$



$$x^2 + 4x\left(\frac{4}{x^2}\right) \qquad \because y = \frac{4}{x^2}$$

$$f(x) = x^2 + 16x^{-1} \Rightarrow f'(x) = 2x + 16(-1)x^{-2}$$

$$f'(x) = 2x - \frac{16}{x^2}$$

$$\Rightarrow f''(x) = 2 + 16(-1)(-2)\,x^{-3}$$

$$f''(x) = 2 + \frac{32}{x^3} \rightarrow (ii)$$

$$put\ f'(x) = 0\ so\ 2x - \frac{16}{x^2} = 0 \Rightarrow \frac{2x^3 - 16}{x^2} = 0$$

$$\Rightarrow 2(x^3 - 8) = 0 \Rightarrow (x)^3 - (2)^3 = 0$$

$$Put\ in\ (ii) f''(x) = 2 + \frac{32}{(2)^2} > 0$$

$\Rightarrow\ f(x)hax\ minimum\ (least)value\ at\ x = 2\ and$

$$put\ in\ (i)\ \Rightarrow y = \frac{4}{4} = 1$$

Hence $2dm, 2dm\ and\ 1\ dm\ is\ the\ dimension\ of\ box$

Question No.8 find the dimensions of a rectangular garden having perimeter 80m.if its area is to b maximum.

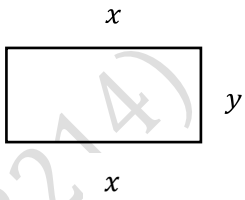
Solution: $\boldsymbol{Let\ x\ and\ be\ required\ dimensions.}$

$perimeter = x + y + x + y$

$80 = 2x + 2y \Rightarrow x + y = 40$

$\Rightarrow \boldsymbol{y = 40 - x \rightarrow (i)}$

$area\ of\ rectangular = f(x) =$



$$f(x) = xy = x(40 - x) = 40x - x^2$$

$f'(x) = 40 - 2x \Rightarrow f''(x) = -2 \rightarrow (ii)$

$put\ f'(x) = 0\ so\ 40 - 2x = 0 \Rightarrow 2x = 40$

$\Rightarrow x - 20\ put\ in\ (ii) \Rightarrow f''(20) = -< 0$

$\Rightarrow\ f(x)hax\ maximum\ value\ at\ x = 2$

$and\ put\ in\ (i)\ \ y = 40 - 20 \Rightarrow y = 20$

So dimensions of rectangular garden are 20m and 20m

Question No.9 An open tank of square base of side x and vertical sides is to be constructed to contain a given quantity of water. Find the depth in terms of x if the tank with lead will be least.

Solution:

Let length, width and height of tank with square base are $x, x, and\ y$ respectively.

$$\because volume = x \times x \times y = x^2y$$

$$y = \frac{v}{x^2} \rightarrow (i)$$

$$\because surface\ area =$$

$$area\ of\ square + area\ of\ four\ verticle\ walls$$

$$\Rightarrow f(x) = x^2 + 4xy$$

$$\Rightarrow f(x) = x^2 + 4x\left(\frac{v}{x^2}\right) = x^2 + \frac{4v}{x}$$

$$f(x) = x^2 + 4vx^{-1}$$

$$\Rightarrow f'(x) = 2x + 4v(-1)(x^{-2}) = 2x - \frac{4v}{x^2}$$

$$\Rightarrow f''(x) = 2 + 4v(-1)(-2)x^{-3} = 2 + \frac{8v}{x^3} \rightarrow (ii)$$

$$Put\ f'(x) = 0 \Rightarrow 2 + 4v(-1)(-2)(x^{-3}) = 2 + \frac{8v}{x^3}$$

$$\rightarrow (ii)$$

$$Put\ f'(x) = 0 \Rightarrow 2x - \frac{4v}{x^2} = 0$$

$\Rightarrow\ \frac{2x^3 - 4v}{x^2} = 0 \Rightarrow 2^3 = 4v$

$\Rightarrow\ x^3 = \frac{4v}{2} = 0 \Rightarrow x = (2v)^{\frac{1}{3}}\ put\ in\ (ii)$

$$f''(2v)^{\frac{1}{3}} = 2 + 8v\left[(2v)^{\frac{1}{3}}\right]^{-3} = 2 + 8v(2v)^{-1}$$

$$= 2 + \frac{8v}{2v} = 2 + 4 = 6 > 0$$

$\Rightarrow\ f(x)has\ minimum\ value\ at\ x = (2v)^{\frac{1}{3}}$

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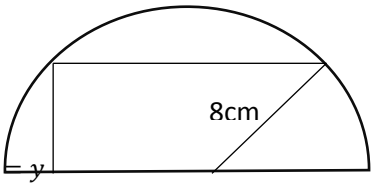
$$and\ put\ in\ (i) y = \frac{\frac{x^3}{2}}{x^2} = \frac{x}{2}$$

$y = \frac{x}{2}$ *which is required height (depth)interms of* x.

Question No.10 find the dimensions of the rectangular of maximum area which fits inside the semi –circle of radius 8cm as shown in figure.

Solution:

Let O be centre of semi − circle
let ABCD be rectangular with fits inside the semi − circle
Of radius 8cm as
Shown in figure.



let $|AC| = |BD| = y$
ALSO let $|AC| = |BD| = x$
so lenghtof rectangual $= 2x$ *and width of rectangual* $= y$
$\because$ *radius* $8cm(given)so\ |OC| = 8cm$
in $\Delta OAC\ \ x^2 + y^2 = (8)^2$ *using pathagourm theorem*

$$\Rightarrow y^2 = 64 - x^2 \Rightarrow y = \pm\sqrt{64 - x^2}$$
$$y = \sqrt{64 - x62} \rightarrow (i)$$
$$y = -\sqrt{64 - x^2}\ rejected\ being\ length$$
$$\because Area\ of\ rectangular = 2xy \Rightarrow A = f(x) = 2xy$$
$$\Rightarrow A^2 = f(x) = 4x^2(64 - x^2) = 256x^2 - 4x^4$$
$$\Rightarrow f'(x) = 519x2x - 16x^3$$
$$f'' = 512 - 48x^2 \rightarrow (ii)$$
$$put\ f'(x) = 0 \Rightarrow 512x - 16x^3 = 0$$
$$\Rightarrow\ \ 16x(32 - x^2) = 0$$
$$\Rightarrow\ \ 16x = 0 \qquad x^2 = 32 \Rightarrow x = \pm\sqrt{32}$$
$$\Rightarrow x = 0,\ \ x = \sqrt{32}\ ,\qquad x = -\sqrt{32}\ rejected\ beinglength$$

So put $x = put\ x = \sqrt{32}\ in\ (ii) \ \Rightarrow f''(x)\left(\sqrt{32}\right)$

$$= 512 - 48\left(\sqrt{32}\right)^2$$
$$\Rightarrow f''\left(\sqrt{32}\right) = 512 - 48(32) = -1024 < 0$$
$$\Rightarrow f(x) has\ \ maximum\ value\ at\ x = \sqrt{32}$$
$$and\ put\ in(i) y = \sqrt{64 - \left(\sqrt{32}\right)^2} = \sqrt{64 - 32} = \sqrt{32}$$
$$y = \sqrt{32}$$

Thus length of rectangular $= 2x = 2\left(4\sqrt{2}\right) = 8\sqrt{2}$
Width of rectangular $= y = 4\sqrt{2}$
Hence dimensions are $8\sqrt{2}$ *and* $4\sqrt{2}$

Question No.11 find the point on the curve $y = x^2 - 1$ that is closed to the point $(3, -1)$

Solution:

$$\because y = x^2 - 1 \rightarrow (i)$$

Let $p(x, y)$*be required point.*

$$Let\ d = distance\ b\backslash w(x, y) and\ (3, -1)$$
$$\Rightarrow\ \ d = \sqrt{(x-3)^2 + (y+1)^2}$$
$$\Rightarrow\ \ d^2 = (x-3)^2 + (y+1)^2 \qquad take\ d^2 = f(x)$$
$$\Rightarrow\ \ f(x) = (x-3)^2 + (y+1)^2$$
$$f(x) = x^2 + 9 - 6x + (x^2 - 1)^2 + 1 + 2(x^2 - 1)$$
$$\because y = x^2 - 1$$
$$= x^2 + 9 - 6x + x^4 + 1 - 2x^2 + 1 + 2x^2 - 2$$
$$f(x) = x^4 + x^2 - 6x + 9$$
$$\Rightarrow f'(x) = 4x^3 + 2x - 6$$
$$\Rightarrow f''(x) = 12x^2 + 2 \rightarrow (ii)$$

Put $f'(x) = 0 \ \Rightarrow 4x^3 + 2x - 6 = 0$
$2x^3 + x - 3 = 0 \ \ put\ x = 1$
$\Rightarrow (x - 1)(2x^2 + 2x + 3) = 0$
$\Rightarrow x = 1, 2x^2 + 2x + 3 = 0 \quad (rejected\ being\ complex)$
so put $x = 1\ in\ (ii)$

	2	0	1	-3
1	↓	2	2	3
	2	2	3	0

$2x^2 + 2x + 3\ and\ x - 1\ are\ factors\ of\ 2x^3 + x - 3 = 0$
$$f''(x) = 12(1)^2 + 2 = 14 > 0$$
$\Rightarrow f(x) has\ minimum\ value\ at\ x = 1\ and\ put\ in\ (i)$
$$\Rightarrow y = (1)^2 - 1 = 0$$
So $(1,0) is\ the\ required\ point\ closest\ to\ (3, -1)$

Question No.12

Find the point on the curve $y = x^2 - 1$ so $(1, 0)$*is the required point closet to* $(18, -1)$

Solution:

$$\because y = x^2 - 1 \rightarrow (i)$$

Let $p)x, y)$ *be the required point.*

$$\because d = distance\ b\backslash w\ p(x, y) amd\ (18,1)$$
$$\Rightarrow d = \sqrt{(x-18)^2 + (y-1)^2}$$
$$\Rightarrow d^2 = \sqrt{(x-18)^2 + (y-1)^2} \qquad take\ d^2 = f(x)$$
$$\Rightarrow f(x) = (x-18)^2 + (y-1)^2$$
$$f(x) = (x-18)^2 + (x^2 + 1 - 1) \because y = x^2 + 1$$
$$= x^2 + 324 - 36x + x^4$$
$$f(x) = x^4 + x^2 - 36x + x^4$$
$$f'(x) = 4x^3 + 2x - 36$$
$$f''(x) = 12x\text{^}2 + 2 \rightarrow (ii)$$
$$Put\ f'(x) = 0 \Rightarrow 4x^3 + 2x - 36 = 0$$
$\Rightarrow 2x^3 + x - 18 = 0 \qquad for\ x = 2$
$\Rightarrow (x - 2)(2x^2 + 4x + 9) = 0$
$x = 2, \ \ 2x^2 + 4x + 9 = 0 \ \ rejected\ being\ complex$
Put$x = 2\ in\ (ii)$

	2	0	1	-18
2	↓	4	8	18
	2	2	9	0

$2x^2 + 4x + 9\ and\ x - 2\ are\ factors\ of$
$$2x^3 + x - 18 = 0$$
$$put\ x = 2\ in\ (ii)$$
$$\Rightarrow f''(x) = 12(2)^2 + 2 > 0$$
$\Rightarrow f(x) has\ minimum\ value\ at\ x = 2\ put\ in\ (i)$
$\Rightarrow y = (2)^2 + 1 = 5$
So $(2,5) is\ the\ required\ point\ which\ is\ closest\ to\ (18,1$

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